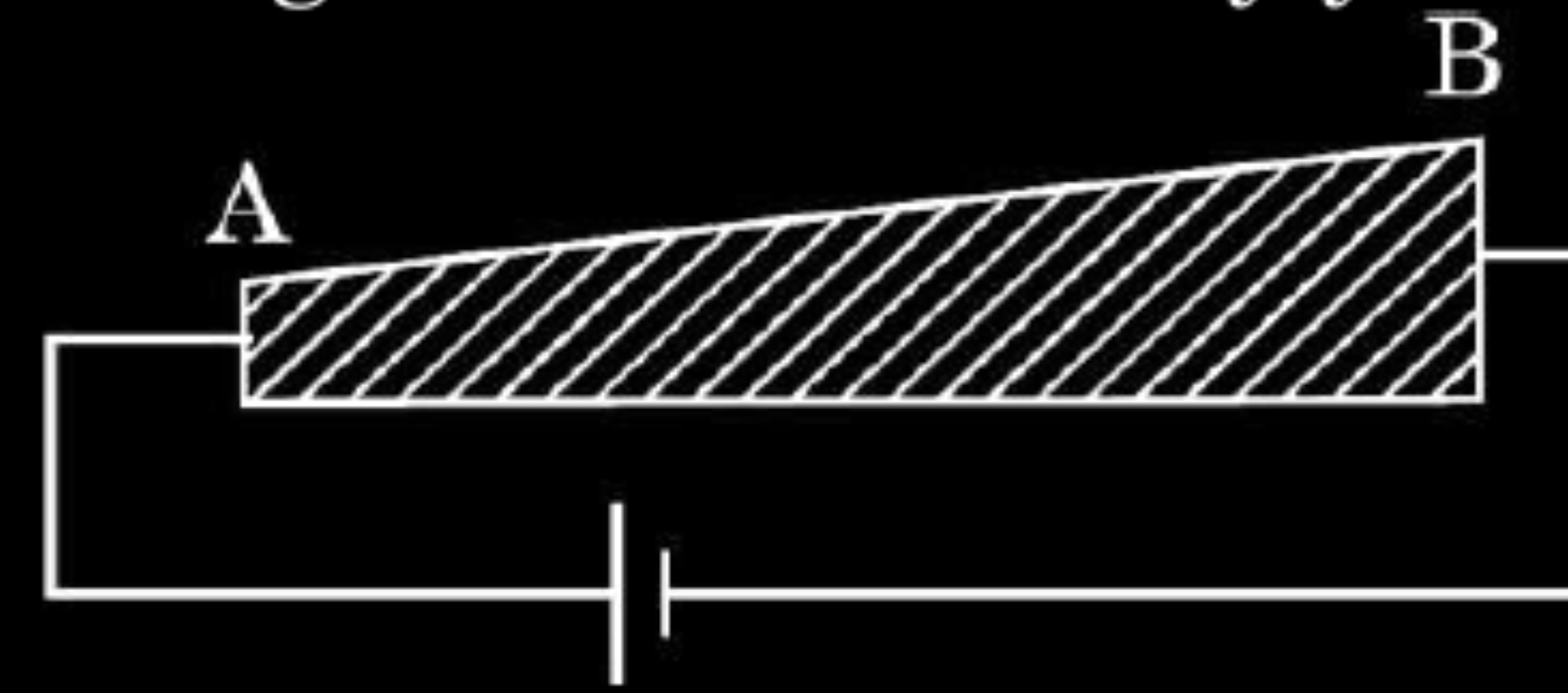
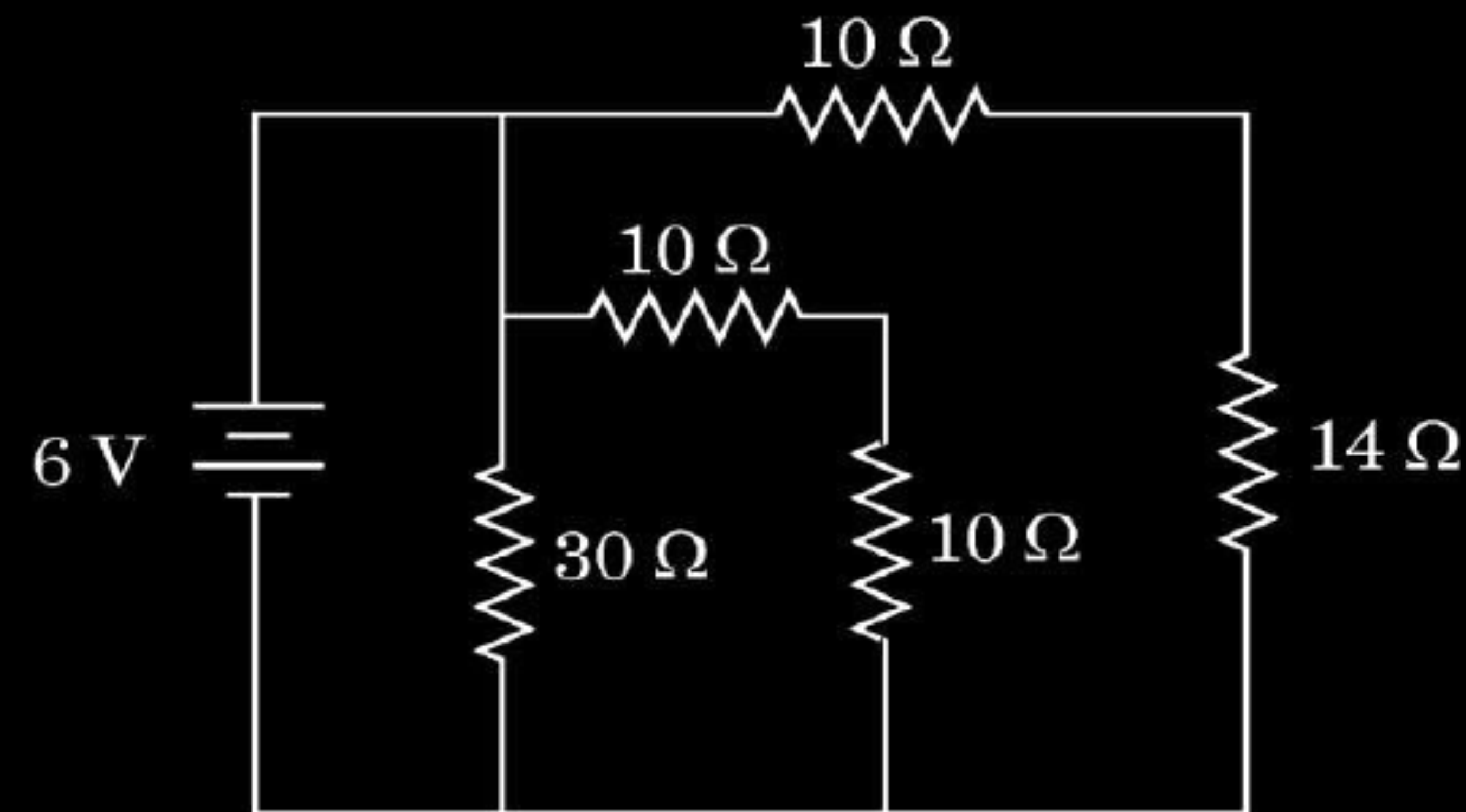


32. (a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire ? Justify your answer.



(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery.



5

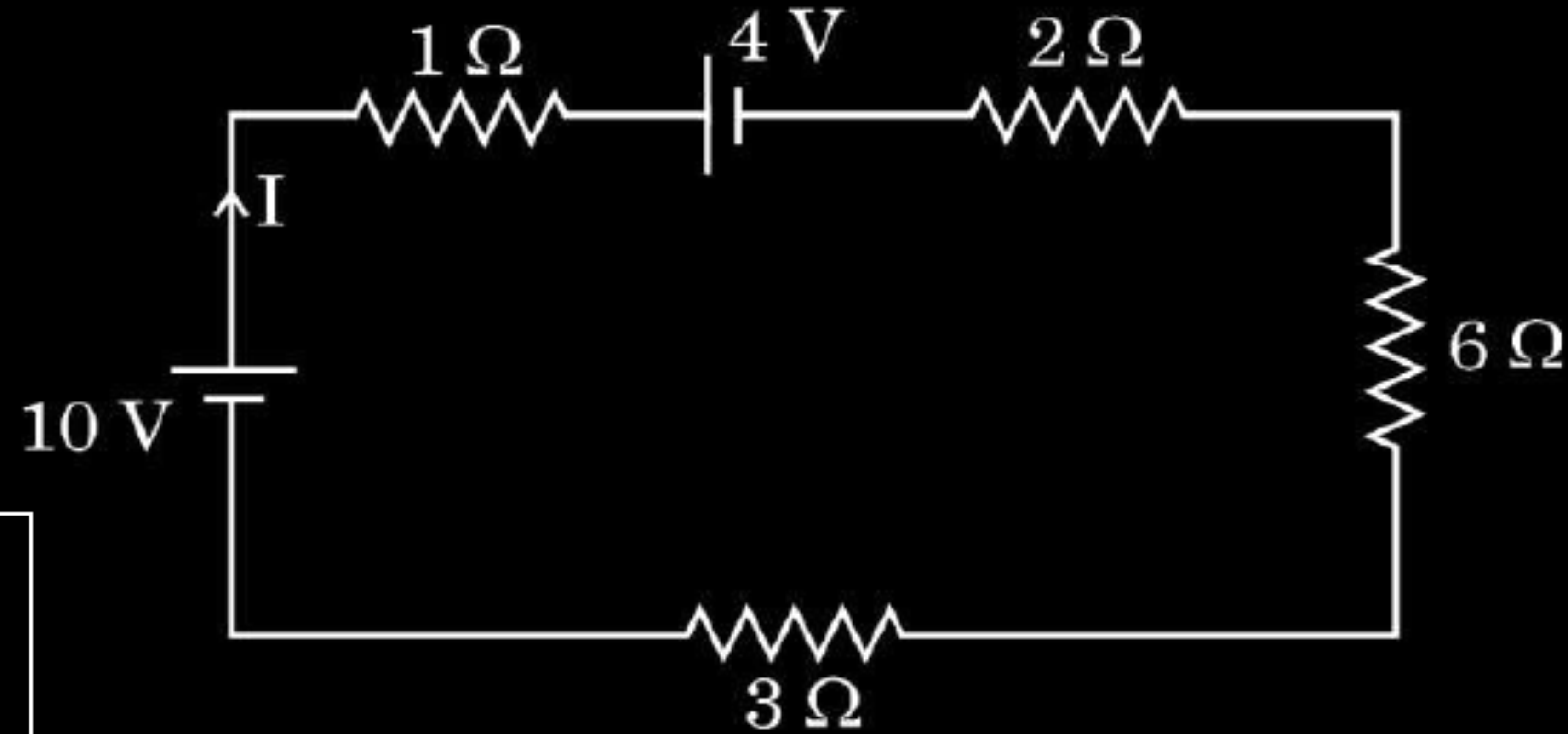
(i) Definition & S.I. Unit	1+ ¹ / ₂
(ii) Change in Electric field and drift velocity along the wire	¹ / ₂ + ¹ / ₂
Justification	¹ / ₂ + ¹ / ₂
Effective resistance and current	1 ¹ / ₂

(i) Mobility: Mobility is defined as the magnitude of the drift velocity per unit electric field. S.I. Unit: $\frac{m^2}{V.s}$ or $\frac{C.s}{kg}$	1
(ii) Both electric field and the drift velocity decreases. Justification: $v_d = \frac{I}{neA}$ As area increases across the wire, drift velocity decreases. $v_d = \frac{eE}{m} \tau$ As drift velocity decreases, electric field decreases (since e ,m and τ are constant).	$\frac{1}{2} + \frac{1}{2}$
(iii) From the diagram 10Ω and 14 Ω are in series $R_1=10 \Omega + 14 \Omega = 24 \Omega$ 10 Ω and 10 Ω are in series $R_2=10 \Omega+ 10 \Omega = 20 \Omega$ 24 Ω, 20 Ω and 30 Ω are in parallel. $\frac{1}{R} = \frac{1}{24} + \frac{1}{20} + \frac{1}{30} = \frac{5 + 6 + 4}{120} = \frac{15}{120}$ $R= 8 \Omega$ Electric current in the circuit $I = \frac{V}{R} = \frac{6}{8} = \frac{3}{4} A$	$\frac{1}{2}$
Note: Full credit of 1 ½ marks is to be awarded for part (iii) even if a student does not attempt this part.	1

OR

- (b) (i) Define electrical conductivity of a wire. Give its SI unit.
- (ii) High current is to be drawn safely from (1) a low-voltage battery, and (2) a high-voltage battery. What can you say about the internal resistance of the two batteries ?
- (iii) Calculate the total energy supplied by the batteries to the circuit shown in the figure, in one minute.

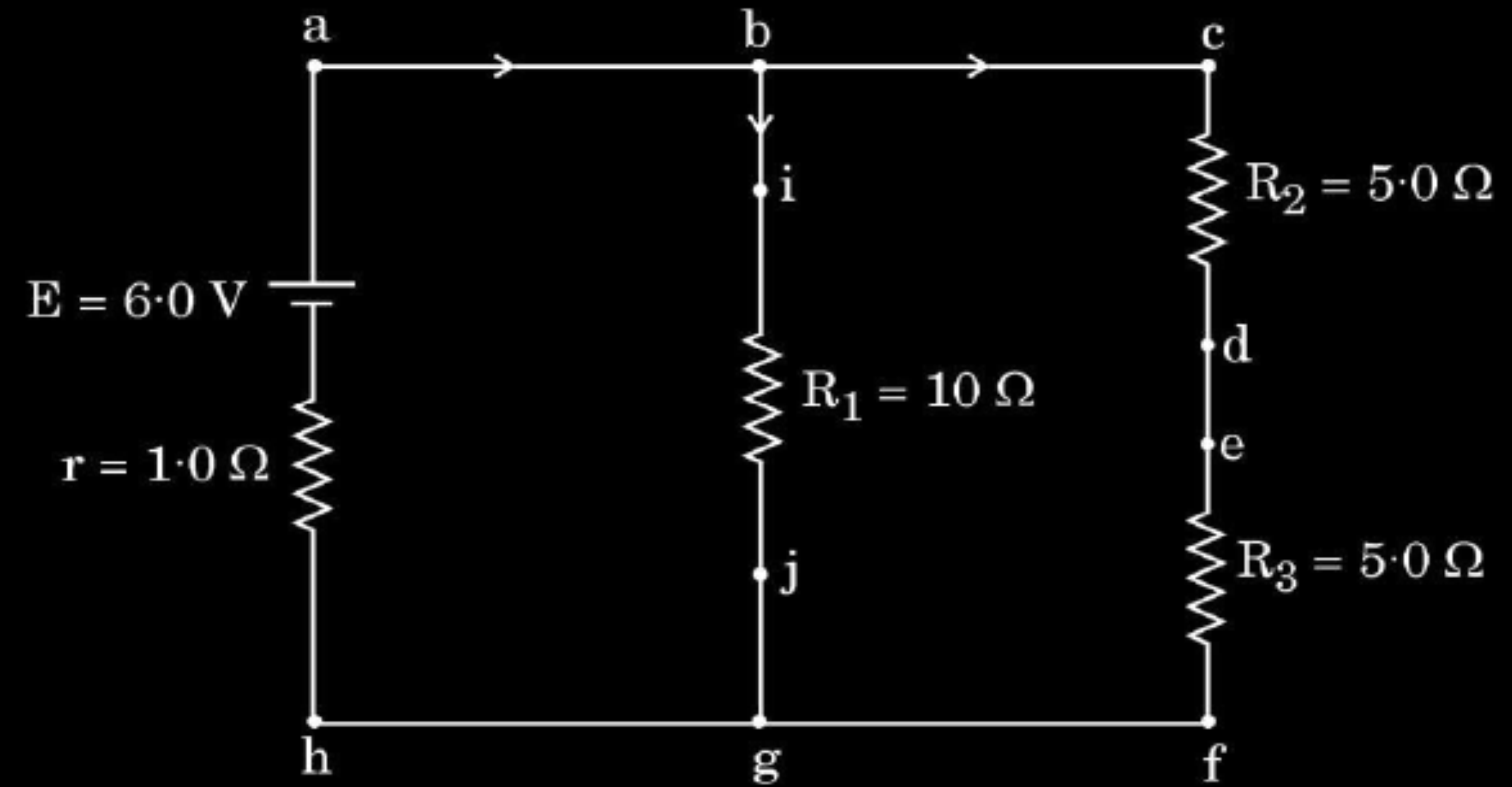
5



(i) Definition & S.I. Unit	1½
(ii) Explanation of internal resistance for low voltage and high voltage battery	1+1
(iii) Total energy	1½

(i) Definition : Electrical conductivity is defined as the measure of a material's ability to carry a current through it. Alternatively: It is the reciprocal of the resistivity. Alternatively: It is defined as the current density per unit electric field.	1
S.I. Unit: $(\text{ohm})^{-1}\text{-m}^{-1}$ or S-m^{-1}	$\frac{1}{2}$
(ii) Low voltage Battery- Internal resistance should be low.	1
High voltage Battery – Internal resistance should be high.	1
(iii) Applying Kirchhoff's loop rule $10 - I \times 1 - 4 - 2I - 6I - 3I = 0$ $12I = 6 \Rightarrow I = 0.5 \text{ A}$ Heat energy $H = I^2 R t$ $H = 0.25 \times 12 \times 60 = 180 \text{ J}$	$\frac{1}{2}$
	$\frac{1}{2}$

34. The following figure shows a circuit diagram. We can find the currents through and potential differences across different resistors using Kirchhoff's rules.



Answer the following questions based on the above :

- | | |
|---|---|
| (a) Which points are at the same potential in the circuit ? | 1 |
| (b) What is the current through arm bg ? | 1 |
| (c) Find the potential difference across resistance R_3 . | 2 |

OR

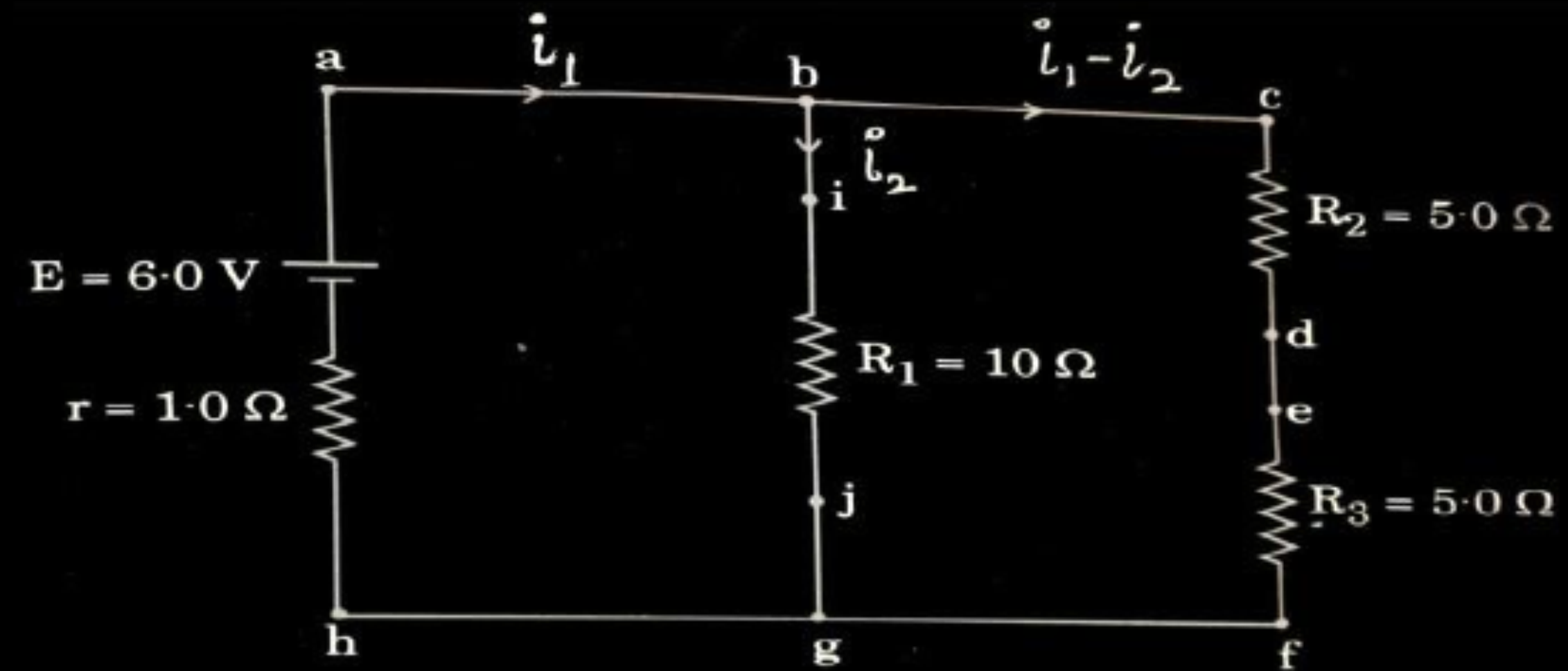
- | | |
|--|---|
| (c) What is the power dissipated in resistance R_2 ? | 2 |
|--|---|

a) Points at same potential	1
b) Current through arm bg	1
c) Potential difference across R_3	
OR	
c) Power dissipated in R_2	2

a) Points (a, b, c)
(d, e)
(j, f, g, h)
are at same potential

Note: Give full credit if a student mentions any two points at same potential from the above.

b)



According to Kirchhoff's loop rule

for closed loop abgha

$$-6 + 10 I_2 + I_1 = 0$$

$$I_1 + 10 I_2 = 6 \quad (i)$$

for closed loop acfha

$$-6 + 10 (I_1 - I_2) + I_1 = 0$$

$$11 I_1 - 10 I_2 = 6 \quad (ii)$$

Adding (i) and (ii)

$$12 I_1 = 12$$

$$I_1 = 1 \text{ A}$$

$$I_2 = 0.5 \text{ A}$$

= current through arm bg

Note: Award 1 mark if a student calculates the current by any other method.

1

$$\begin{aligned} c) V_{R3} &= (I_1 - I_2) \times R_3 \\ &= 0.5 \times 5 \\ &= 2.5 \text{ V} \end{aligned}$$

OR

$$\begin{aligned} (c) P &= (I_1 - I_2)^2 \times R_2 = (0.5)^2 \times 5 \\ &= 1.25 \text{ W} \end{aligned}$$

1

1

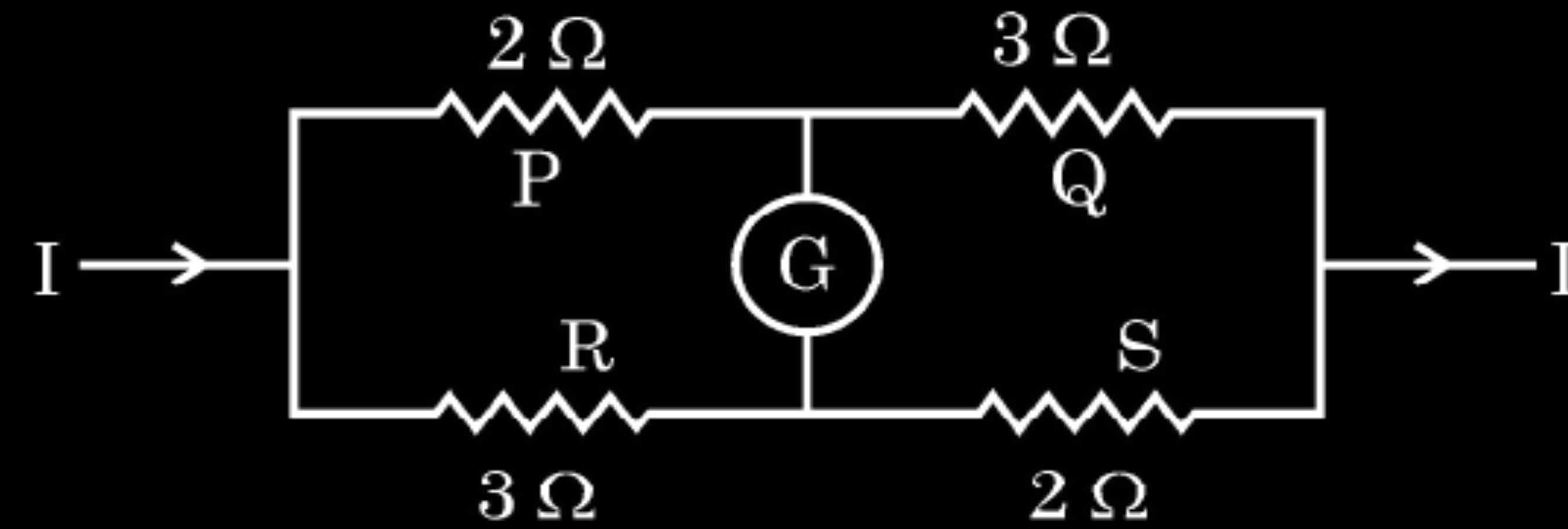
1

1

1/2

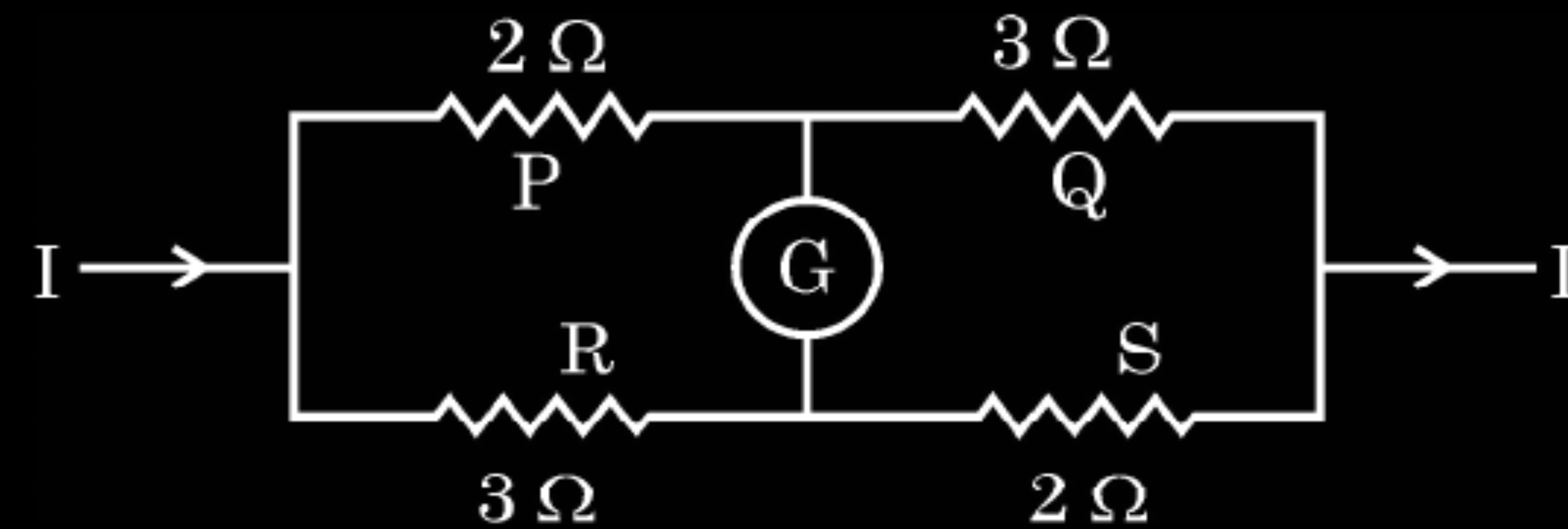
1/2

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.



Reason (R) : For a balanced bridge small current should flow through the galvanometer.

17. **Assertion (A)** : The given figure does not show a balanced Wheatstone bridge.



Reason (R) : For a balanced bridge small current should flow through the galvanometer.

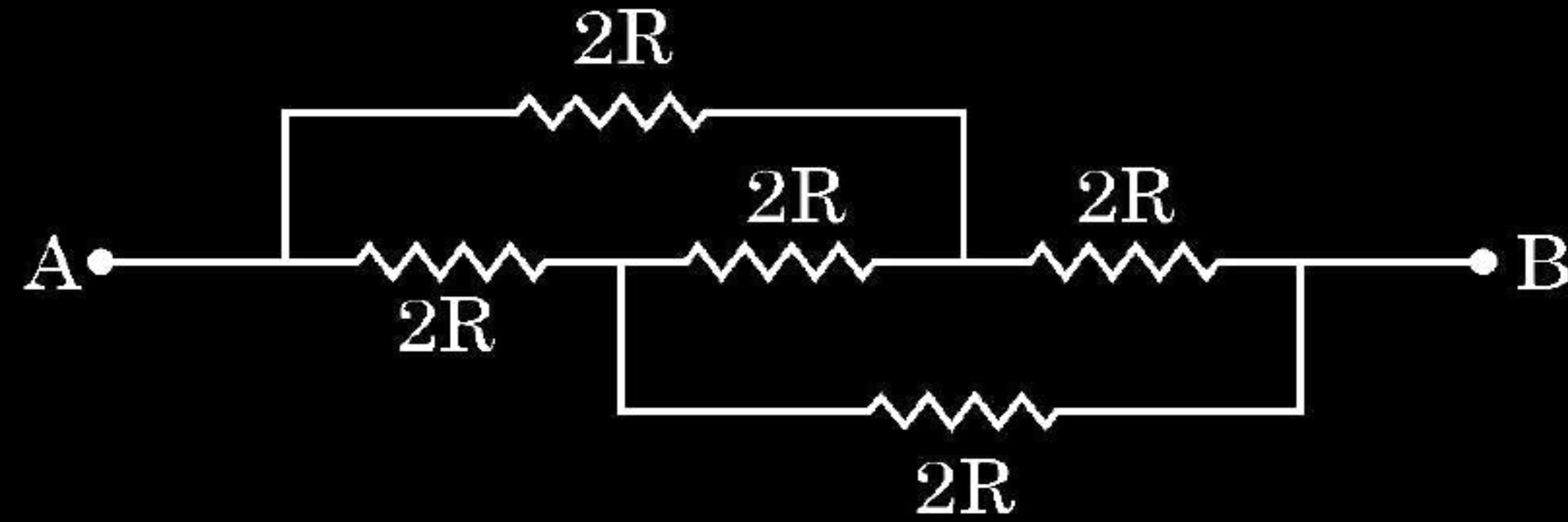
(a)

Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

17. **Assertion (A)** : The equivalent resistance between points A and B in the given network is $2R$.

1

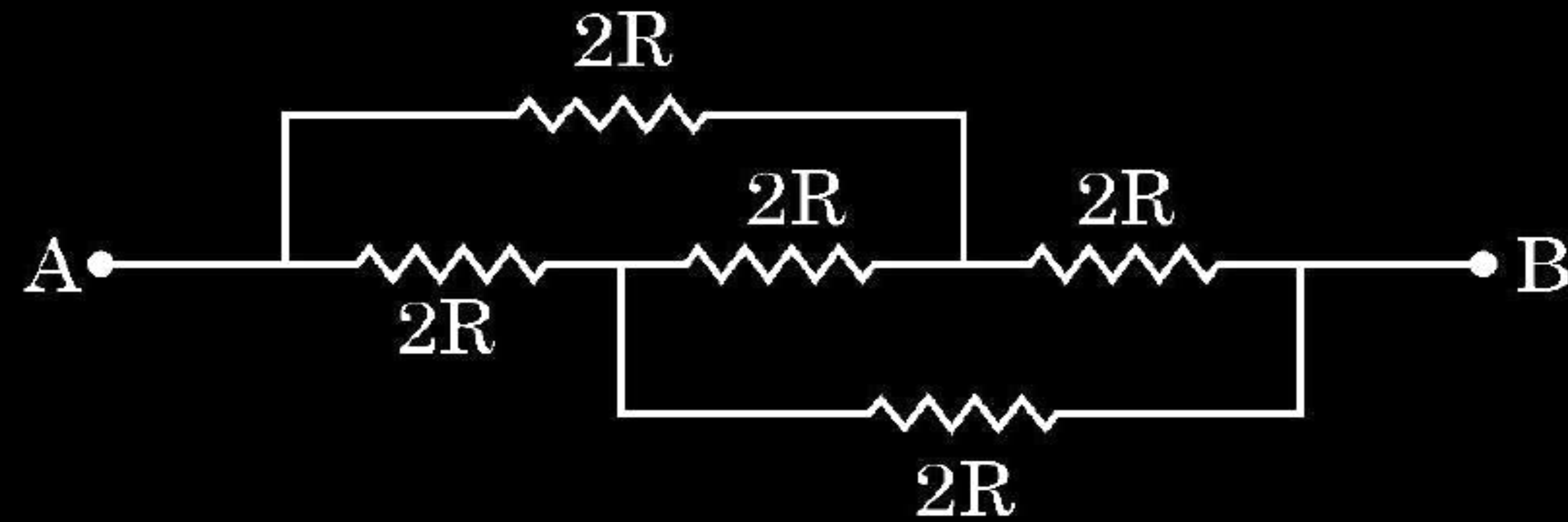
Reason (R) : All the resistors are connected in parallel



17. **Assertion (A)** : The equivalent resistance between points A and B in the given network is $2R$.

1

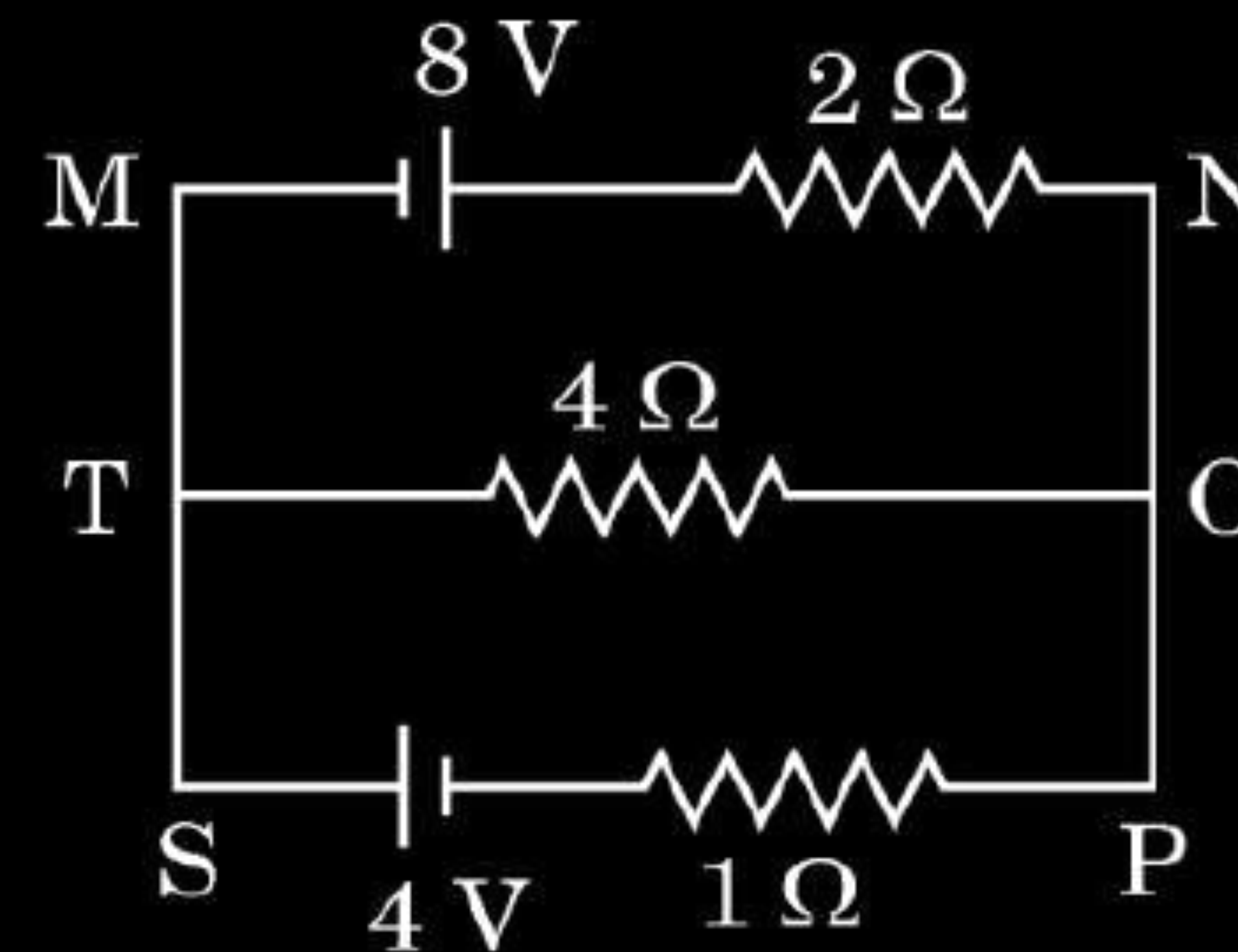
Reason (R) : All the resistors are connected in parallel



(C) Assertion (A) is true and reason (R) is false

- (b) (i) State Kirchhoff's rules. Use them to obtain the condition of balance for a Wheatstone Bridge.
- (ii) Use Kirchhoff's rule to determine the currents flowing through the branches MN, TO and SP in the circuit shown in the figure.

5



(i) Statement of Kirchhoff two rules	$\frac{1}{2} + \frac{1}{2}$
Obtaining the balanced condition	2
(ii) Finding current in branches MN, TO and SP	2

(i) Kirchhoff's junction rule - at any junction, the sum of the current entering the junction is equal to the sum of currents leaving the junction.

Kirchhoff second rule:

The algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

In balanced bridge $I_g=0$,

Hence $I_1=I_3$ and $I_2=I_4$

Using Kirchhoff's loop rule for closed loops ADBA and CBDC

$$-I_1R_1 + 0 + I_1R_1 = 0 \quad (I_g=0) \quad \text{-----(1)}$$

In the second loop $I_3 = I_1$, $I_4 = I_2$

$$I_2R_4 + 0 - I_1R_3 = 0 \quad \text{-----(2)}$$

From equation (1) and (2)

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \text{ and } \frac{I_1}{I_2} = \frac{R_4}{R_3}$$

$$\frac{R_2}{R_1} = \frac{R_4}{R_3}$$

This is the condition for balanced Wheatstone bridge

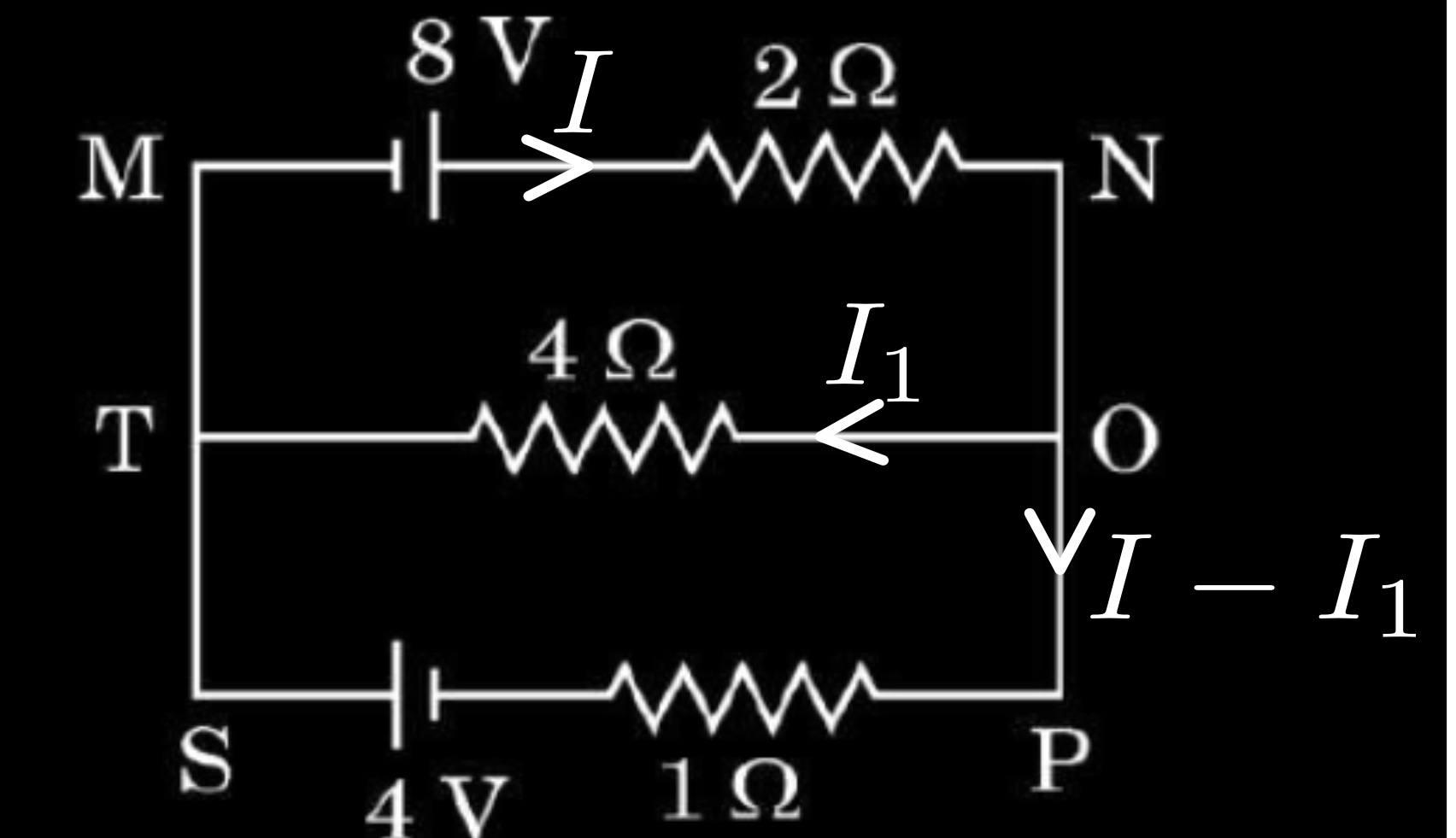
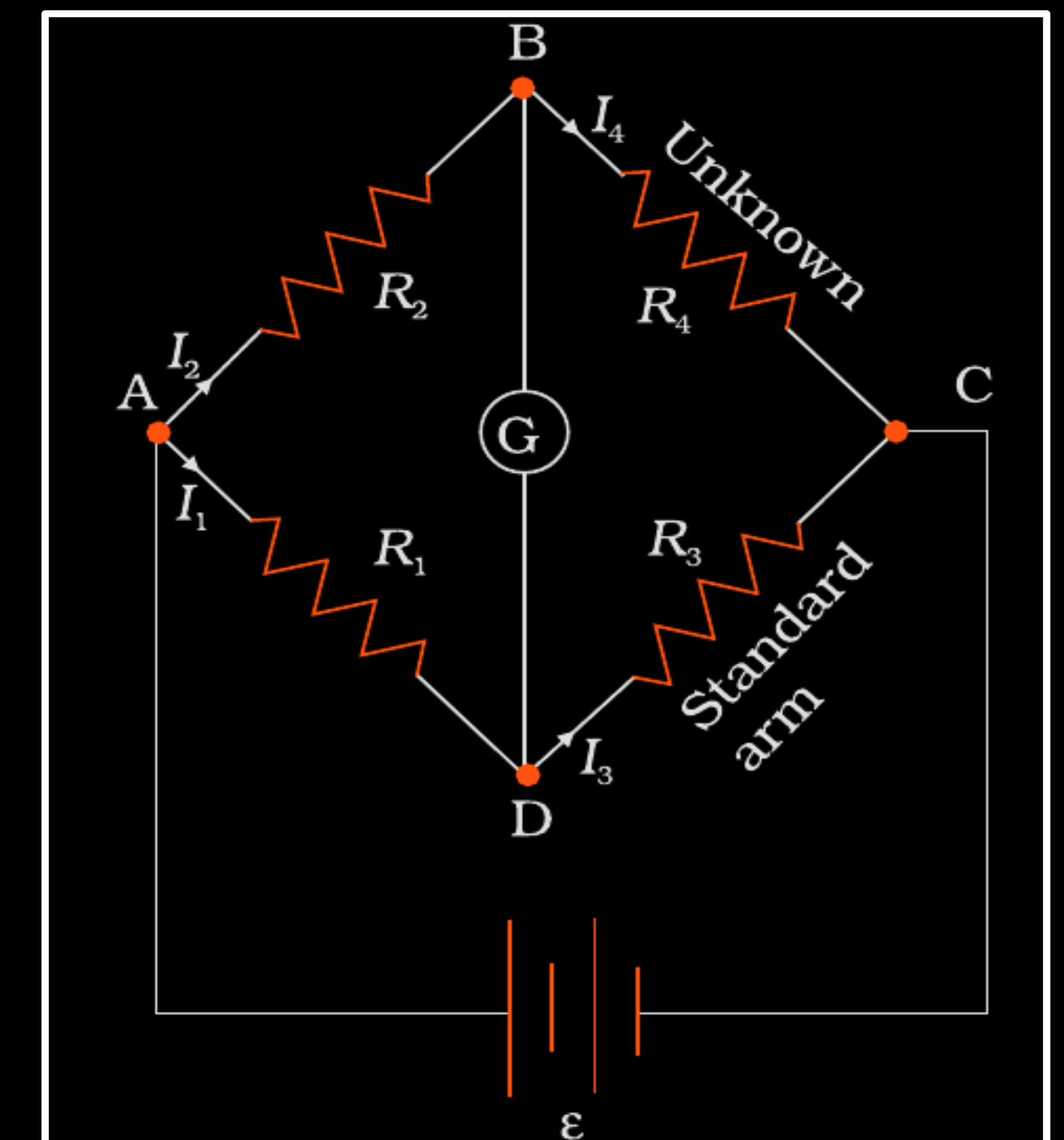
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$



(ii) In loop MNOTM

$$2I + 4I_1 = 8 \quad \text{-----(1)}$$

Loop OPSTO

$$-I + 5I_1 = -4 \quad \text{-----(2)}$$

On solving

Current in MN, $I = 4\text{A}$

Current in TO, $I_1 = 0\text{A}$

Current in SP, $I - I_1 = 4\text{A}$

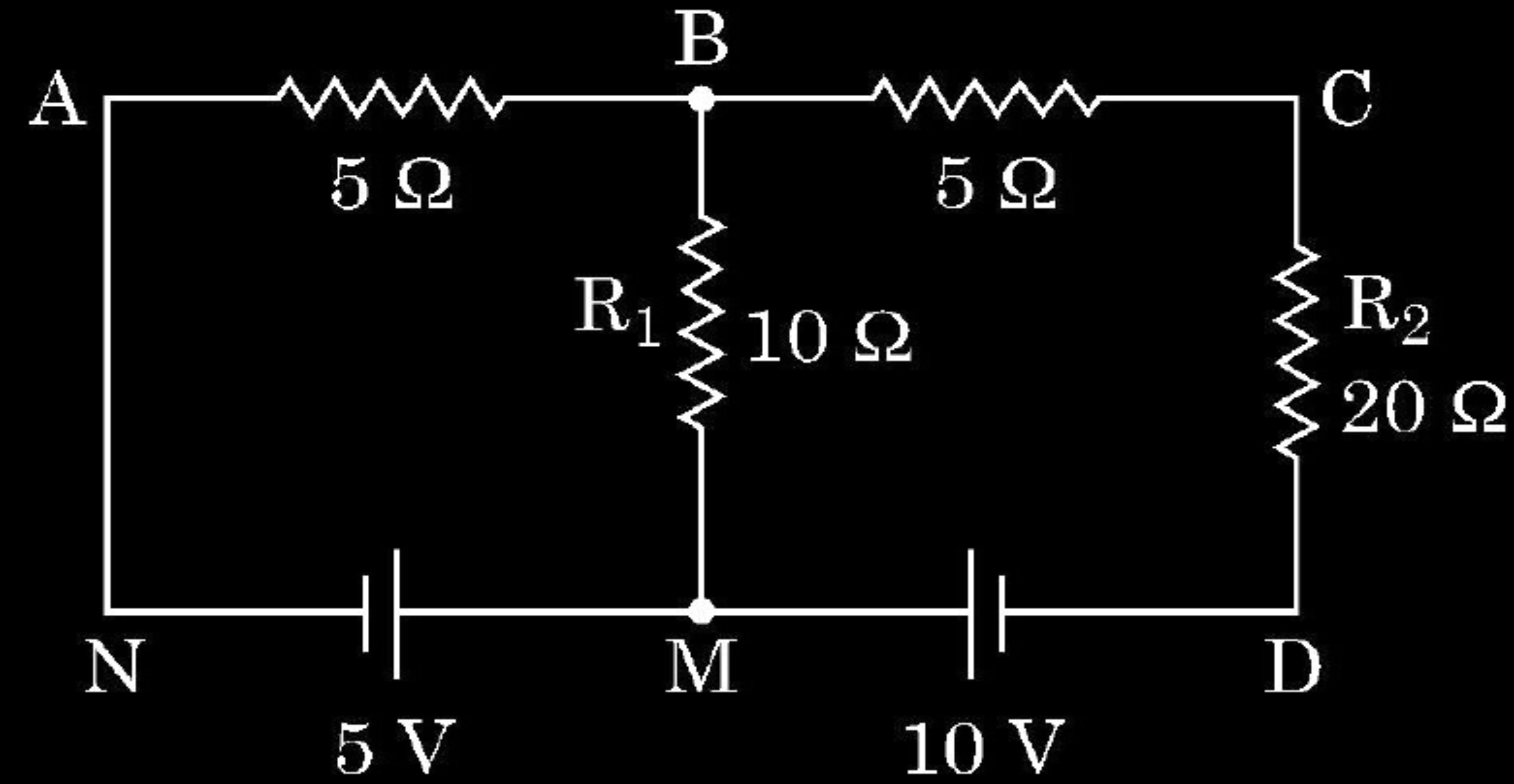
$\frac{1}{2}$

$\frac{1}{2}$

1

27. Find the currents flowing through the branches AB and BC in the network shown.

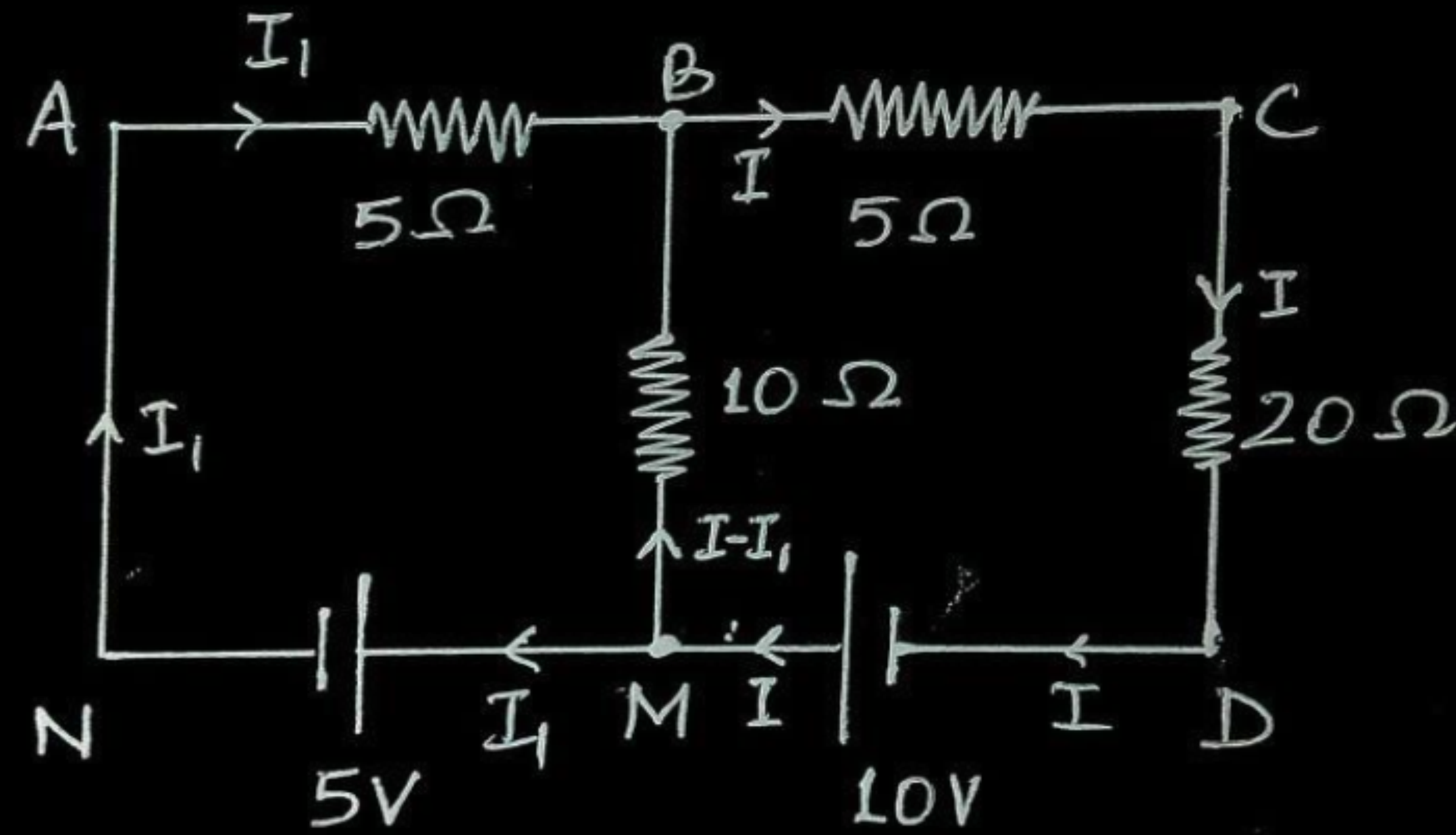
3



Finding current in the arm AB
Finding current in the arm BC

$1\frac{1}{2}$
 $1\frac{1}{2}$

Circuit diagram with distribution of current



Using Kirchhoff's voltage rule

In closed loop ABMNA,

$$-5I_1 + 10(I - I_1) - 5 = 0 \quad \dots\dots\dots (1)$$

In closed loop ACDNA

$$-5I - 20I + 10 - 5 - 5I_1 = 0 \quad \dots\dots\dots (2)$$

Solving eq (1) and (2)

$$I_1 = -\frac{3}{17} \text{ A and } I = \frac{4}{17}$$

Magnitude of current in arm AB = $\frac{3}{17}$ A

Magnitude of current in arm BC = $\frac{4}{17}$ A

1

$\frac{1}{2}$

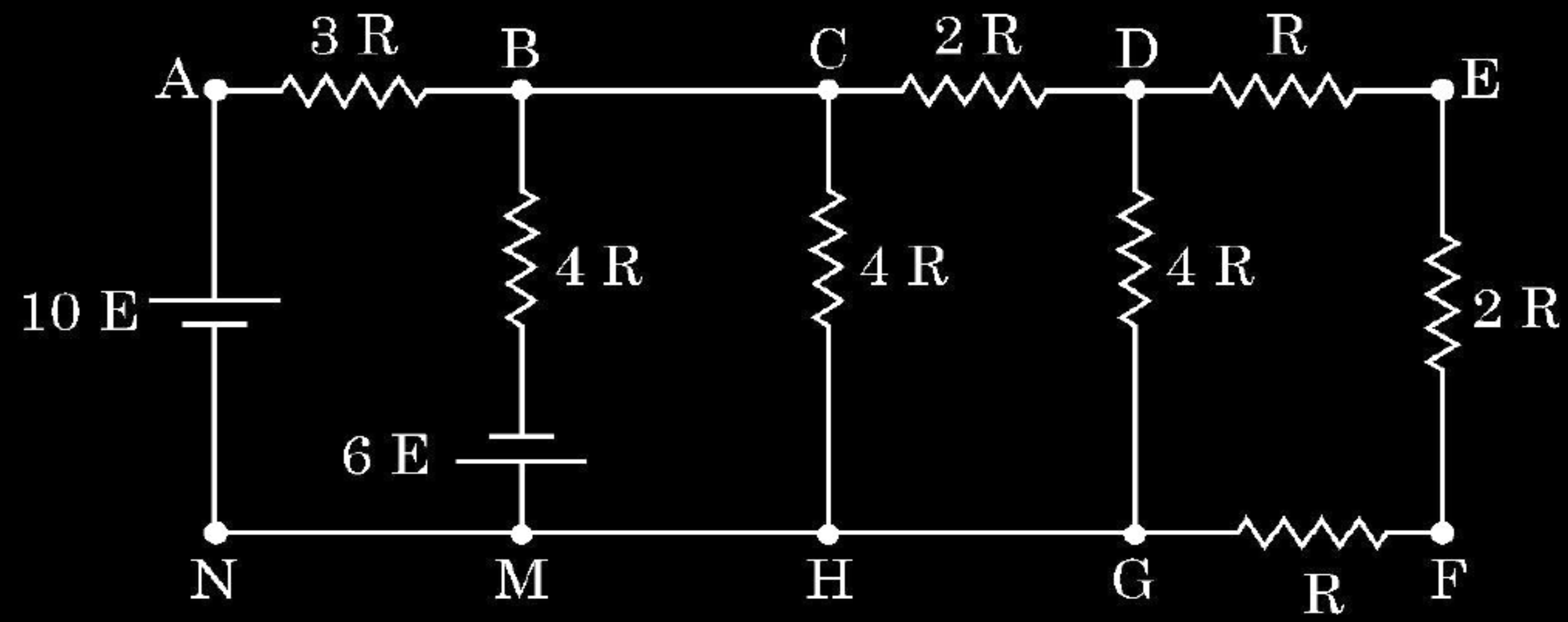
$\frac{1}{2}$

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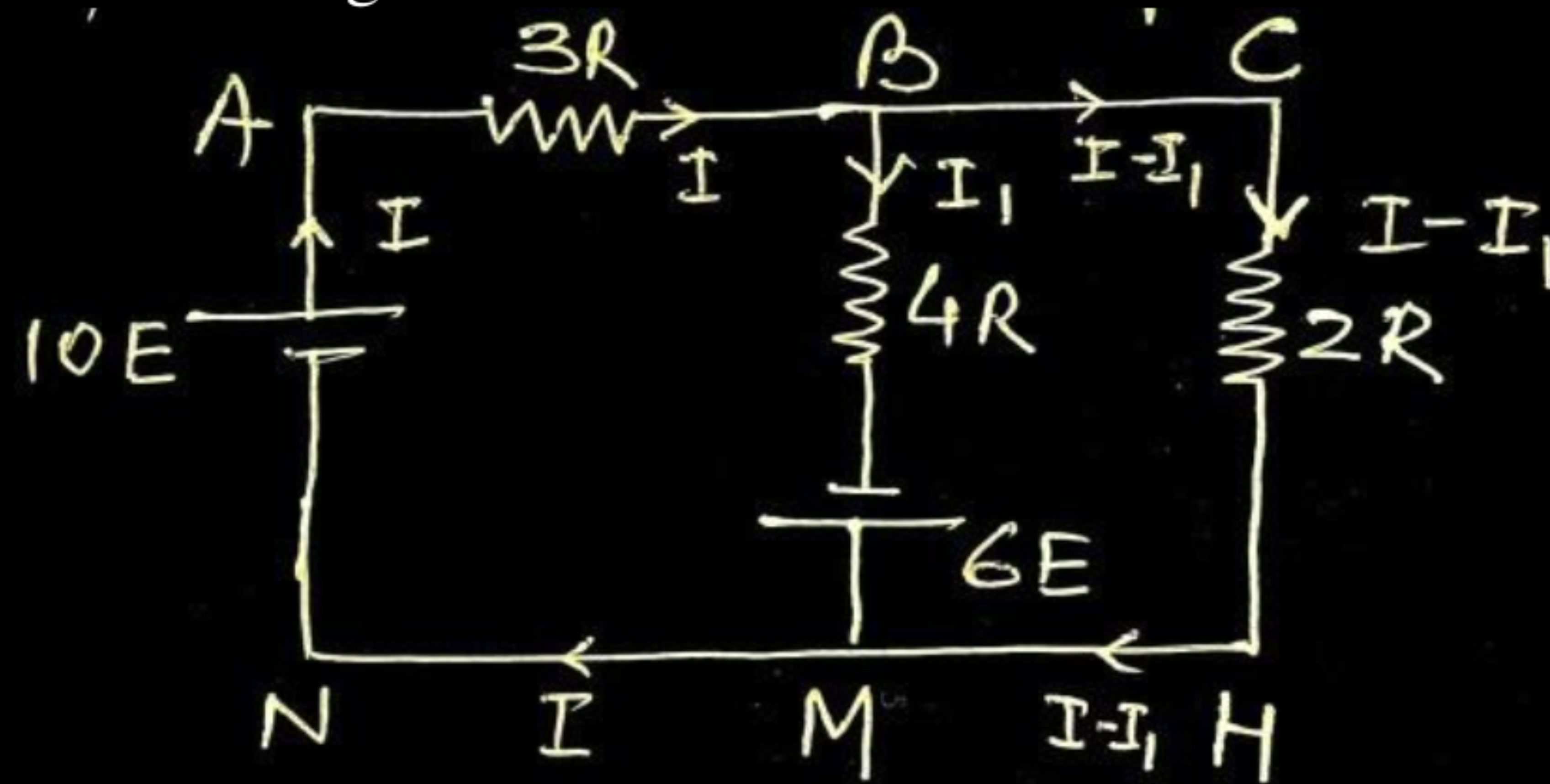
25. Find the current in branch BM in the network shown :

3



Finding equivalent resistance across CH, $R_{CH} = 2R$

Equivalent circuit diagram



In closed loop ABMNA

$$-3IR - 4I_1R + 16E = 0 \quad \dots\dots\dots(1)$$

In closed loop BCHMB

$$-2R(I - I_1) - 6E + 4I_1R = 0 \quad \dots\dots\dots(2)$$

On solving equations (1) and (2)

$$I_1 = \frac{25E}{13R}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

- 17.** (a) What is meant by ‘relaxation time’ of free electrons in a conductor ?
 Show that the resistance of a conductor can be expressed by

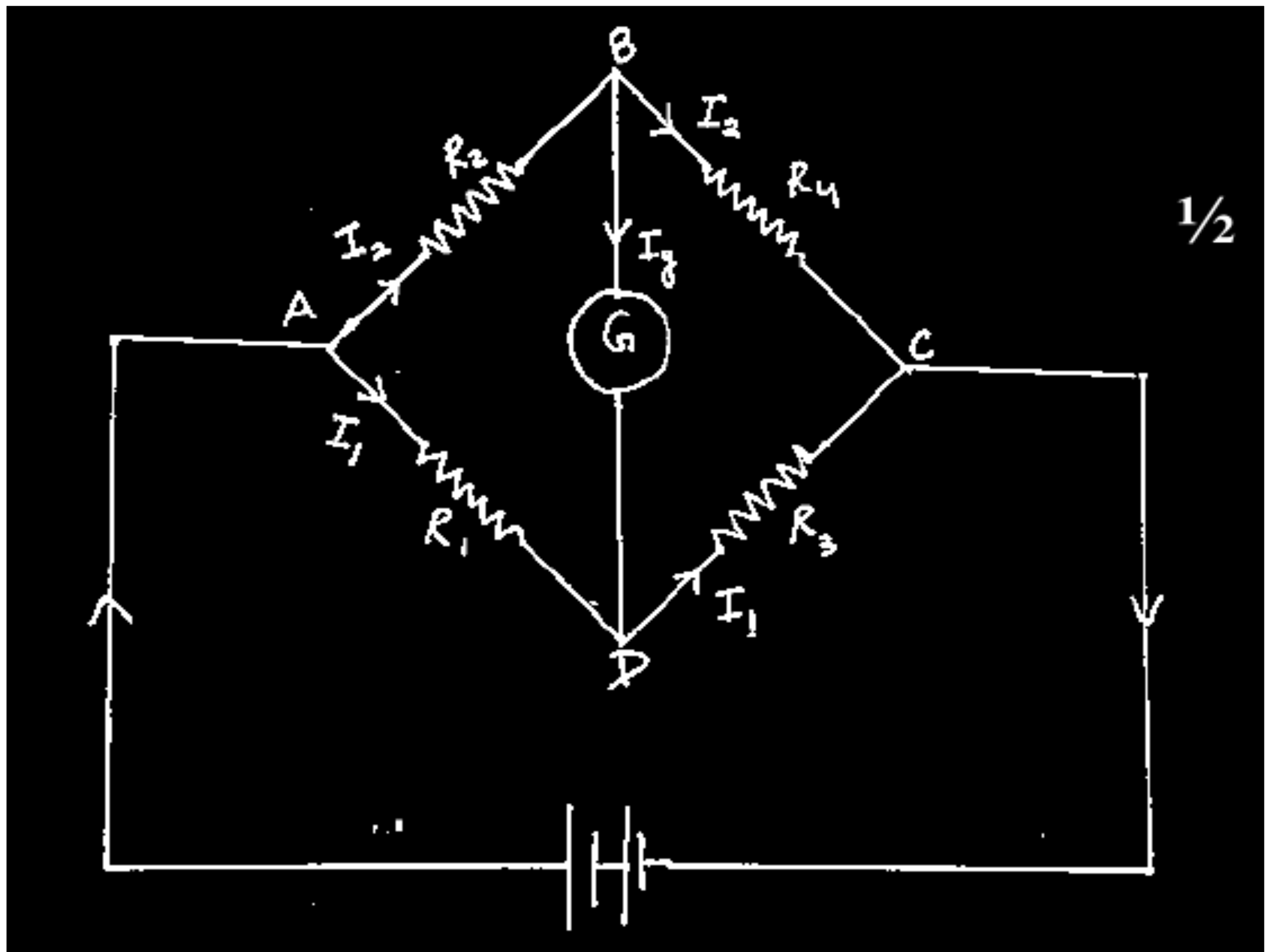
$$R = \frac{m l}{n e^2 \tau A}$$
, where symbols have their usual meanings. 2
- OR**
- (b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. 2

Meaning of relaxation time	1/2
Derivation of R	1 1/2

Circuit diagram of Wheatstone bridge	1/2
Obtaining the condition when current flows through galvanometer	1 1/2

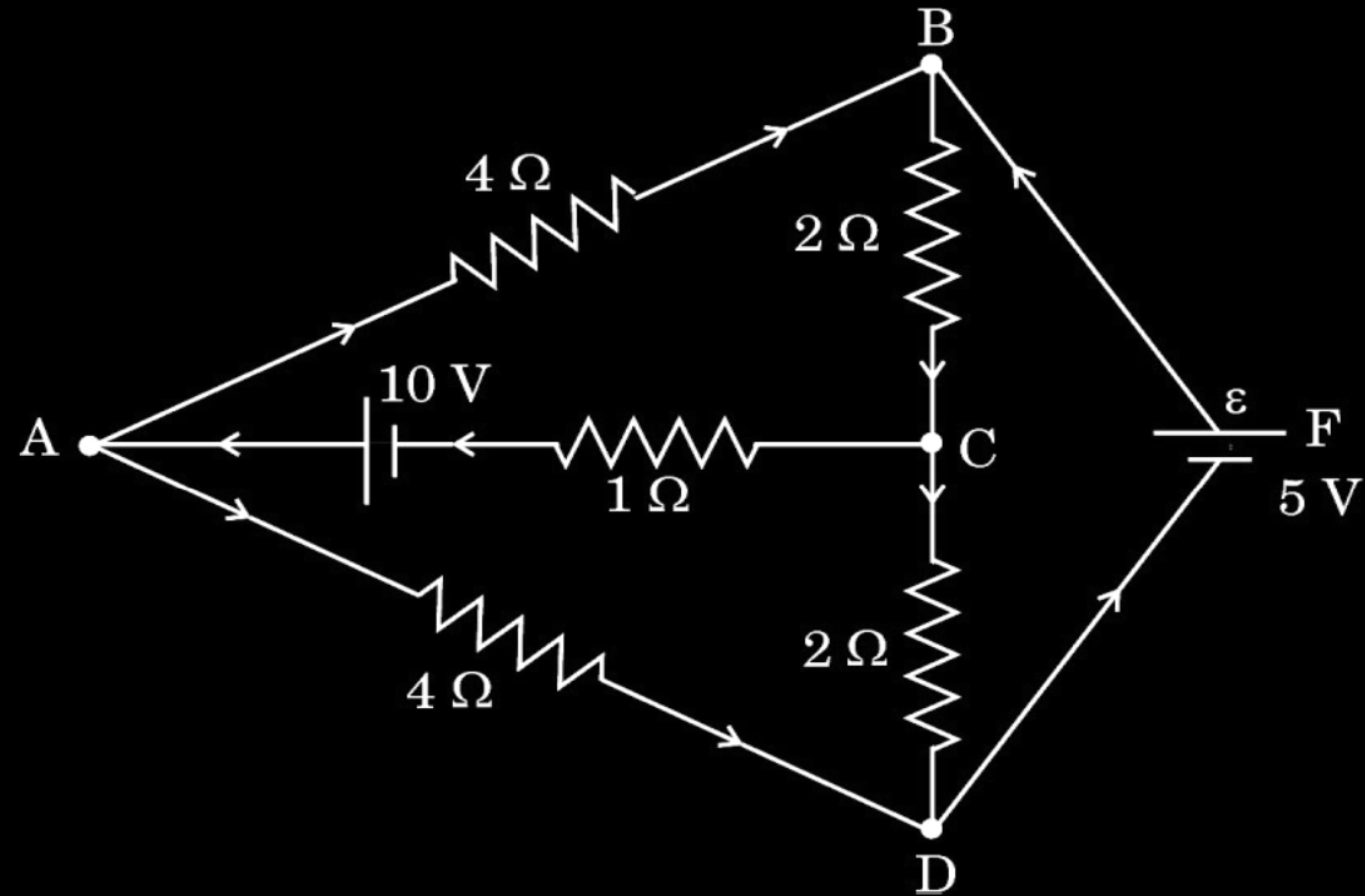
Average time between two successive collisions of electron in presence of electric field.	1/2
Drift velocity of an electron	
$v_d = \frac{eE}{m} \tau \quad \text{--- (i)}$	1/2
Current flowing through a conductor of length l and area of cross section A	
$I = neAv_d \quad \text{--- (ii)}$	
$I = \frac{ne^2 AE \tau}{m} = \frac{ne^2 A \tau V}{ml}$	1/2
$R = \frac{V}{I} = \frac{ml}{ne^2 \tau A}$	1/2

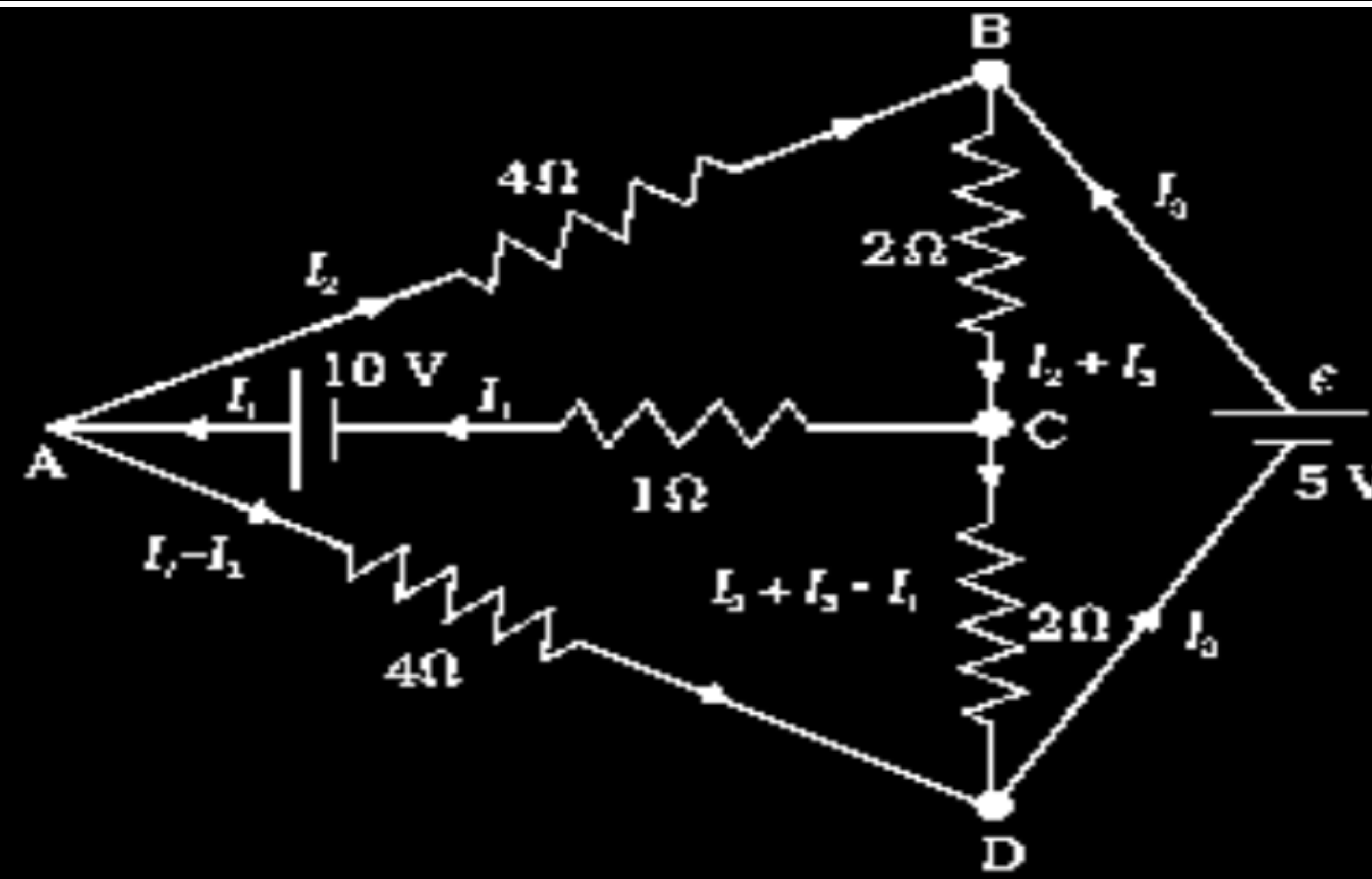
By applying Kirchoff's loop rule to closed loops ADBA and CBDC	
$-I_1 R_1 + 0 + I_2 R_2 = 0 \quad \text{----- (i) } [I_g = 0]$	1/2
$I_2 R_4 + 0 - I_1 R_3 = 0 \quad \text{----- (ii)}$	
From eq (i) -	
$\frac{I_1}{I_2} = \frac{R_2}{R_1}$	1/2
From eq (ii) -	
$\frac{I_1}{I_2} = \frac{R_4}{R_3}$	
Hence,	
$\frac{R_2}{R_1} = \frac{R_4}{R_3}$	1/2



22. Determine the current in branches AB, AC and BC of the network shown in figure.

3





For closed loop ADCA ,

$$10 - 4(I_1 - I_2) + 2(I_2 + I_3 - I_1) - I_1 = 0$$

$$7I_1 - 6I_2 - 2I_3 = 10 \text{ -----(i)}$$

For closed loop ABCA ,

$$10 - 4I_2 - 2(I_2 + I_3) - I_1 = 0$$

$$I_1 + 6I_2 + 2I_3 = 10 \text{ -----(ii)}$$

For closed loop BCDED ,

$$5 - 2(I_2 + I_3) - 2(I_2 + I_3 - I_1) = 0$$

$$2I_1 - 4I_2 - 4I_3 = -5 \text{ -----(iii)}$$

$$\text{Current in branch AB} = I_2 = \frac{5}{8} A$$

$$\text{Current in branch AC} = I_1 = 2.5 A$$

$$\text{Current in branch BC} = I_2 + I_3 = 2.5 A$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

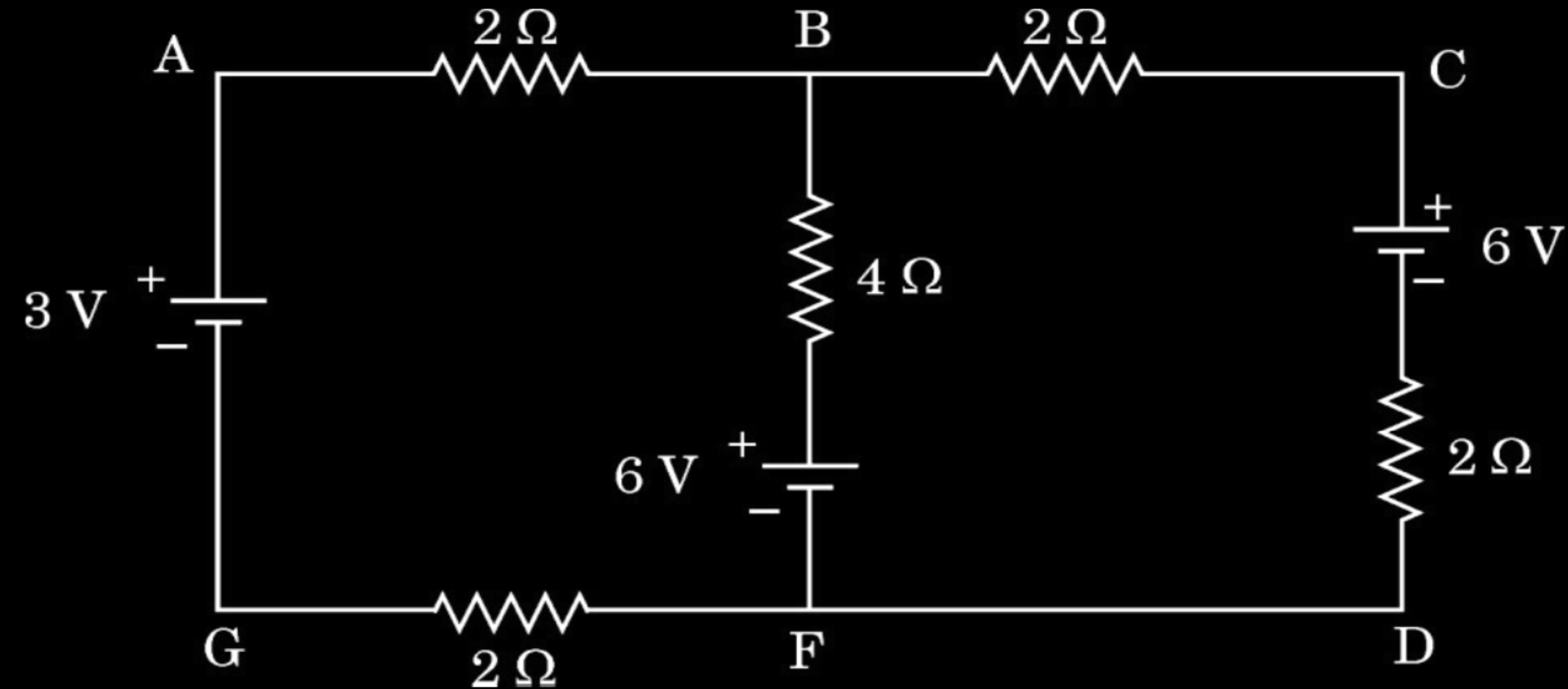
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$\frac{1}{2}$

25. The figure shows a circuit with three ideal batteries. Find the magnitude and direction of currents in the branches AG, BF and CD.

3



Finding magnitude and direction of current in AG, BF and CD 1+1+1

By Kirchhoff's Laws (at point B)

$$I_1 + I_2 = I_3 \quad \dots\dots(1)$$

In the closed loop AGFBA

$$3 + 2I_3 - 6 + 4I_2 + 2I_3 = 0$$

$$I_2 + I_3 = \frac{3}{4} \quad \dots\dots(2)$$

From (i)

$$2I_1 + I_2 = \frac{3}{4} \quad \dots\dots(3)$$

In closed loop BFDCB

$$-4I_2 + 6 + 2I_1 - 6 + 2I_1 = 0$$

$$I_2 - I_1 = 0$$

$$I_2 = I_1 \quad \dots\dots(4)$$

Putting in (3)

$$I_1 = \frac{1}{4} A$$

From (4)

$$I_2 = \frac{1}{4} A$$

$$\text{From (2)} \quad I_3 = \frac{1}{2} A$$

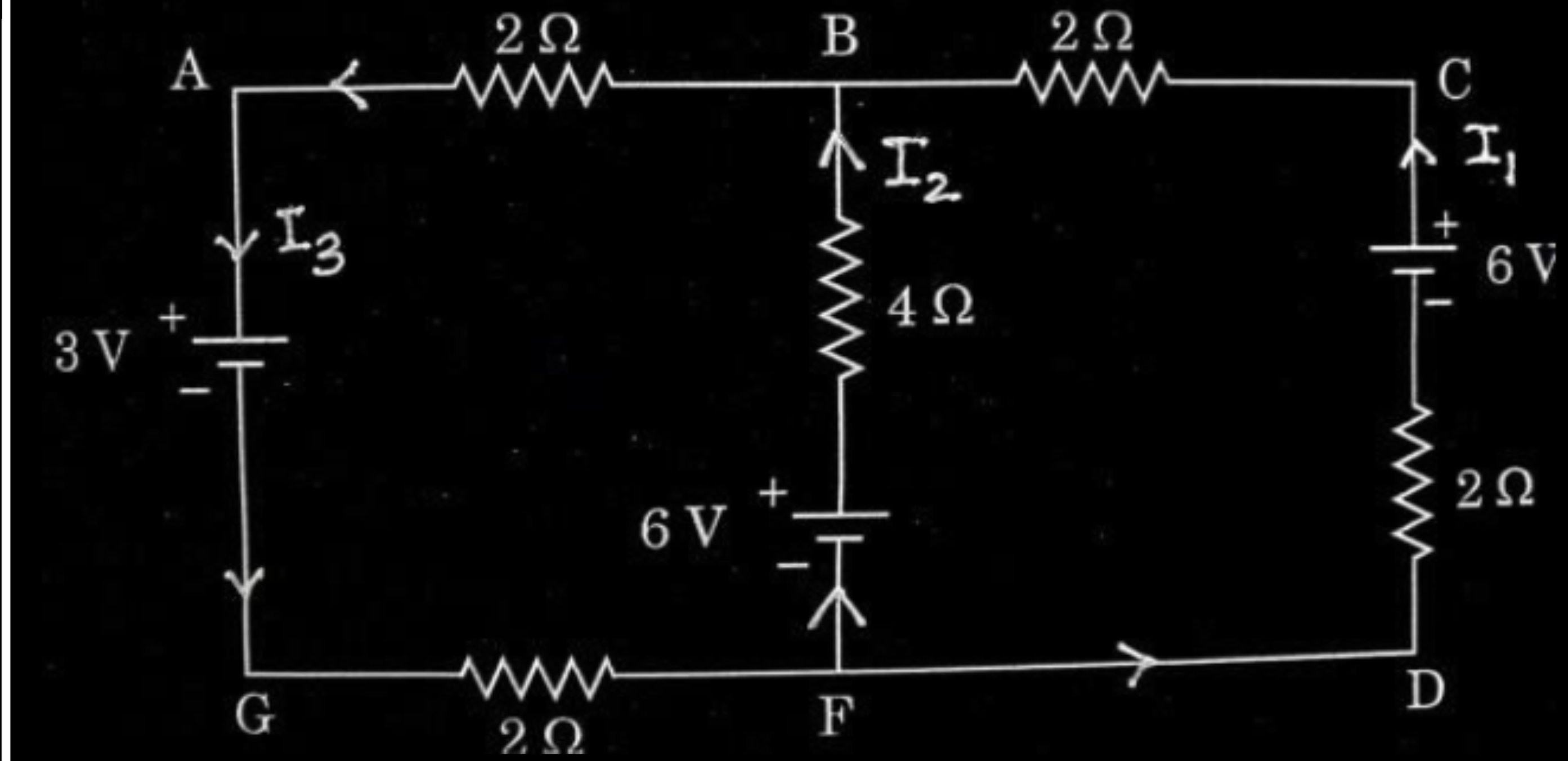
1/2

1/2

1/2

1/2

1/2



1/2

- 23.** (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m , charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\frac{ne^2}{m} \right) \tau$. 3

OR

- (b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced. 3

- | | |
|--------------------------------------|-----|
| • Defining current density | 1/2 |
| • Whether scalar or vector | 1/2 |
| • Showing $\vec{j} = \alpha \vec{E}$ | 2 |

- | | |
|--------------------------------|---|
| Defining Wheatstone bridge | 1 |
| Obtaining balancing conditions | 2 |

Current density is the amount of charge flowing per second per unit area normal to the flow.

Alternatively:

$$\vec{j} = \frac{I}{A}$$

It is a vector quantity.

The amount of charge crossing the area A in time Δt is $I \Delta t$, where I is the magnitude of the current. Hence,

$$I \Delta t = ne A |v_d| \Delta t$$

$$I \Delta t = \frac{e^2 A}{m} \tau n \Delta t |E|$$

$$I = |j|A$$

$$|j| = \frac{ne^2}{m} \tau |E|$$

$$\vec{j} = \alpha \vec{E}$$

1/2

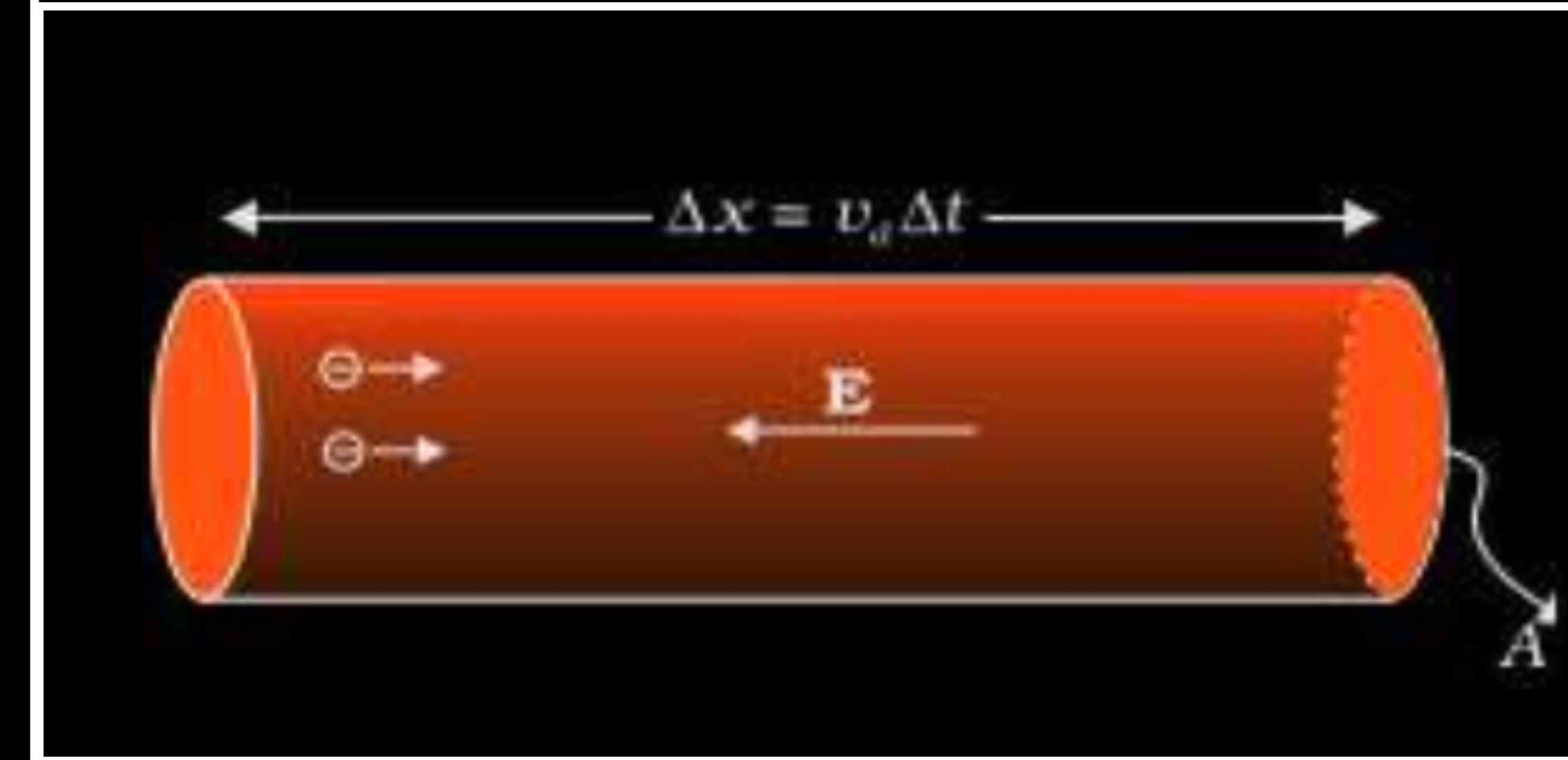
1/2

1/2

1/2

1/2

1/2



Alternatively:

If the figure is explained in words full credit to be given.

For loop ADBA:

$$-I_1 R_1 + I_2 R_2 + I_g G = 0 \quad (1)$$

For loop CBDC:

$$I_4 R_4 - I_3 R_3 - I_g G = 0 \quad (2)$$

For balanced wheatstone bridge, $I_g = 0$

And by applying Kirchhoff's junction rule to junction D and B,

$$I_1 = I_3 \text{ \& } I_2 = I_4$$

From eqn (1) and (2)

$$\frac{I_1}{I_2} = \frac{R_2}{R_1} \text{ and } \frac{I_1}{I_2} = \frac{R_4}{R_3}$$

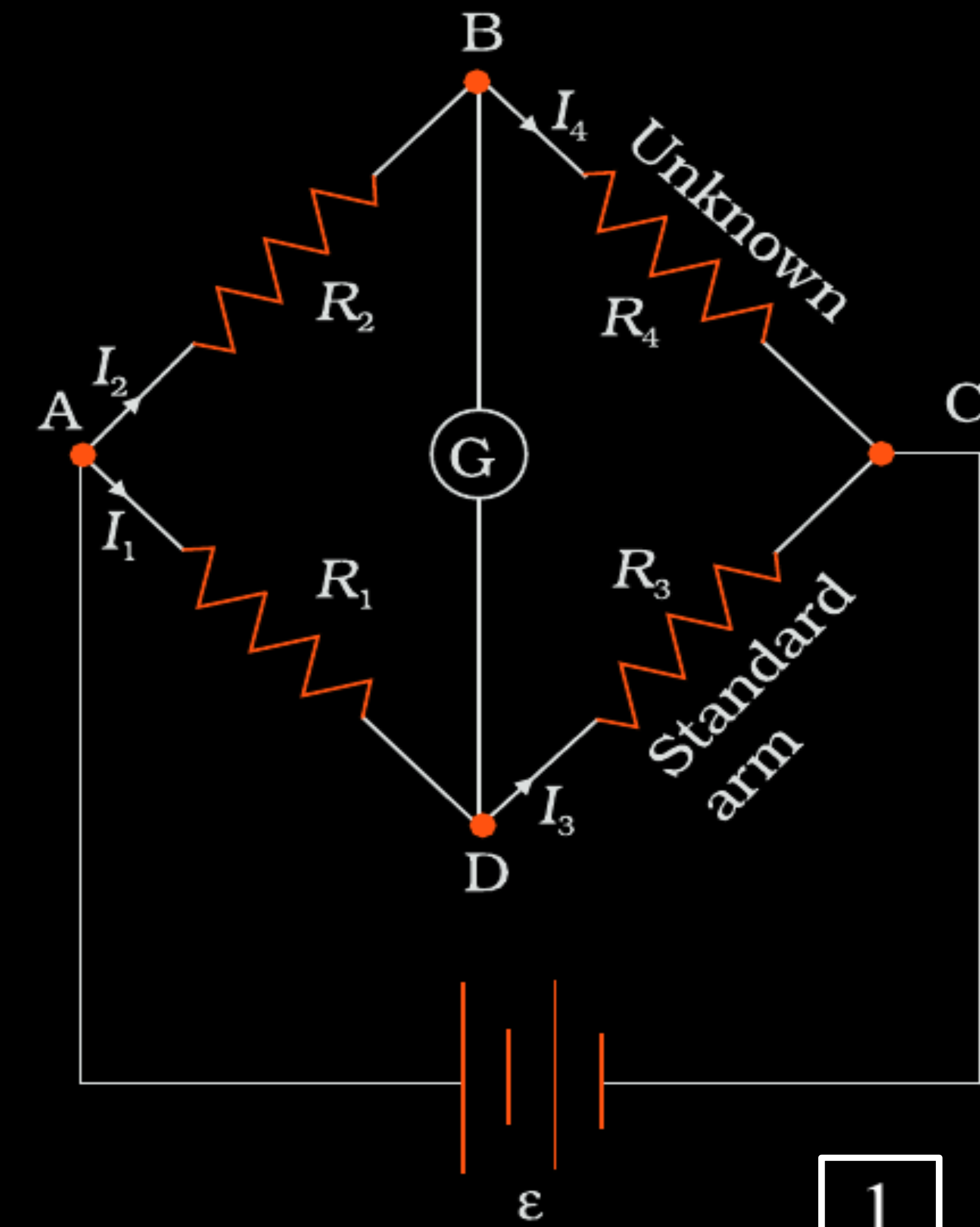
$$\Rightarrow \frac{R_2}{R_1} = \frac{R_4}{R_3}$$

$\frac{1}{2}$

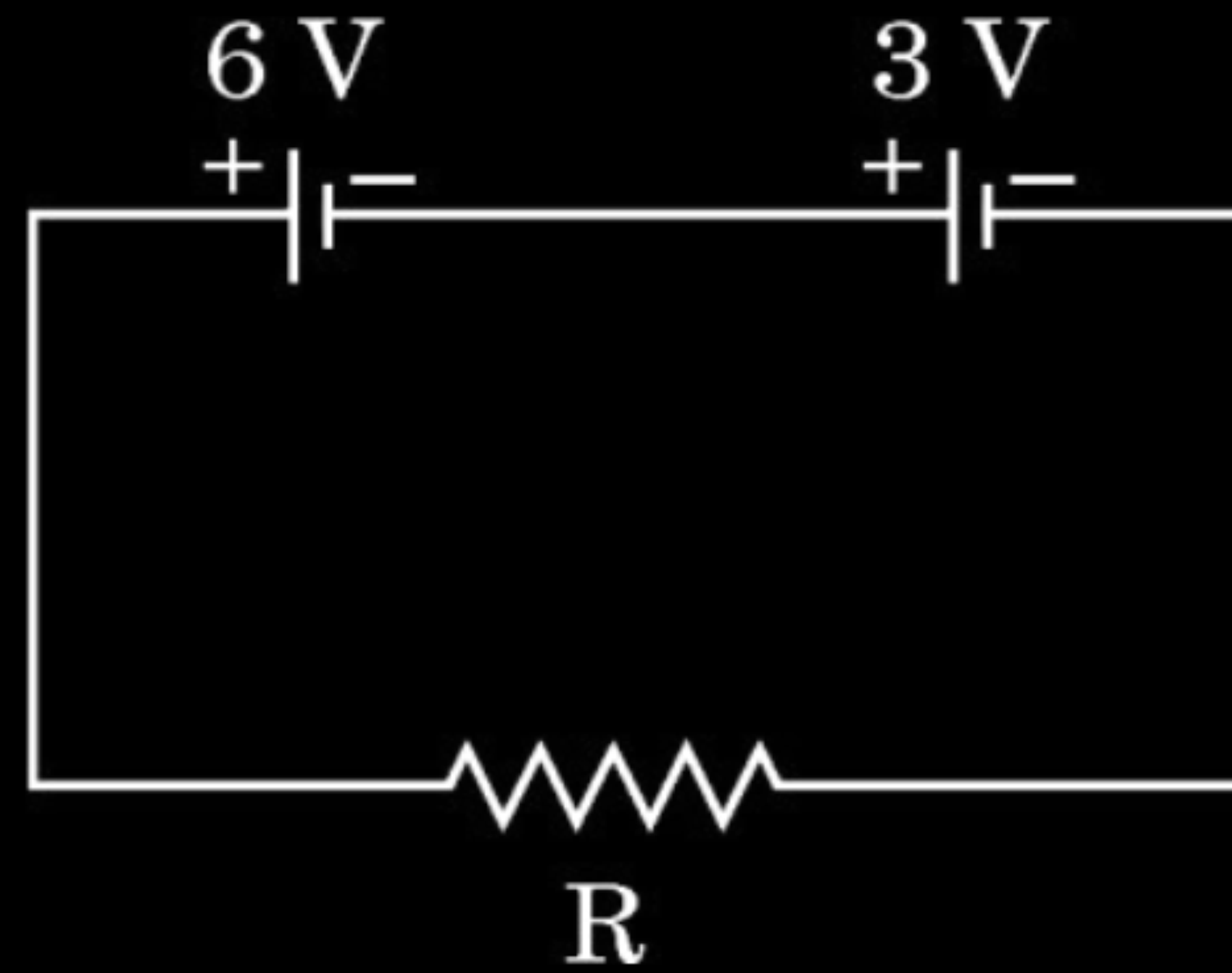
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17. Two batteries of emfs 6 V and 3 V and internal resistances $0.8\ \Omega$ and $0.2\ \Omega$ respectively are connected in series to an external resistance R , as shown in figure. Find the value of R so that the potential difference across the 6 V battery be zero.



Finding the value of R

2

$$i = \frac{9}{R+1}$$

As potential difference across 6V is zero;

$$6 - ir = 0 \Rightarrow 6 - \left(\frac{9}{R+1} \right) (0.8) = 0$$

On solving;

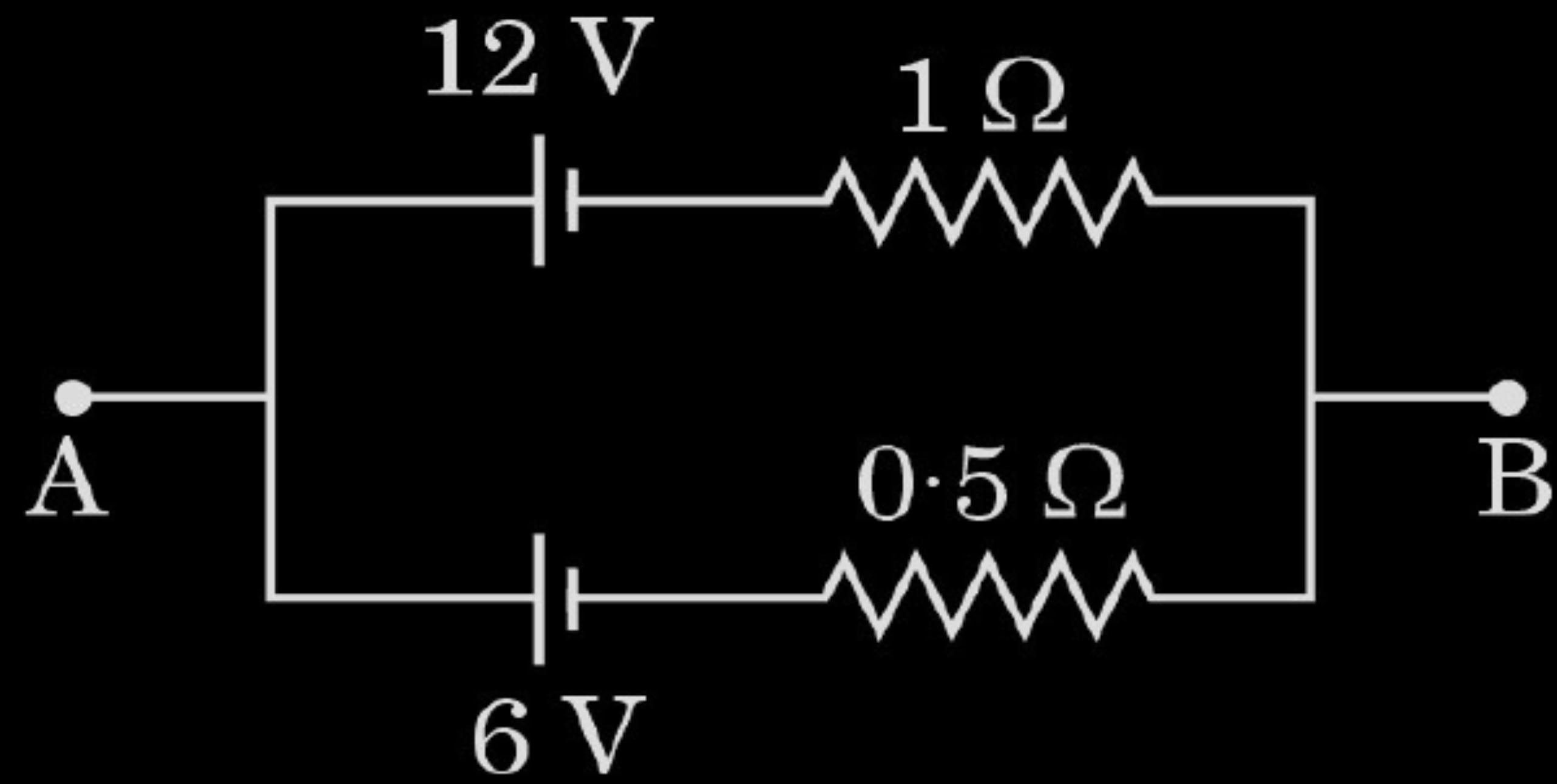
$$R = 0.2\Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

1

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



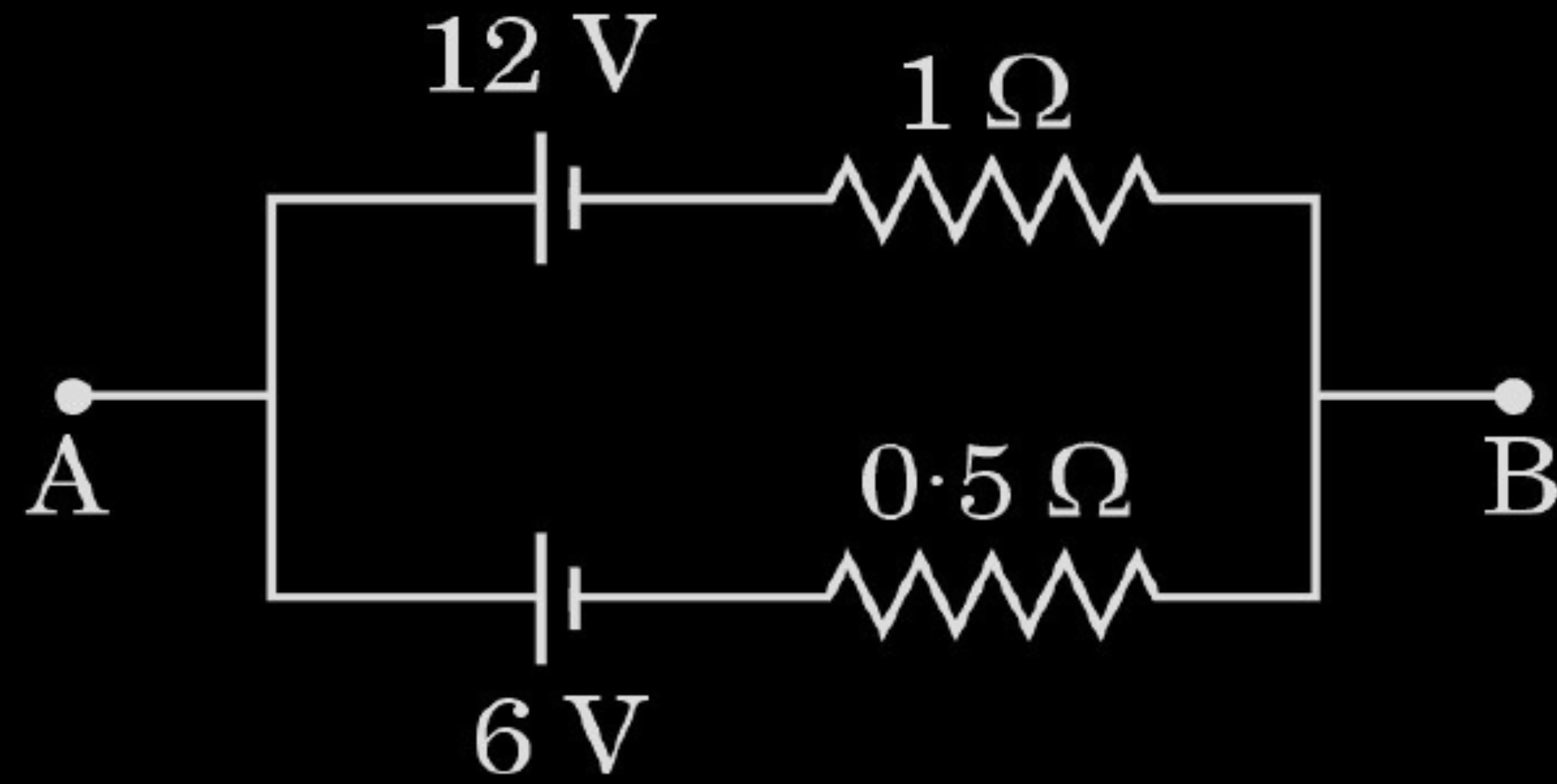
(A) 6 V

(B) 8 V

(C) 9 V

(D) 12 V

3. Consider the circuit shown in the figure. The potential difference between points A and B is :



(A) 6 V

(B) 8 V

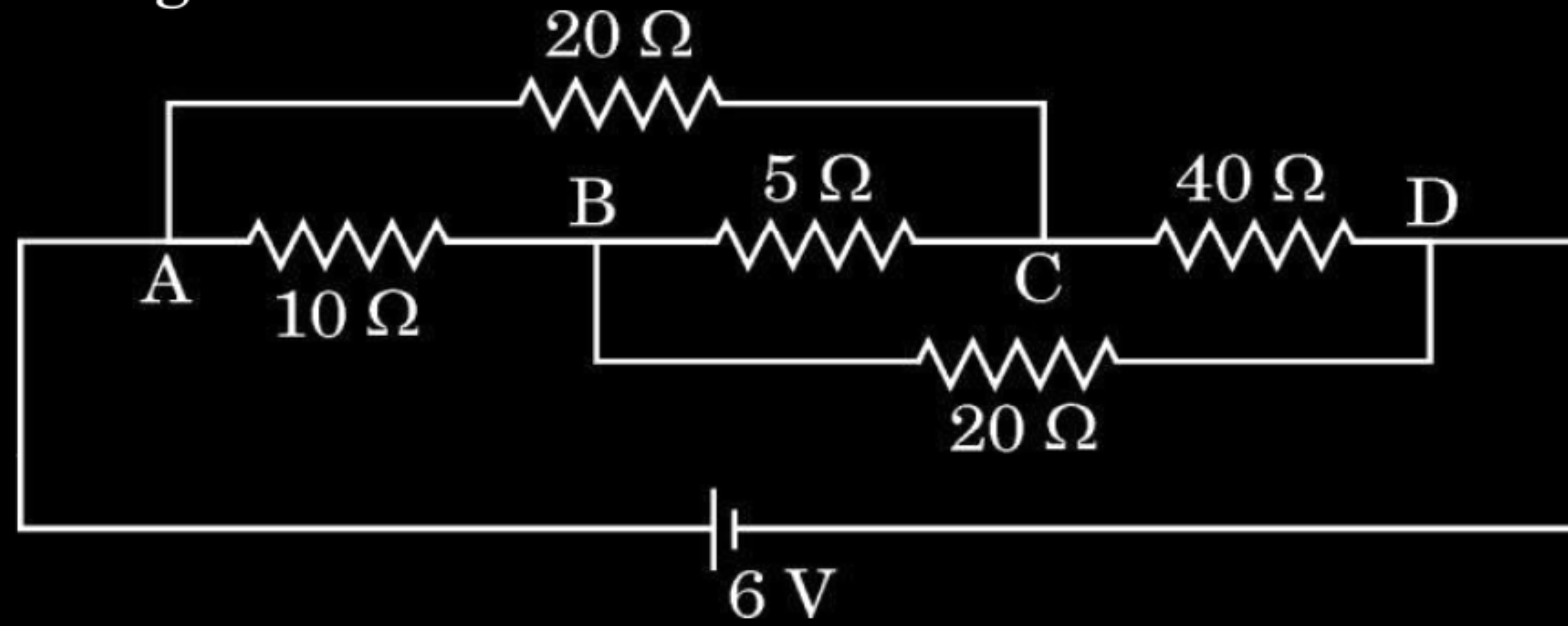
(C) 9 V

(D) 12 V

(B) 8V

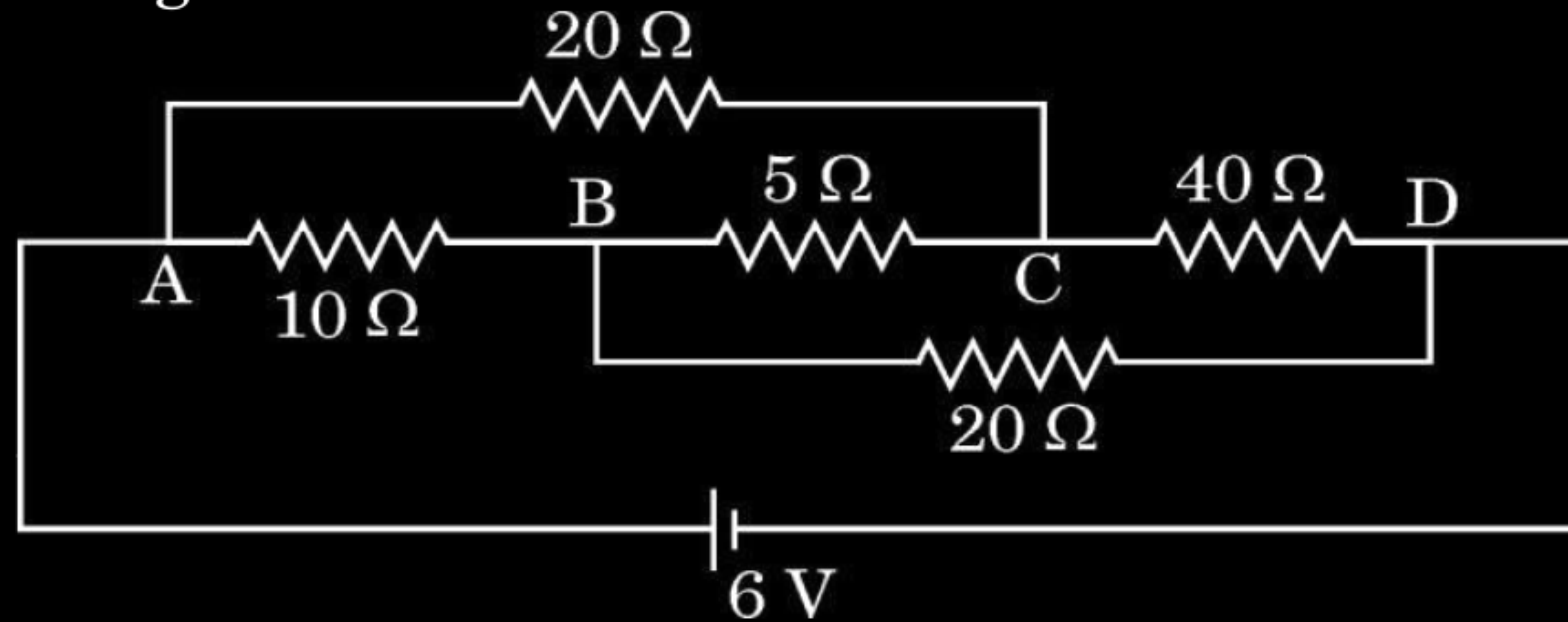
17. Calculate the value of the current passing through the battery in the given circuit diagram.

2



17. Calculate the value of the current passing through the battery in the given circuit diagram.

2



The given circuit forms a balanced Wheatstone bridge so no current flows through 5Ω resistor.

$$R_{eq} = \frac{30 \times 60}{30 + 60}$$

$$= 20 \Omega$$

$$I = \frac{V}{R_{eq}}$$

$$= \frac{6}{20} = \frac{3}{10} A$$

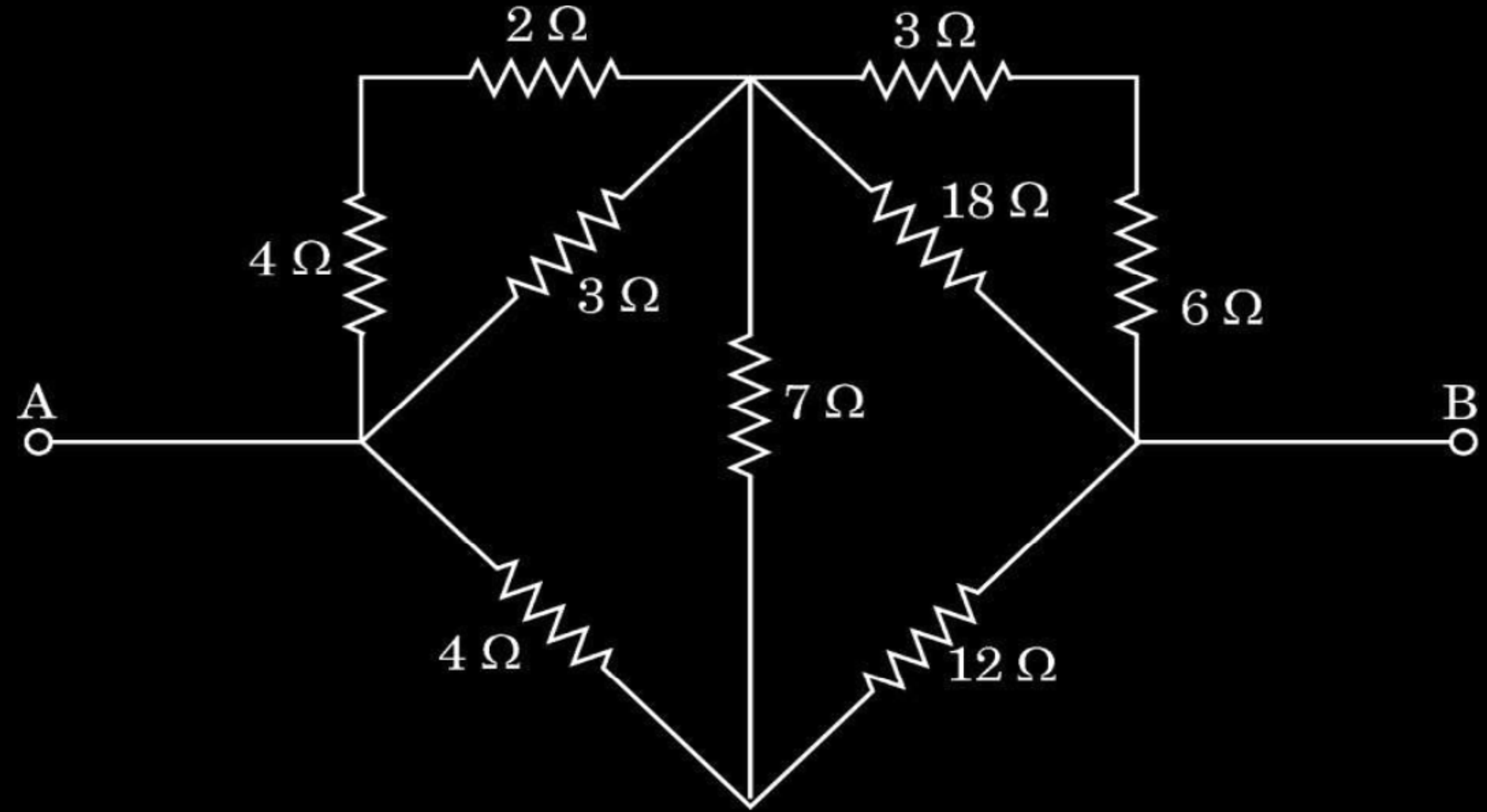
$\frac{1}{2}$

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$\frac{1}{2}$

7. The effective resistance between points A and B in the given circuit is :



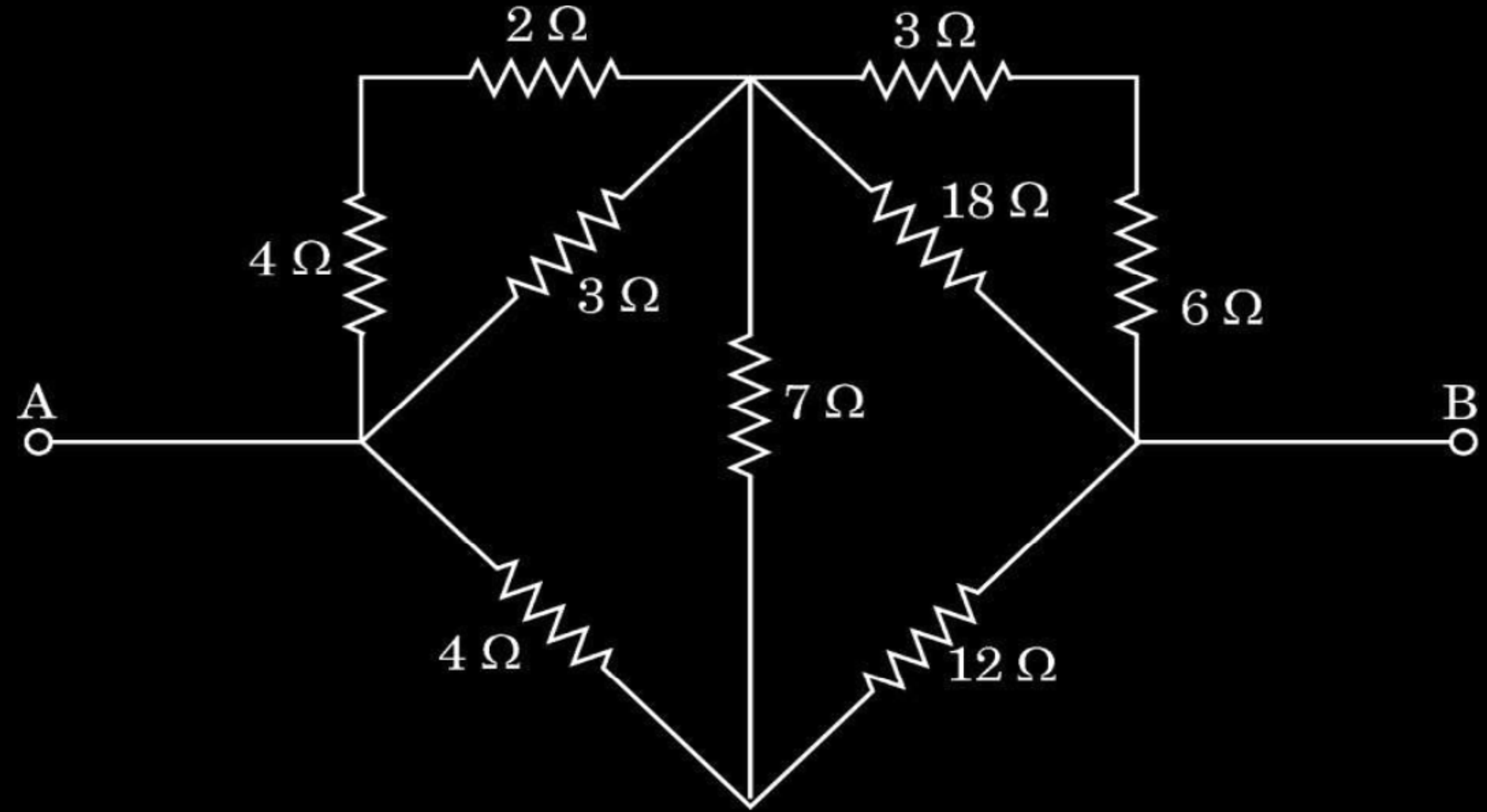
(A) $6\ \Omega$

(B) $\frac{8}{3}\ \Omega$

(C) $\frac{16}{3}\ \Omega$

(D) $2\ \Omega$

7. The effective resistance between points A and B in the given circuit is :



(A) 6Ω

(B) $\frac{8}{3} \Omega$

(C) $\frac{16}{3} \Omega$

(D) 2Ω

- 22.** Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are $5\ \Omega$, $5\ \Omega$ and $1\ \Omega$ respectively. Calculate the currents flowing through the cells A, B and C. 3

- 22.** Three cells A, B and C of emfs 2 V, 3 V and 5 V respectively are connected in parallel to each other. Their internal resistances are $5\ \Omega$, $5\ \Omega$ and $1\ \Omega$ respectively. Calculate the currents flowing through the cells A, B and C. 3

In loop ABEFA,

$$5 - 3 - 5I_1 - I = 0$$

$$2 = 5I_1 + I \quad \text{-----(1)}$$

In loop CBEDC,

$$2 - 3 - 5I_1 + 5I - 5I_1 = 0$$

$$-1 = 10I_1 - 5I \quad \text{-----(2)}$$

Solving equation (1) and (2)

$$I = \frac{5}{7} \text{ A} \quad \text{in arm AF/through the cell of 5V(A)}$$

$$I_1 = \frac{9}{35} \text{ A} \quad \text{in arm BE/through the cell of 3V(B)}$$

$$I - I_1 = \frac{16}{35} \text{ A} \quad \text{in arm CD/through the cell of 2V(C)}$$

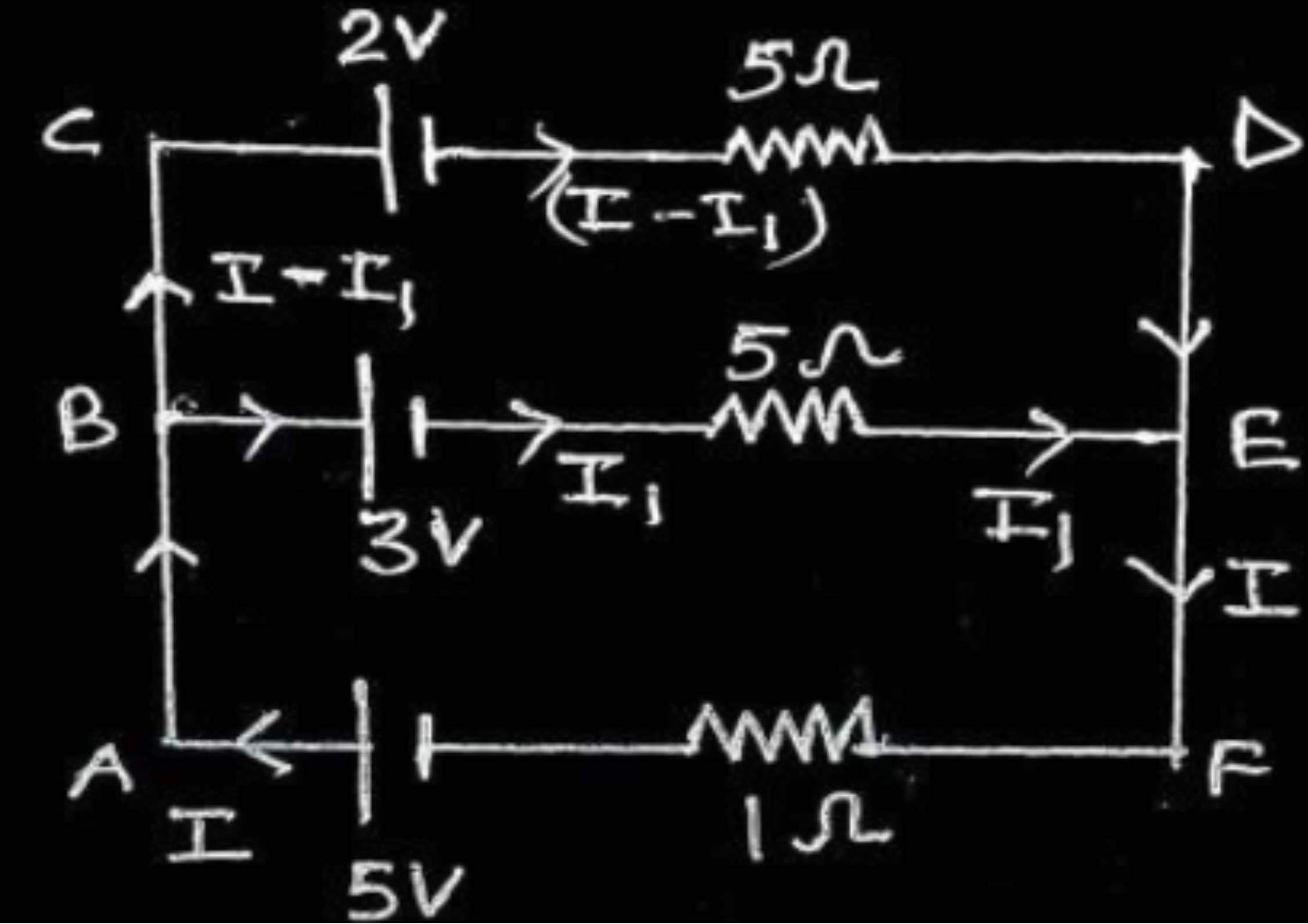
$$\frac{1}{2}$$

$$\frac{1}{2}$$

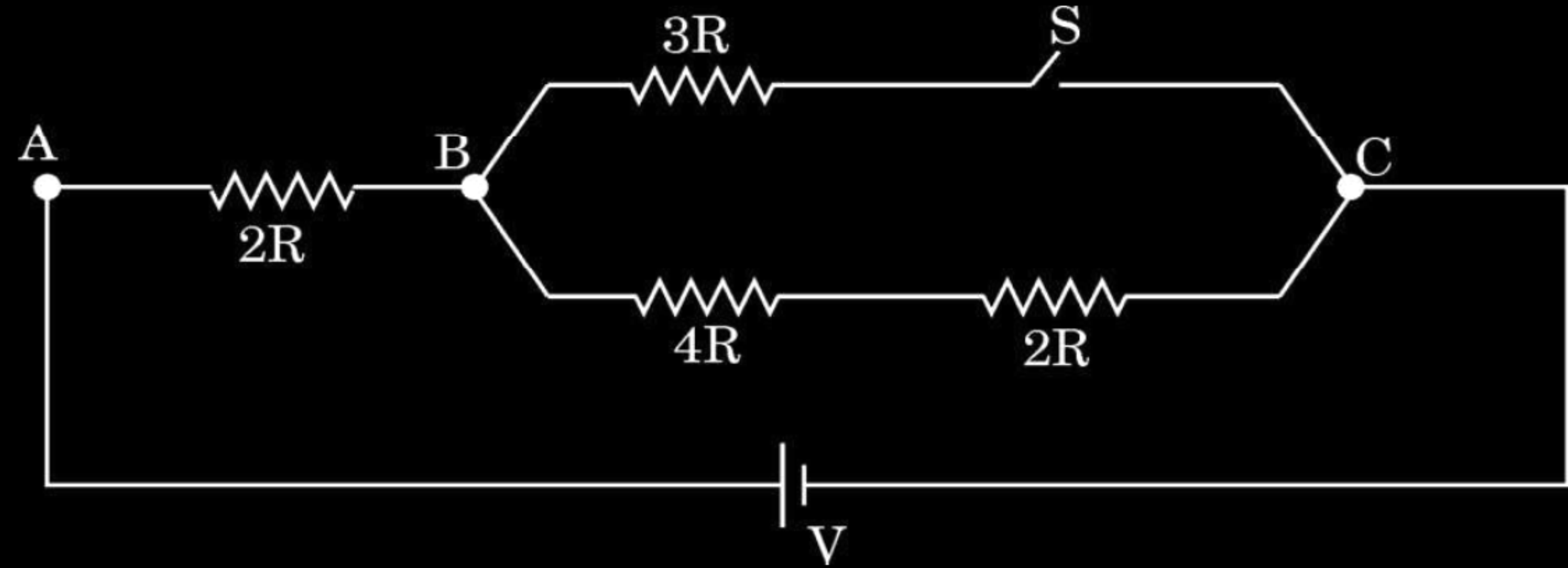
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1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



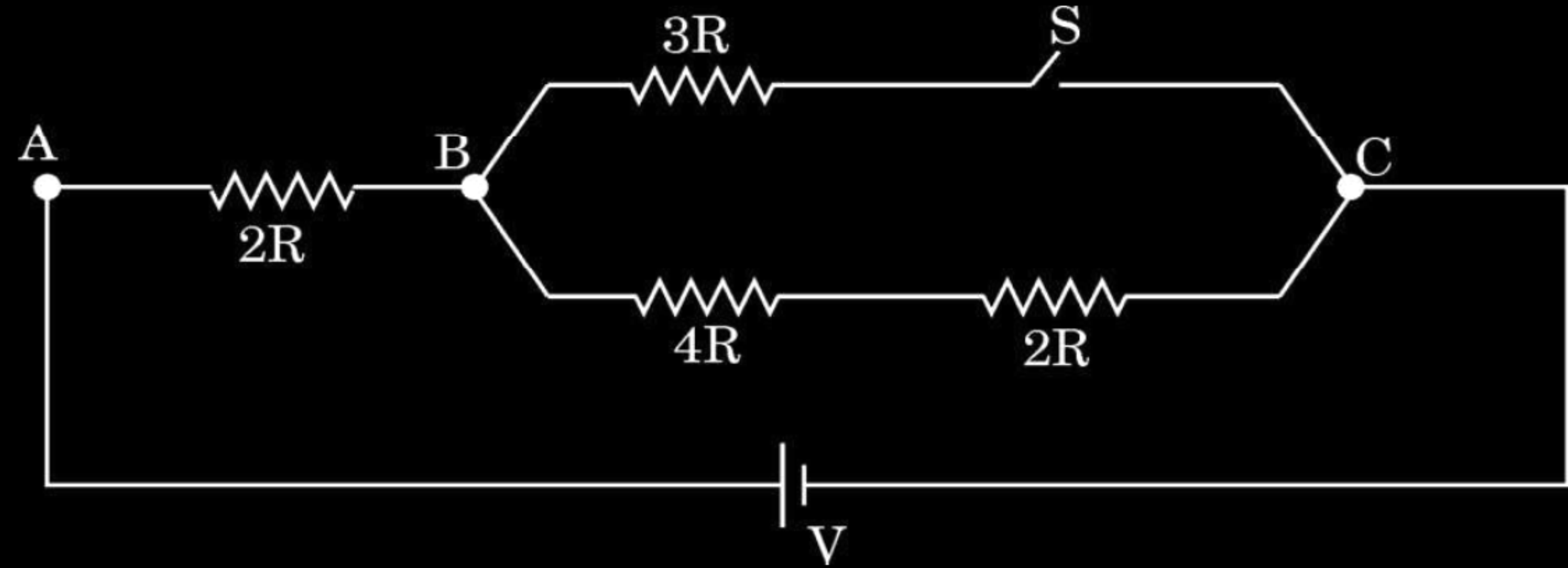
(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

1. The ratio of potential difference across AB in the circuit shown for the case (i) when switch S is closed and (ii) when S is open is :



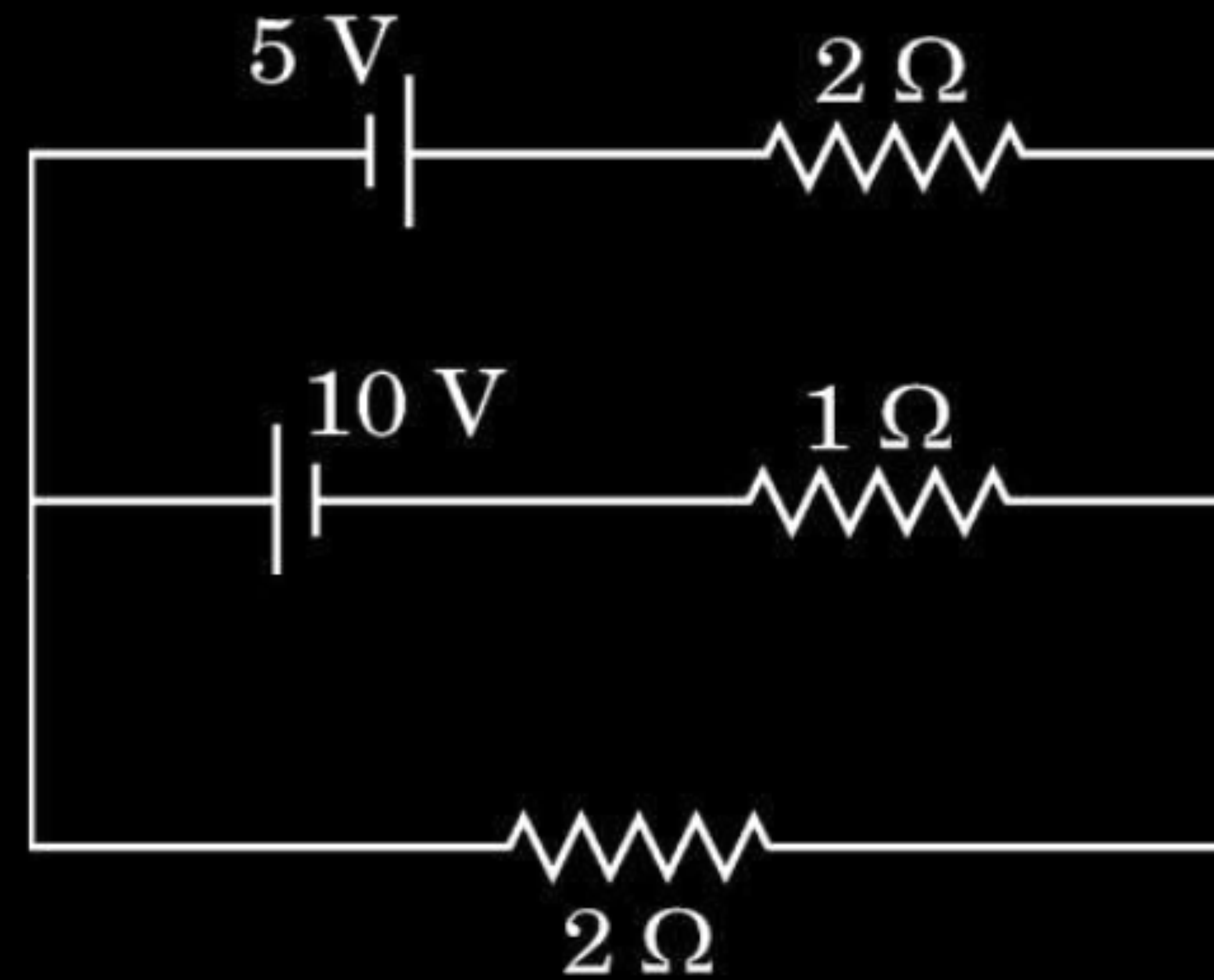
(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 1

(D) 2

26. State Kirchhoff's laws. Apply these laws to find the values of current flowing in the three branches of the given circuit. 3



Stating Kirchhoff's laws

$\frac{1}{2} + \frac{1}{2}$

Finding the values of current in all the three branches

2

Junction rule: At any junction, the sum of the currents entering the junction is equal to the sum of the currents leaving the junction.

1/2

Loop rule: the algebraic sum of changes in potential around any closed loop involving resistors and cells in the loop is zero.

1/2

In closed loop $BCDEB$ $2I_1 + 3I_2 = 10$ ----- (1)

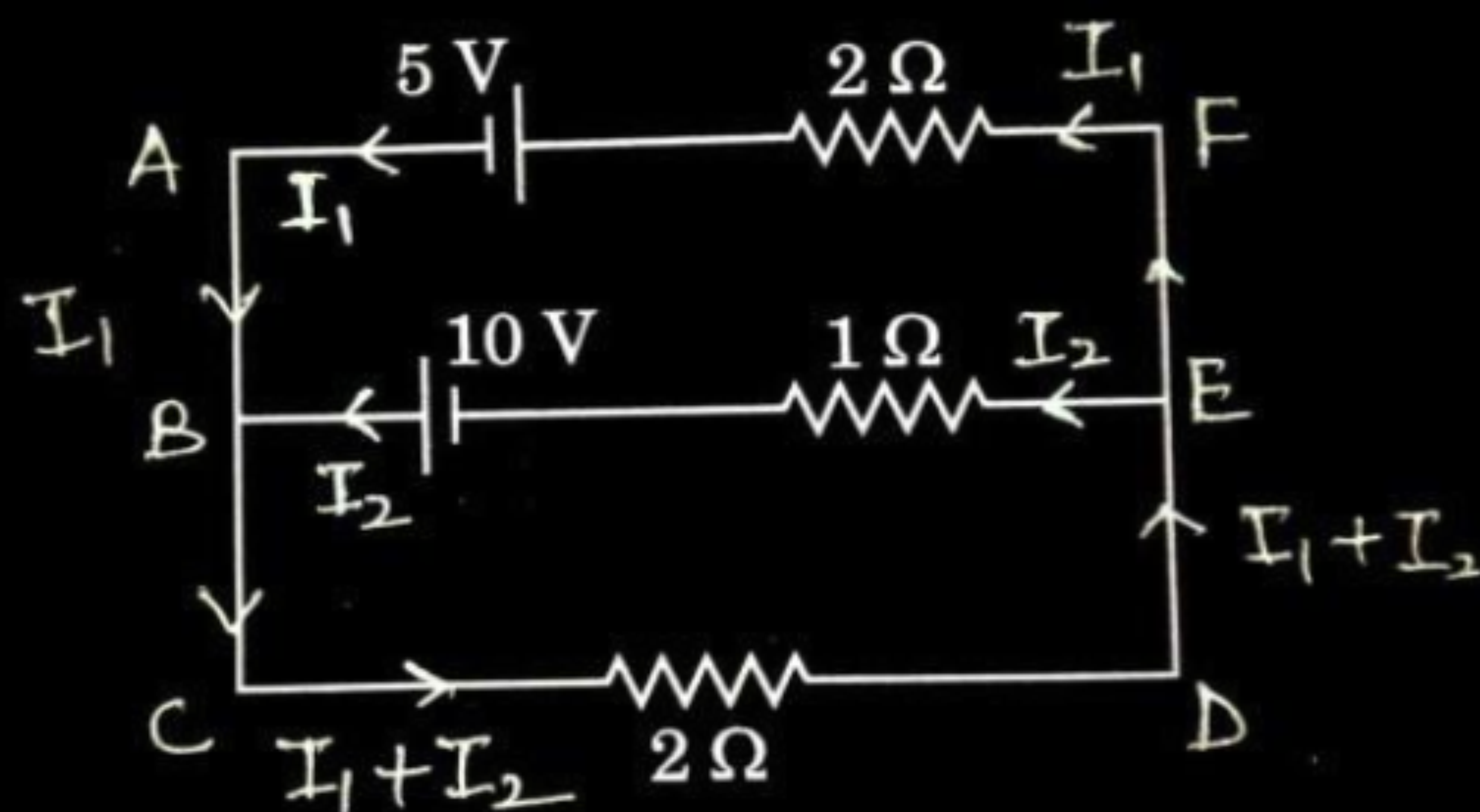
In closed loop $ABCDEFA$ $4I_1 + 2I_2 = -5$ ----- (2)

On solving eq. (1) and (2)

$$I_1 = -\frac{35}{8} \text{ A} \quad \text{in arm AF}$$

$$I_2 = \frac{25}{4} \text{ A} \quad \text{in arm BE}$$

$$I_1 + I_2 = -\frac{15}{8} \text{ A} \quad \text{in arm CD}$$



1/2

1/2

1/2

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

2. If R_s and R_p are the equivalent resistances of n resistors, each of value R , in series and parallel combinations respectively, then the value of $(R_s - R_p)$ is :

(A) $\left(\frac{n^2 - 1}{n^2}\right)R$

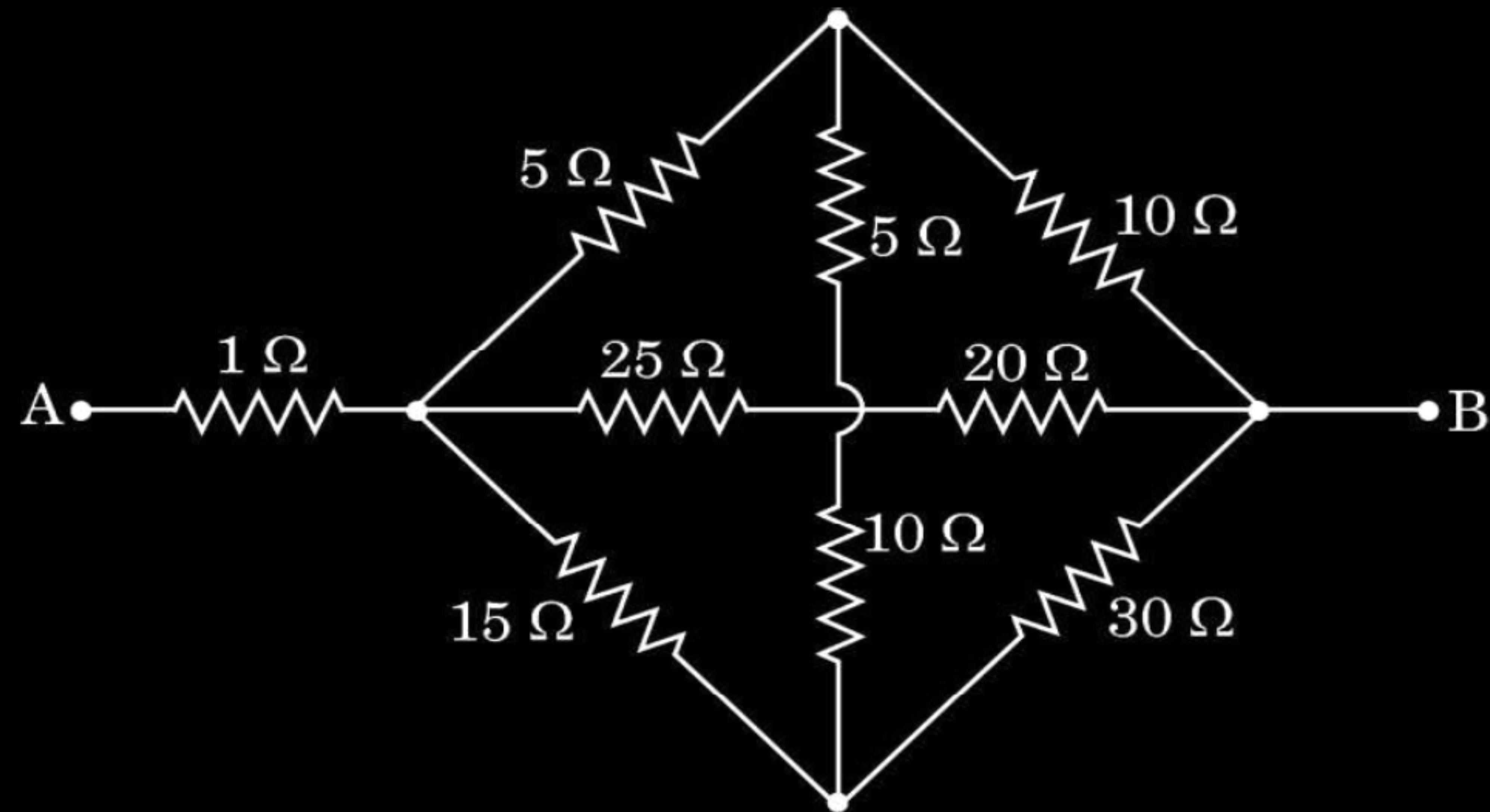
(B) $\left(\frac{n^2 + 1}{n^2 - 1}\right)R$

(C) $\left(\frac{n^2 - 1}{n}\right)R$

(D) $\frac{(n^2 + 1)R}{n^2}$

17. Find the equivalent resistance between points A and B for the network shown in the figure.

2



Resistance between points C and B

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

$$\frac{1}{R} = \frac{1}{15} + \frac{1}{45} + \frac{1}{45}$$

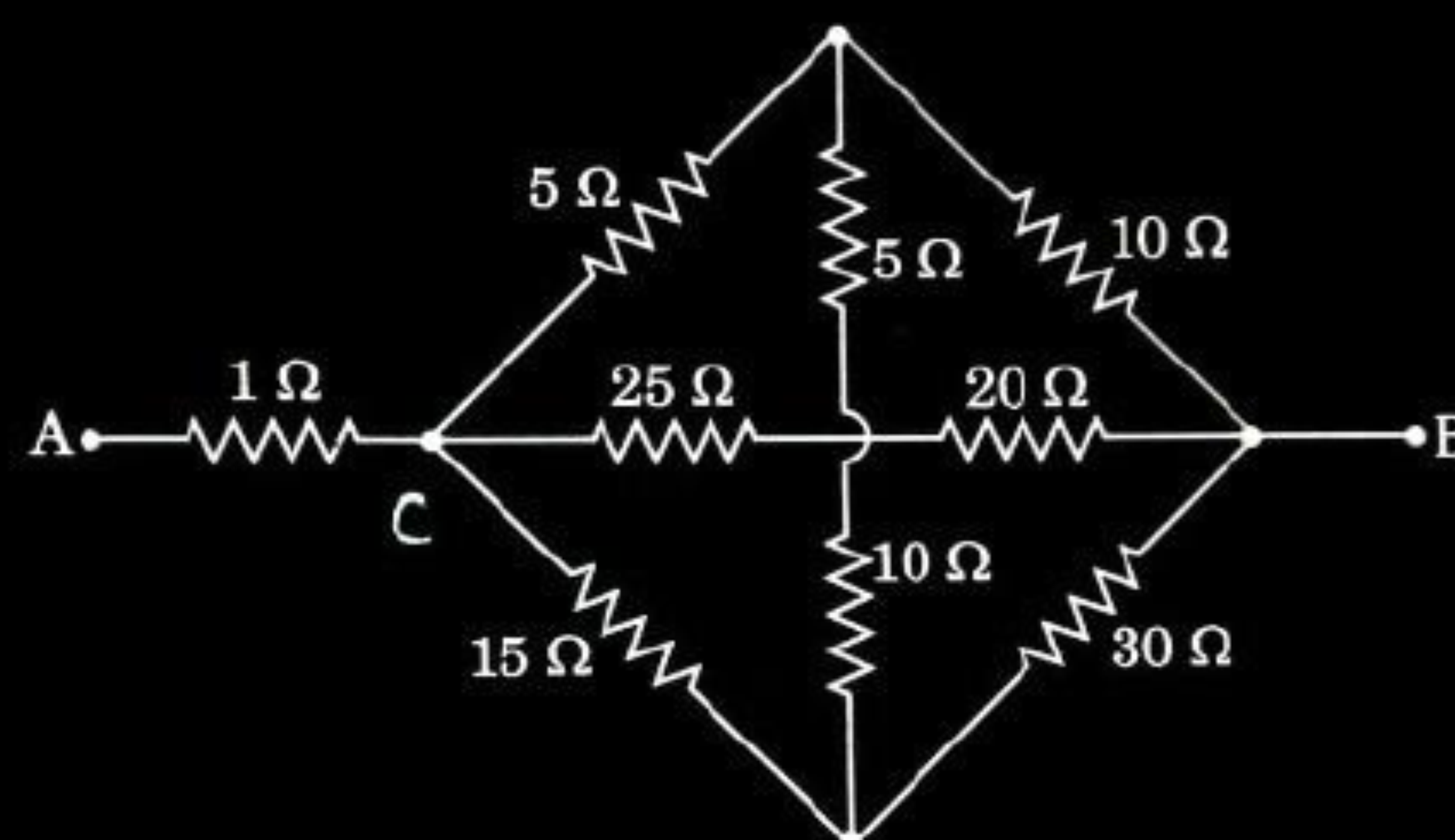
$$R = 9 \Omega$$

Equivalent resistance between points A and B

$$R_{eq} = R_1 + R_2$$

$$R_{eq} = 1 + 9$$

$$= 10 \Omega$$



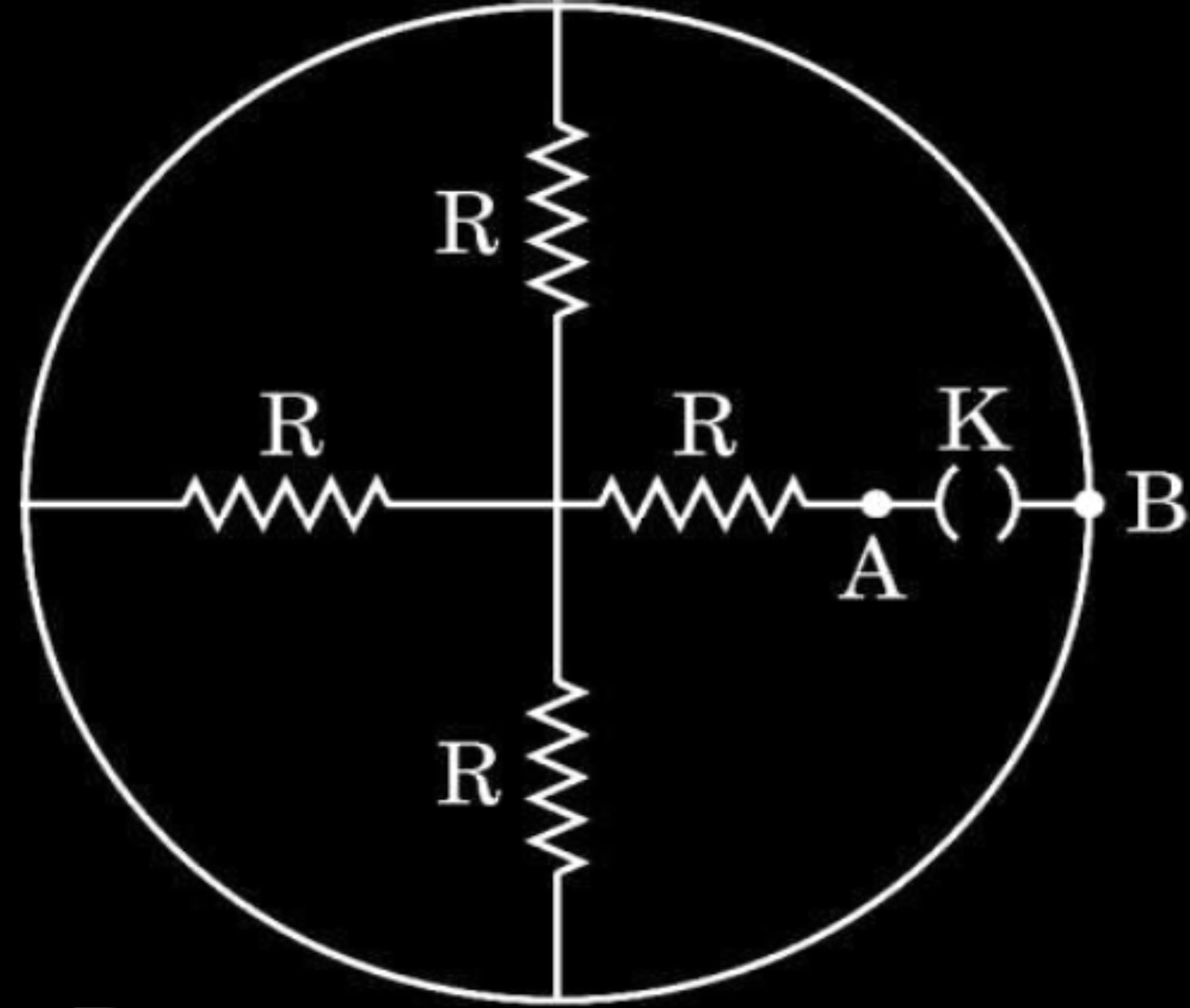
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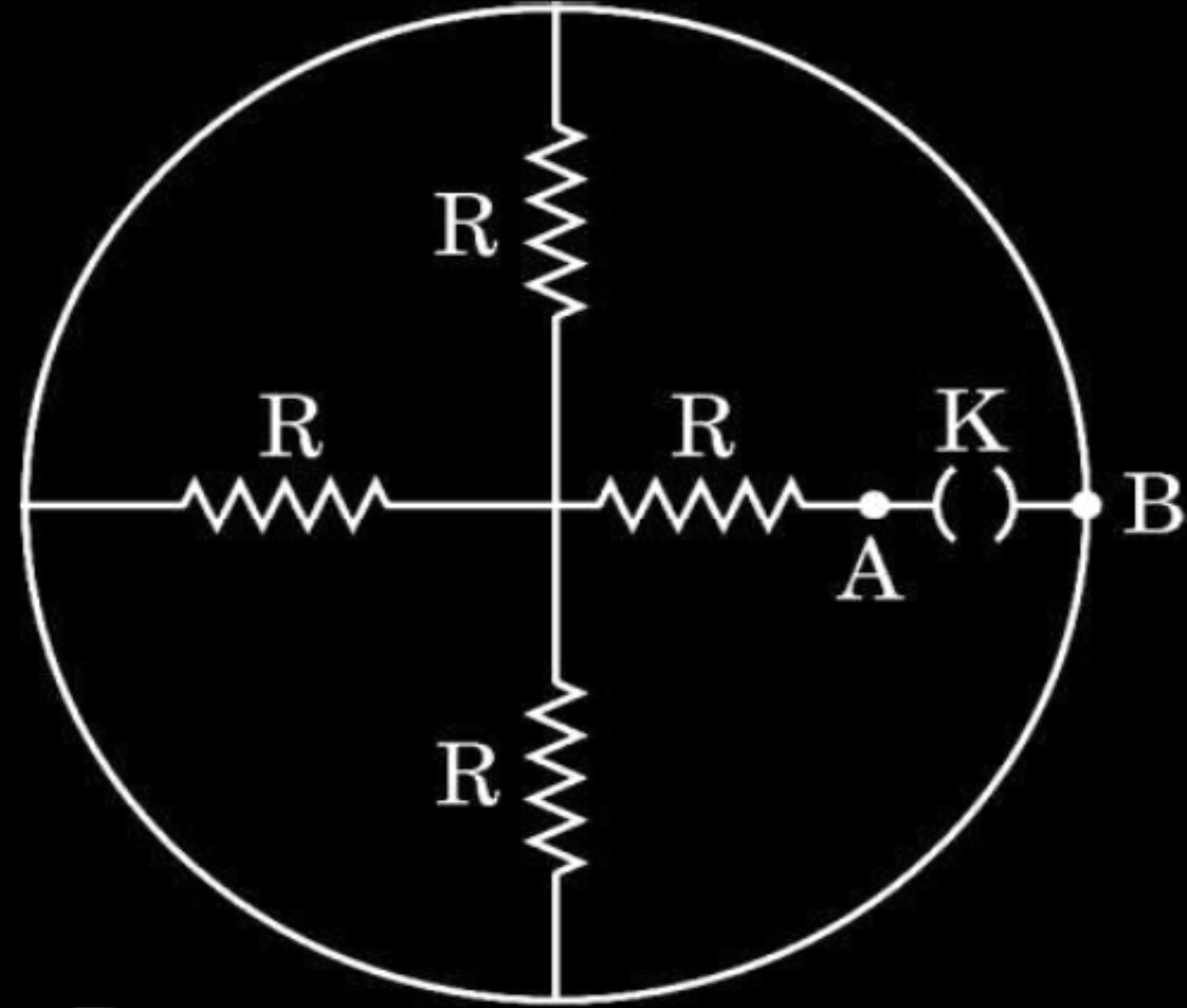
5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :



- (A) $4R$
(C) $\frac{R}{4}$

- (B) ∞
(D) $\frac{4R}{3}$

5. Four resistors, each of resistance R and a key K are connected as shown in the figure. The equivalent resistance between points A and B when key K is open, will be :

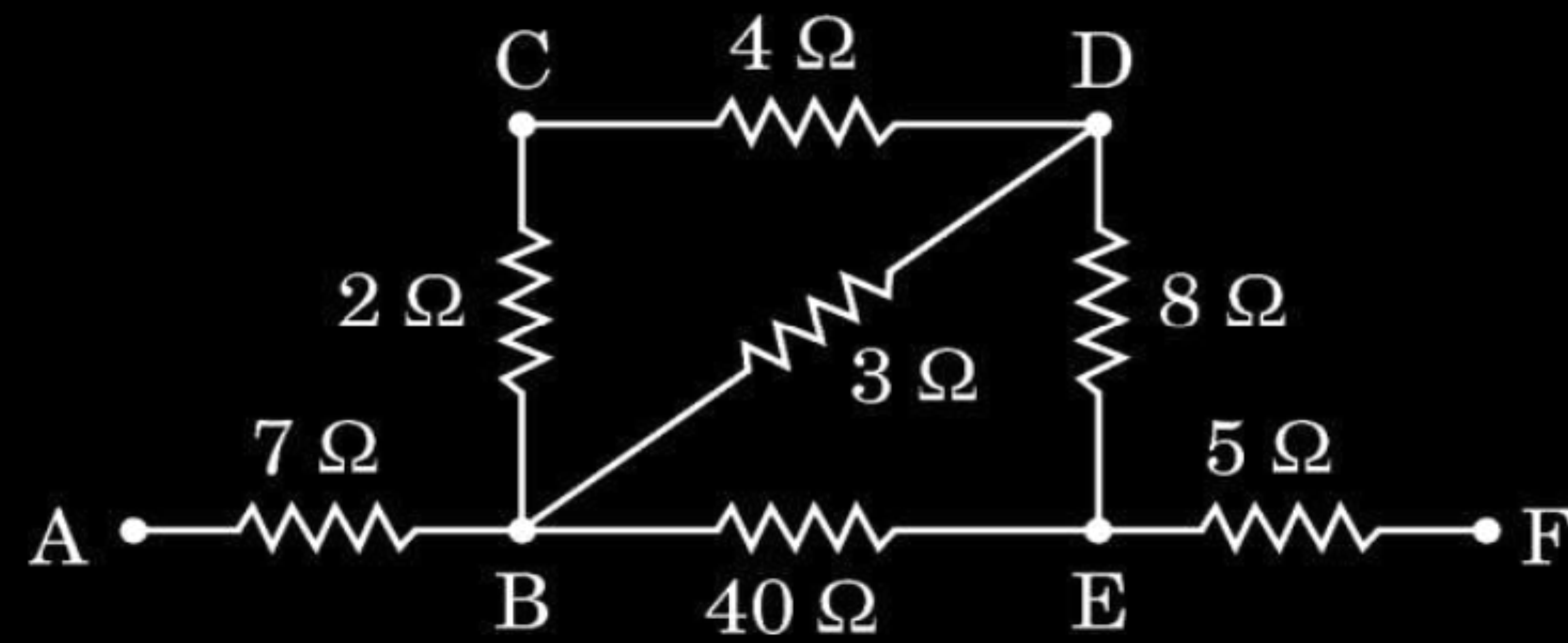


- (A) $4R$
(C) $\frac{R}{4}$

- (B) ∞
(D) $\frac{4R}{3}$

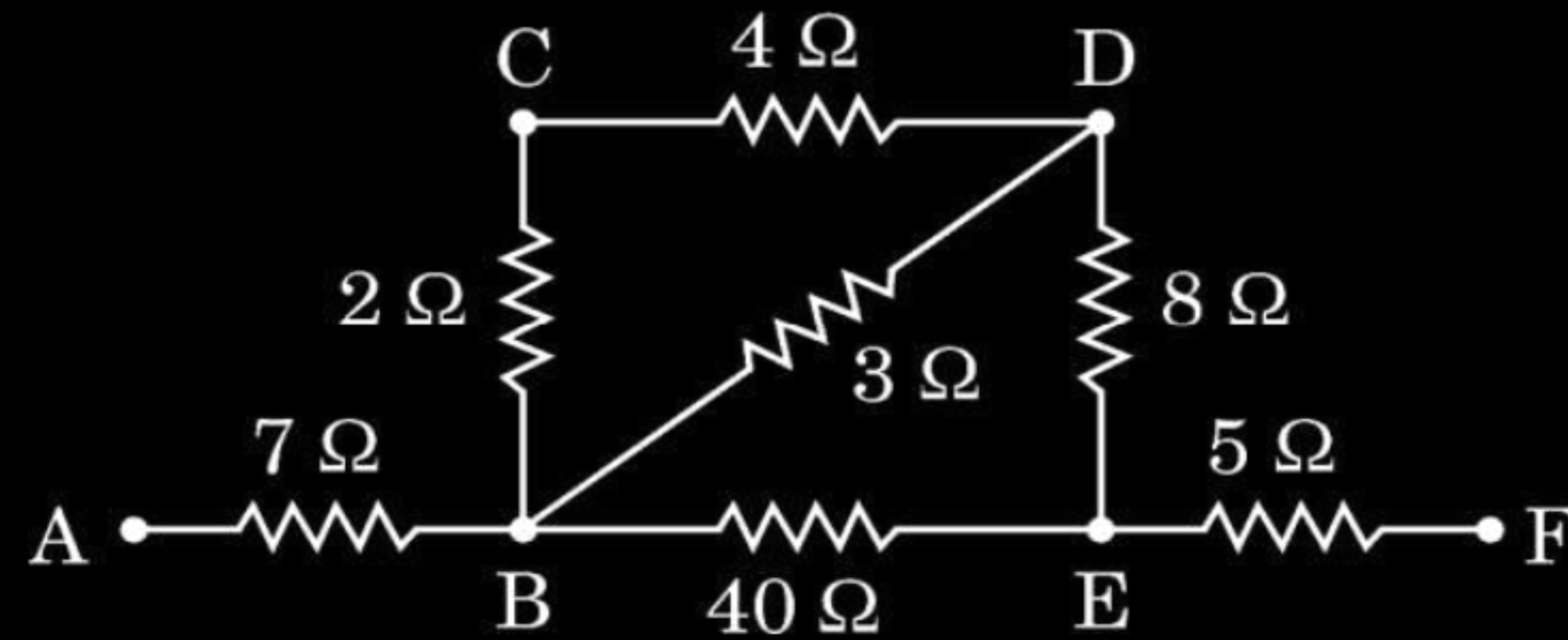
17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

2



17. Find the effective resistance of the network of resistors between points A and F as shown in the figure.

2



$$R_{BCD} = 2 + 4 = 6\Omega$$

$$R_{BD} = \frac{3 \times 6}{6 + 3}$$

$$= 2\Omega$$

$$R_{BDE} = 2 + 8$$

$$= 10\Omega$$

$$R_{BE} = \frac{40 \times 10}{40 + 10}$$

$$= 8\Omega$$

$$R_{AF} = 7 + 8 + 5$$

$$= 20\Omega$$

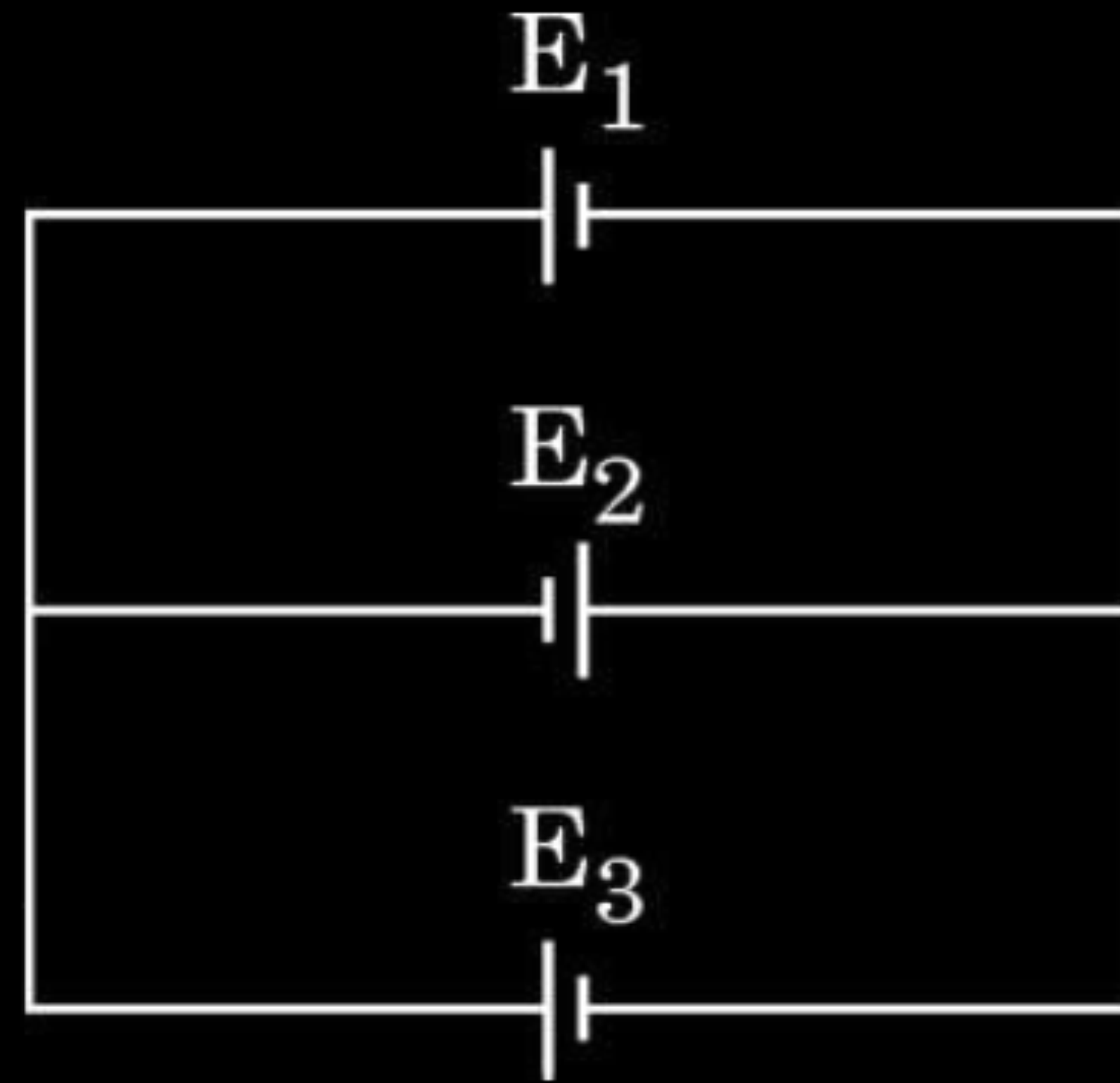
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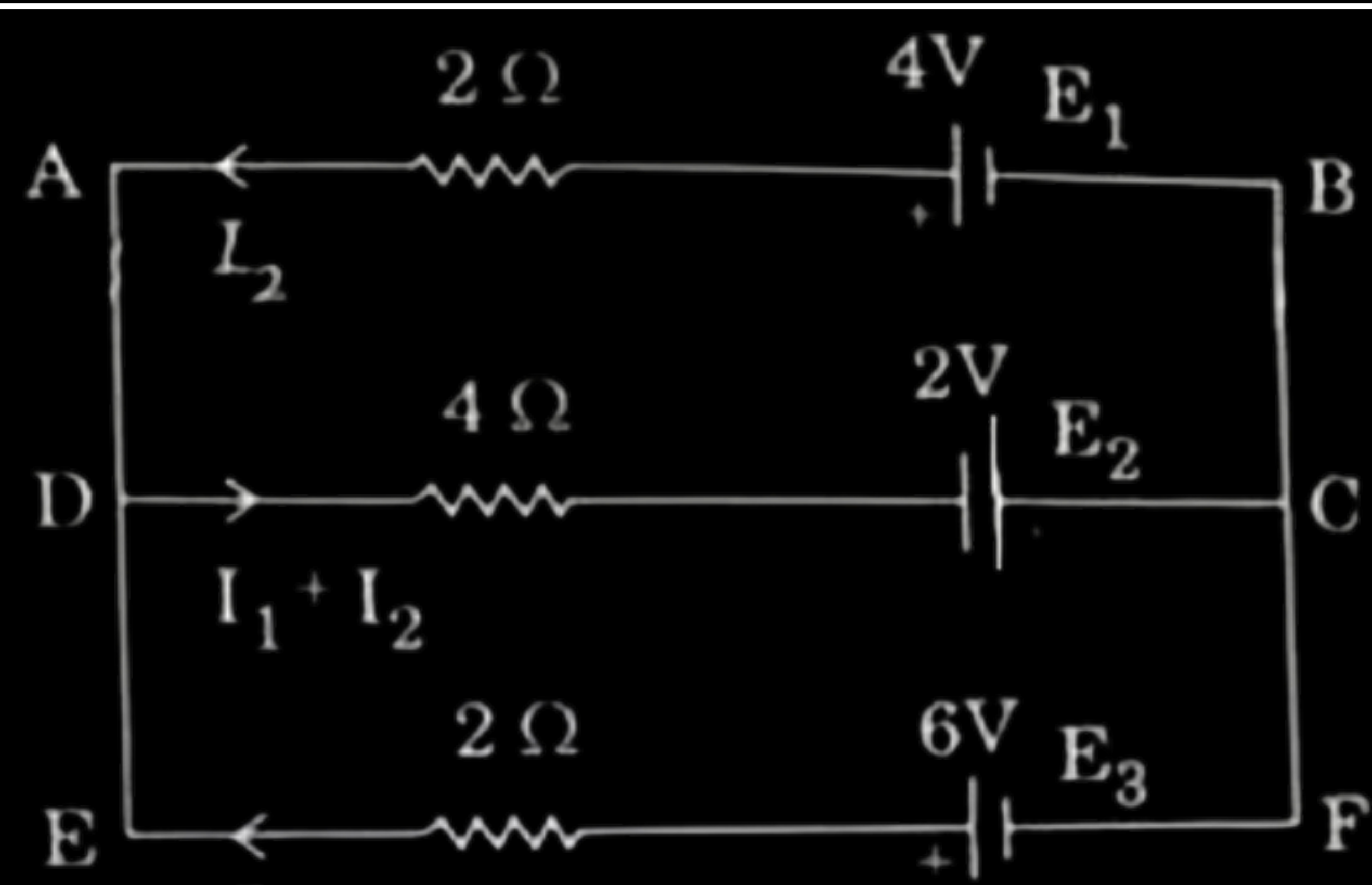
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$\frac{1}{2}$

- 32.** (a) (i) Three batteries E_1 , E_2 and E_3 of emfs and internal resistances (4 V, $2\ \Omega$), (2 V, $4\ \Omega$) and (6 V, $2\ \Omega$) respectively are connected as shown in the figure. Find the values of the currents passing through batteries E_1 , E_2 and E_3 .



- | | | |
|------|---|---|
| (i) | Finding current through batteries E_1 , E_2 and E_3 | 3 |
| (ii) | Finding effective resistance | 2 |



i)

In closed loop ABCD, using Kirchhoff's loop law

$$4I_1 + 6I_2 = 6 \dots\dots\dots(1)$$

Similarly In closed loop CDFE

$$6I_1 + 4I_2 = 8 \dots\dots\dots(2)$$

Solving eqn. (1) and (2)

$$I_2 = \frac{1}{5}A$$

$$I_1 = \frac{6}{5}A$$

$$I_1 + I_2 = \frac{7}{5}A$$

1/2

1/2

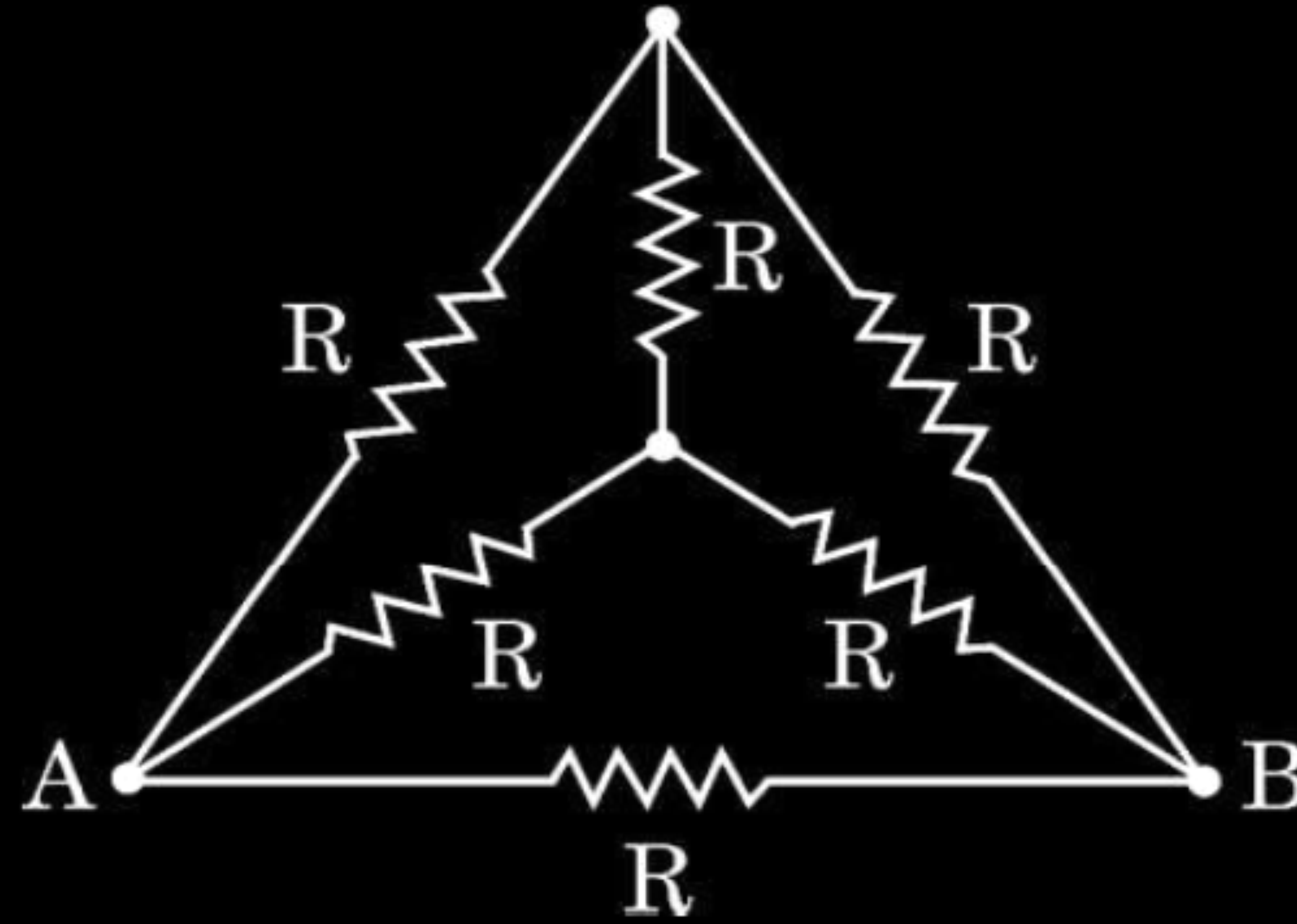
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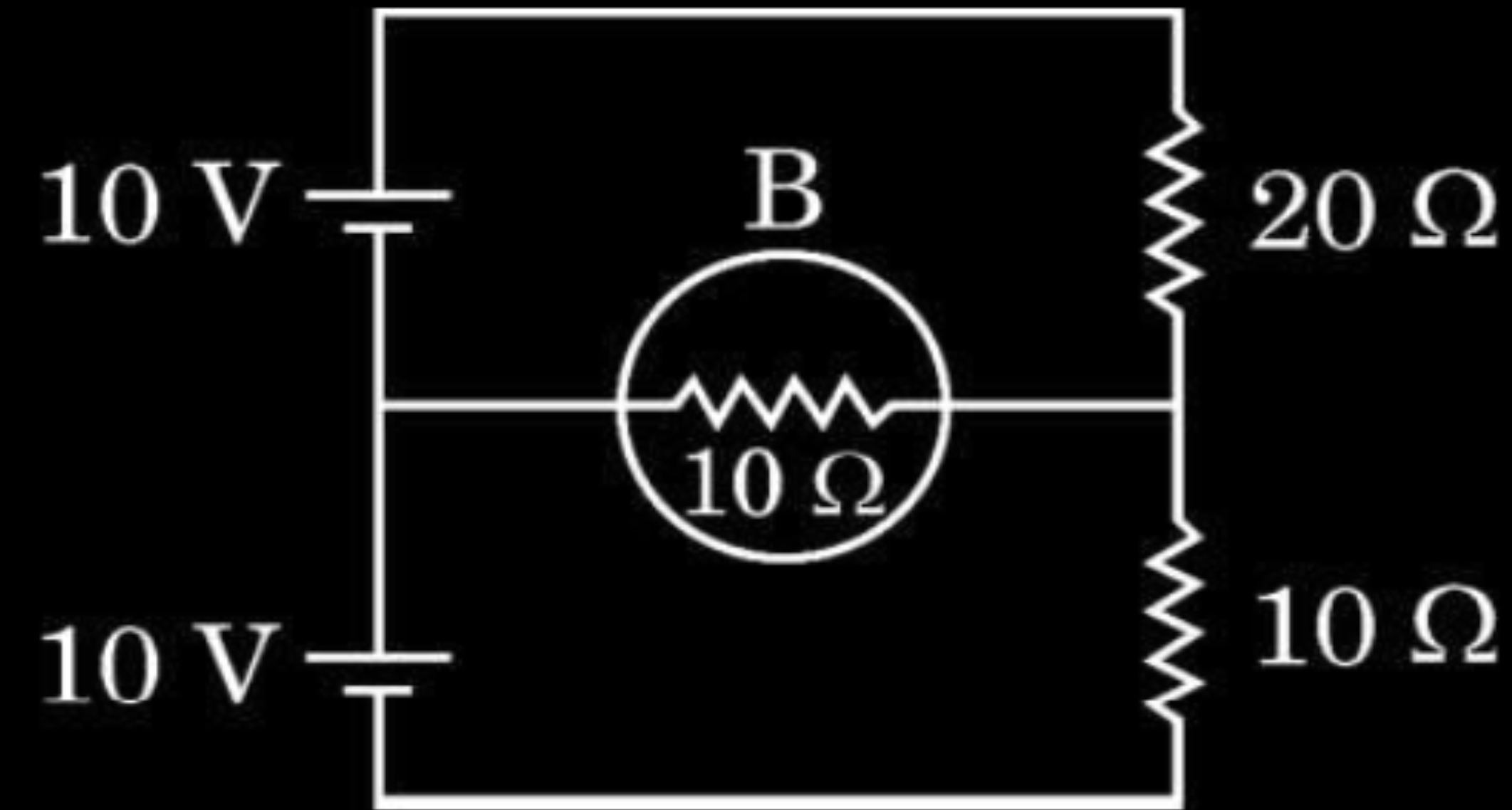
- (ii) The ends of six wires, each of resistance R ($= 10 \Omega$) are joined as shown in the figure. The points A and B of the arrangement are connected in a circuit. Find the value of the effective resistance offered by it to the circuit.

5



OR

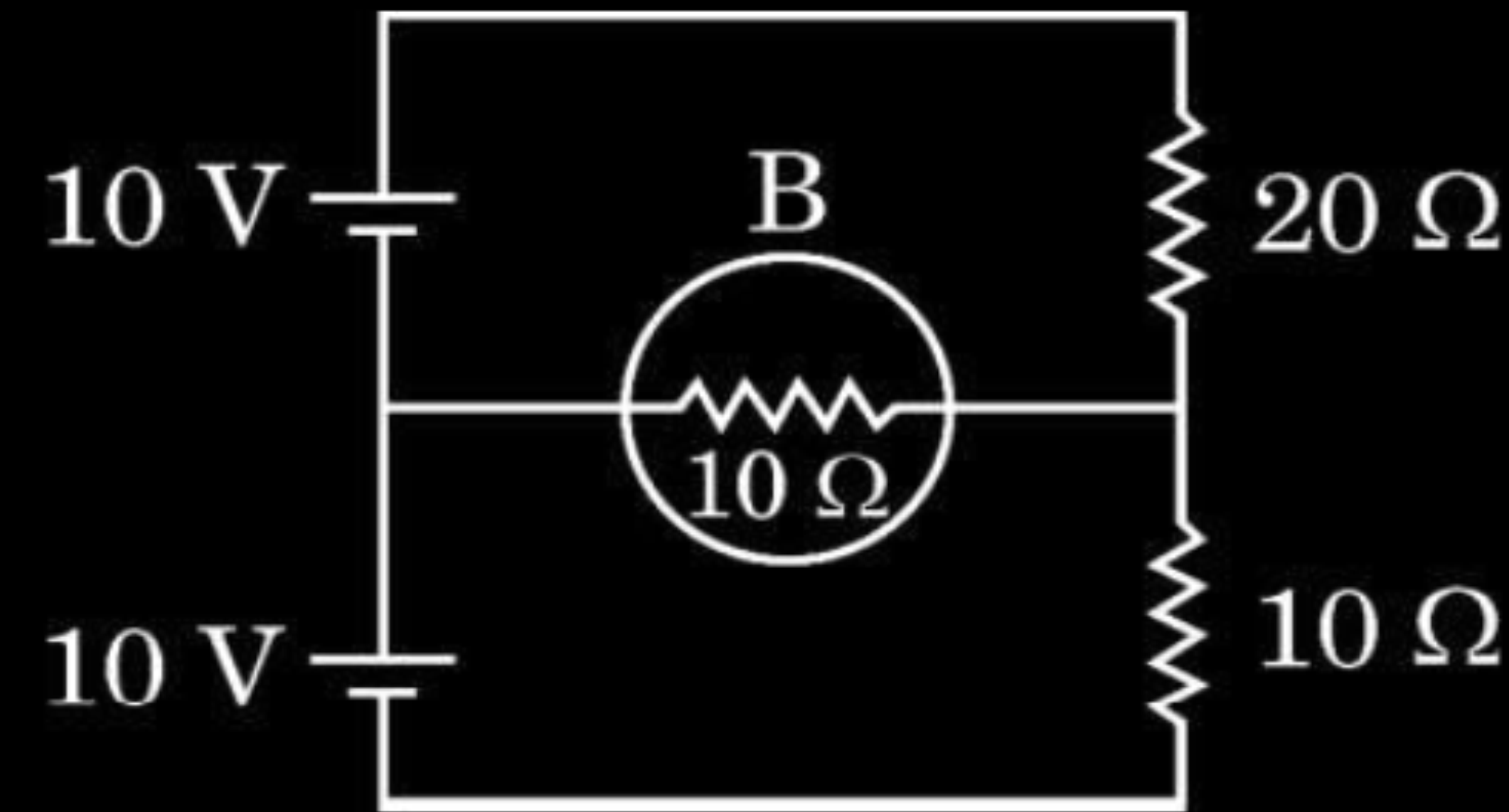
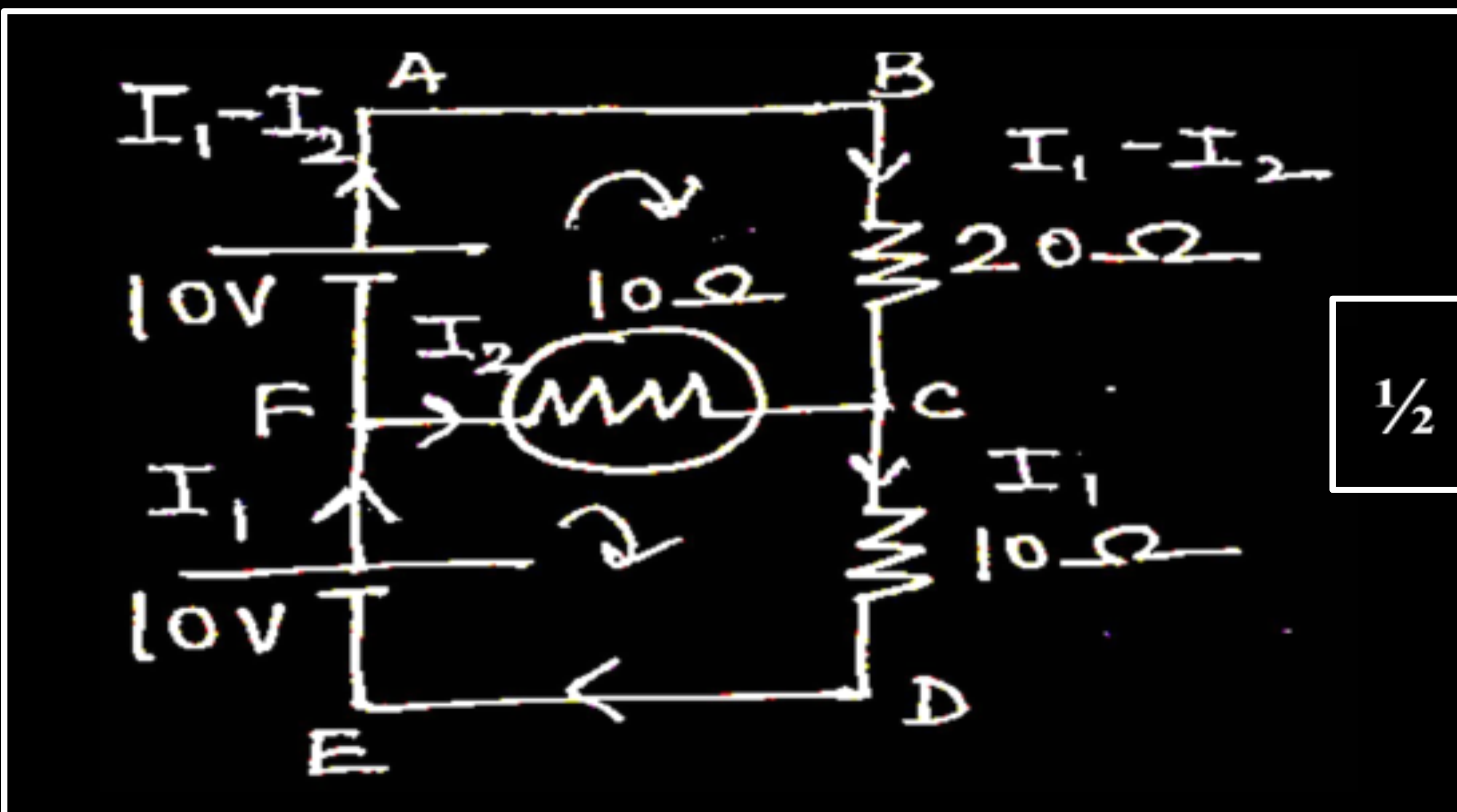
- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.



OR

- (b) Two cells of emf 10 V each, two resistors of $20\ \Omega$ and $10\ \Omega$ and a bulb B of $10\ \Omega$ resistance are connected together as shown in the figure. Find the current that flows through the bulb.

2



By applying Kirchhoff's loop rule to closed loops ABCFA and FCDEF

$$2I_1 - 3I_2 = 1 \text{ --- (1)}$$

$$I_1 + I_2 = 1 \text{ ---- (2)}$$

On solving,

Current through the bulb,

$$I_2 = \frac{1}{5} \text{ A}$$

$$\frac{1}{2}$$

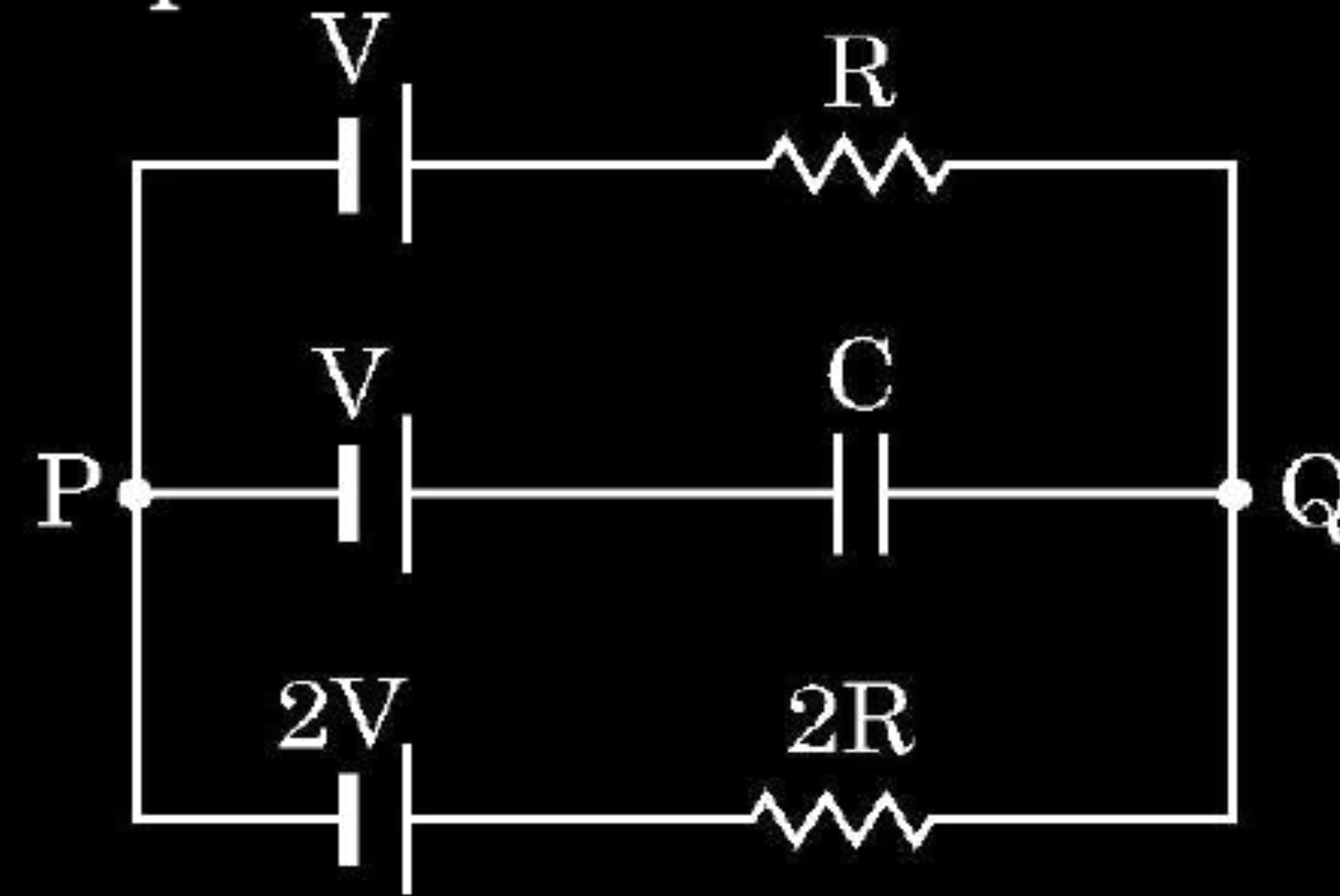
$$\frac{1}{2}$$

$$\frac{1}{2}$$

21. (a) Two wires of the same material and the same radius have their lengths in the ratio $2 : 3$. They are connected in parallel to a battery which supplies a current of 15 A . Find the current through the wires. **2**

OR

- (b) In the circuit three ideal cells of e.m.f. V , V and $2V$ are connected to a resistor of resistance R , a capacitor of capacitance C and another resistor of resistance $2R$ as shown in figure. In the steady state find (i) the potential difference between P and Q and (ii) potential difference across capacitor C .



(a)

Finding current

2

$$R_1 = \frac{\rho l_1}{A}; R_2 = \frac{\rho l_2}{A}$$

$$\frac{l_1}{l_2} = \frac{2}{3} \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}$$

$$I \propto \frac{1}{R}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{3}{2}$$

$$\Rightarrow I_1 = \frac{3}{5} \times 15 = 9A$$

$$\Rightarrow I_2 = \frac{2}{5} \times 15 = 6A$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(b)

Finding the potential difference

(i) between P and Q

1

(ii) across capacitor C

1

In steady state,

$$2V - V = i(2R + R)$$

$$i = \frac{V}{3R}$$

$$(i) \quad V_P - V_Q = -V - iR$$

$$= -V - \frac{V}{3}$$

$$V_P - V_Q = -\frac{4V}{3}$$

1

$$(ii) \quad V_P - V_Q = -V + V_C$$

$$-\frac{4V}{3} = -V + V_C$$

1

$$V_C = -\frac{V}{3}$$

2. A student has three resistors, each of resistance R . To obtain a resistance of $\frac{2}{3} R$, she should connect

1

- (A) all the three resistors in series.
- (B) all the three resistors in parallel.
- (C) two resistors in series and then this combination in parallel with the third resistor.
- (D) two resistors in parallel and then this combination in series with the third resistor.

2. A student has three resistors, each of resistance R . To obtain a resistance of $\frac{2}{3} R$, she should connect

1

(A) all the three resistors in series.

(B) all the three resistors in parallel.

(C) two resistors in series and then this combination in parallel with the third resistor.

(D) two resistors in parallel and then this combination in series with the third resistor.

2. A uniform wire of resistance $2R$ is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

- (a) $2R$ (b) R (c) $\frac{R}{2}$ (d) $\frac{R}{4}$ **1**

2. A uniform wire of resistance $2R$ is bent in the form of a circle. The effective resistance between the ends of any diameter of the circle is :

(a) $2R$

(b) R

(c) $\frac{R}{2}$

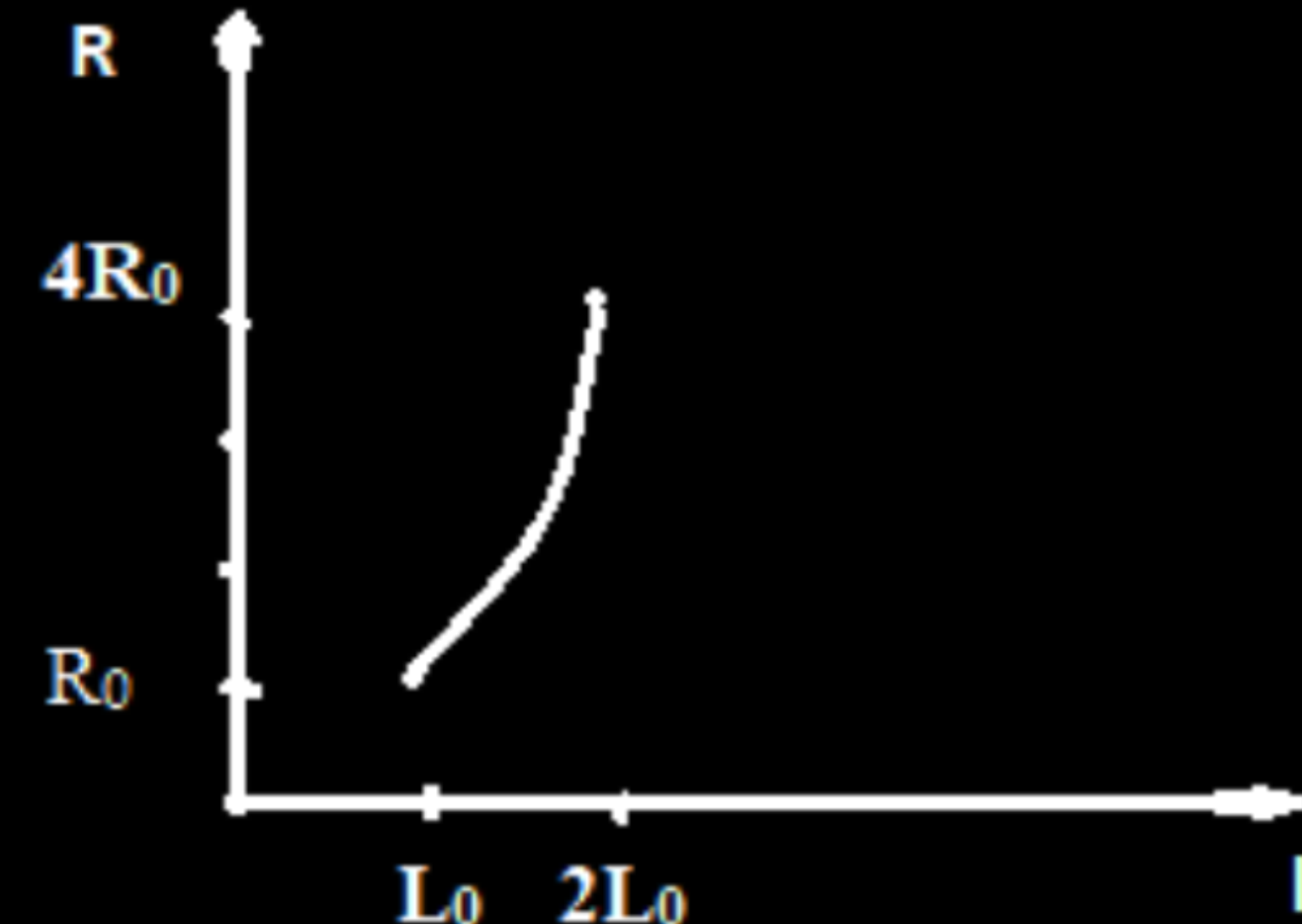
(d) $\frac{R}{4}$

22. A wire of length L_0 has a resistance R_0 . It is gradually stretched till its length becomes $2 L_0$.
- (a) Plot a graph showing variation of its resistance R with its length l during stretching.
 - (b) What will be its resistance when its length becomes $2 L_0$?

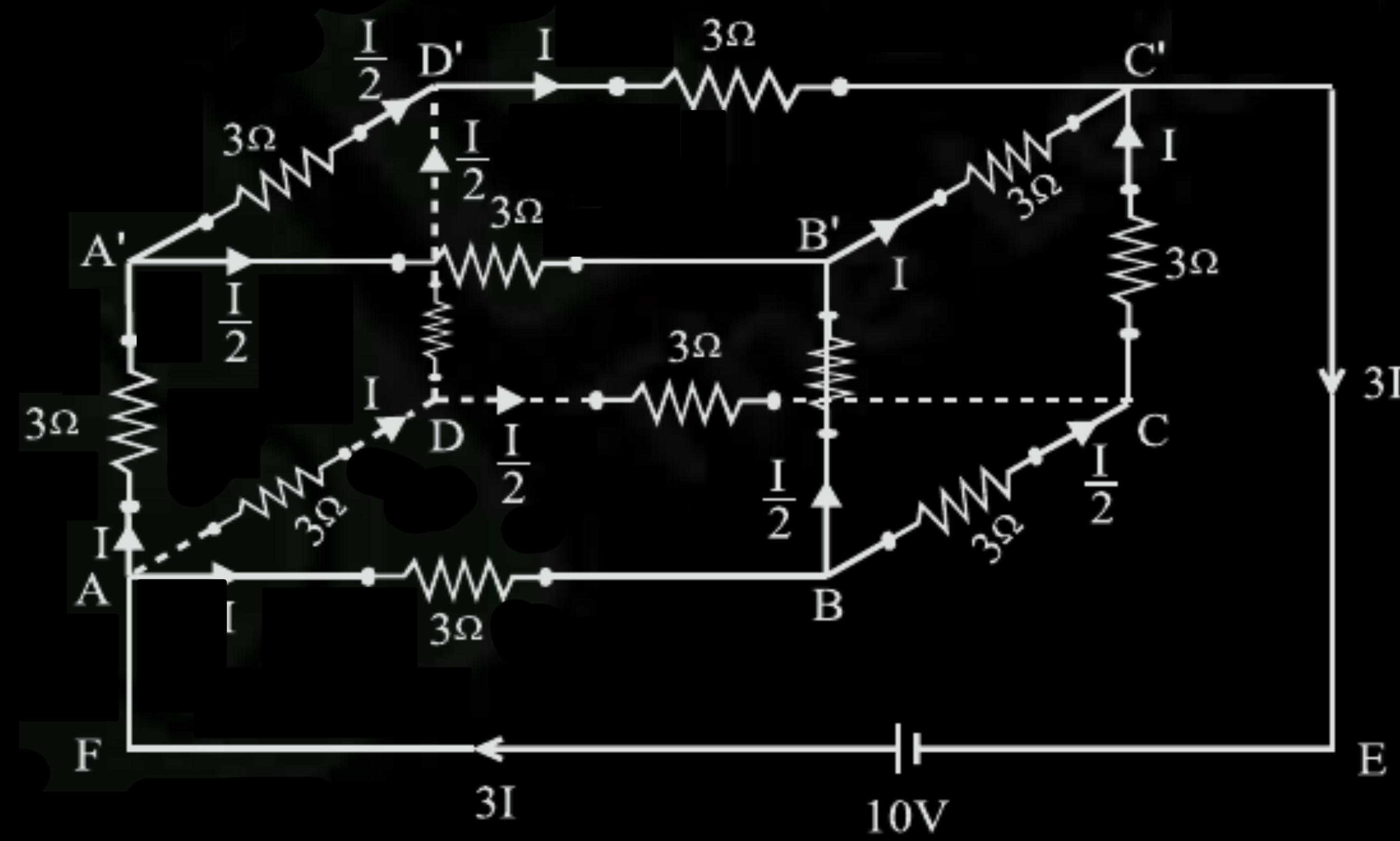
22. A wire of length L_0 has a resistance R_0 . It is gradually stretched till its length becomes $2L_0$.

- (a) Plot a graph showing variation of its resistance R with its length l during stretching.
 (b) What will be its resistance when its length becomes $2L_0$?

2

(a) Graph (b) Final resistance 1 1	
(a) $R = S \frac{l}{A} = S \frac{l^2}{V}$ 	$\frac{1}{2}$ $\frac{1}{2}$
[Note if a student draws only the graph correctly, award full 1 mark] (b) Resistance becomes $4R_0$ (Hint :- As $R = \frac{\rho l}{A} = \frac{\rho l^2}{Al} = \frac{\rho l^2}{V}$) ($V = \text{Volume}$) $R \propto l^2$	1

Twelve wires each having a resistance of $3\ \Omega$ are connected to form a cubical network. A battery of $10\ \text{V}$ and negligible internal resistance is connected across the diagonally opposite corners of this network. Determine its equivalent resistance and the current along each edge of the cube.



$\frac{1}{2}$

Applying loop rule to ABCC'EFA

$$3I + 3\frac{I}{2} + 3I - 10 = 0$$

$\frac{1}{2}$

$$\frac{15}{2} I = 10$$

$$I = \frac{2 \times 10}{15} = \frac{20}{15} \text{ A} = \frac{4}{3} \text{ A}$$

$\frac{1}{2}$

$$R_{\text{tot}} = \frac{V}{3I} = \frac{10 \times 15}{3 \times 20} = 2.5 \Omega$$

$\frac{1}{2}$

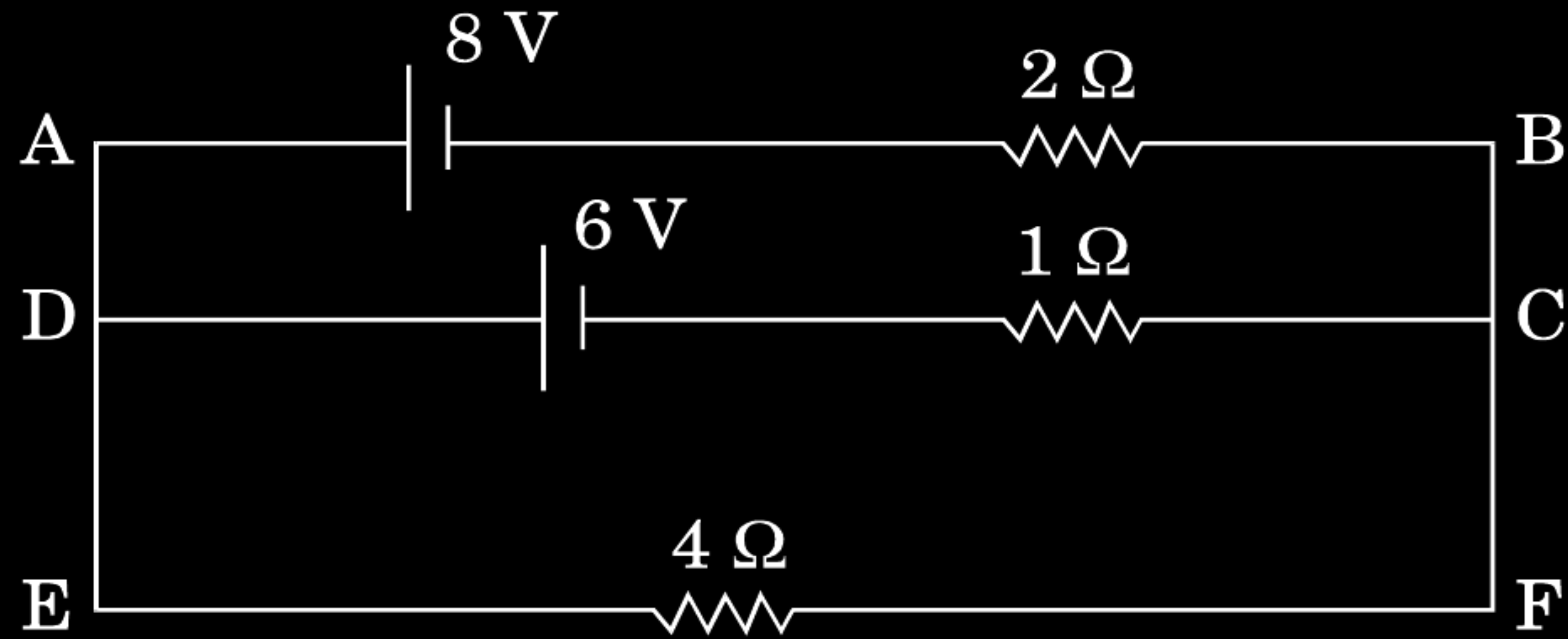
$$\text{Current} = I_{AB} (= I_{AA'} = I_{AD} = I_{D'C'} = I_{B'C'} = I_{CC'}) = \frac{4}{3} \text{ A}$$

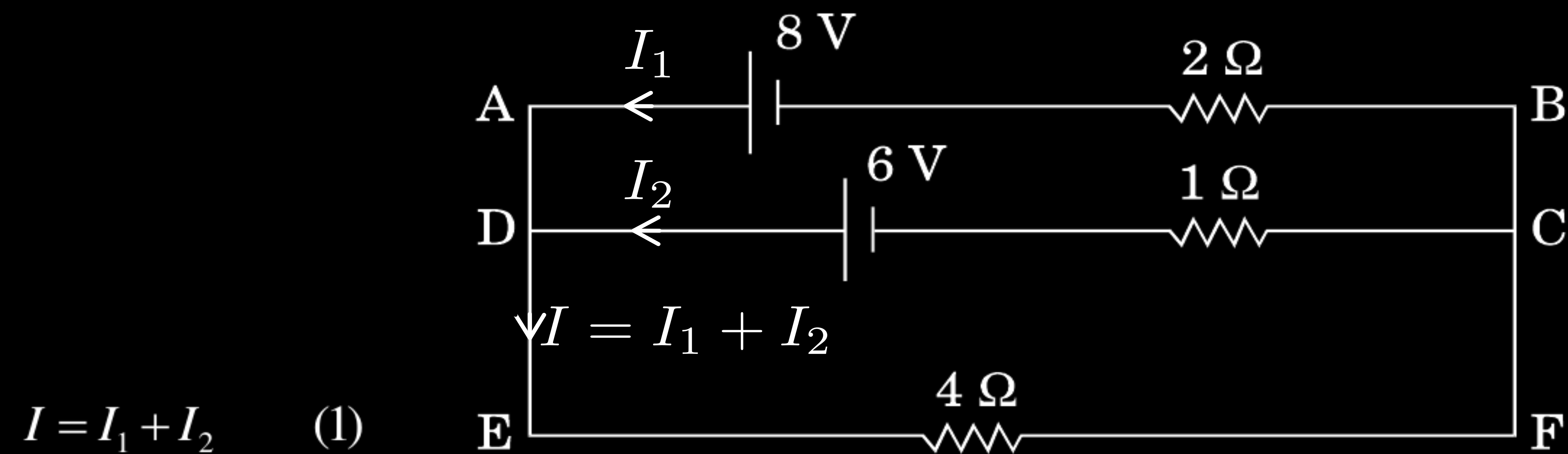
$\frac{1}{2}$

$$I_{DD'} (= I_{A'B'} = I_{A'D'} = I_{DC} = I_{BC} = I_{BB'}) = \frac{2}{3} \text{ A}$$

$\frac{1}{2}$

- (c) Calculate the potential difference across the $4\ \Omega$ resistor in the given electrical circuit, using Kirchhoff's rules.





$$I = I_1 + I_2 \quad (1)$$

In loop ABCDA

$$-8 + 2I_1 - 1 \times I_2 + 6 = 0 \quad (2)$$

In loop DEFCD

$$-4I - 1 \times I_2 + 6 = 0$$

$$4I + I_2 = 6$$

$$4(I_1 + I_2) + I_2 = 6$$

$$4I_1 + 5I_2 = 6 \quad (3)$$

From equations (1) and (2) we get

$$I_1 = \frac{8}{7} \text{ A}, \quad I_2 = \frac{2}{7} \text{ A}, \quad I = \frac{10}{7}$$

Potential difference across resistor 4Ω is:

$$V = \frac{10}{7} \times 4 = \frac{40}{7} \text{ volt}$$

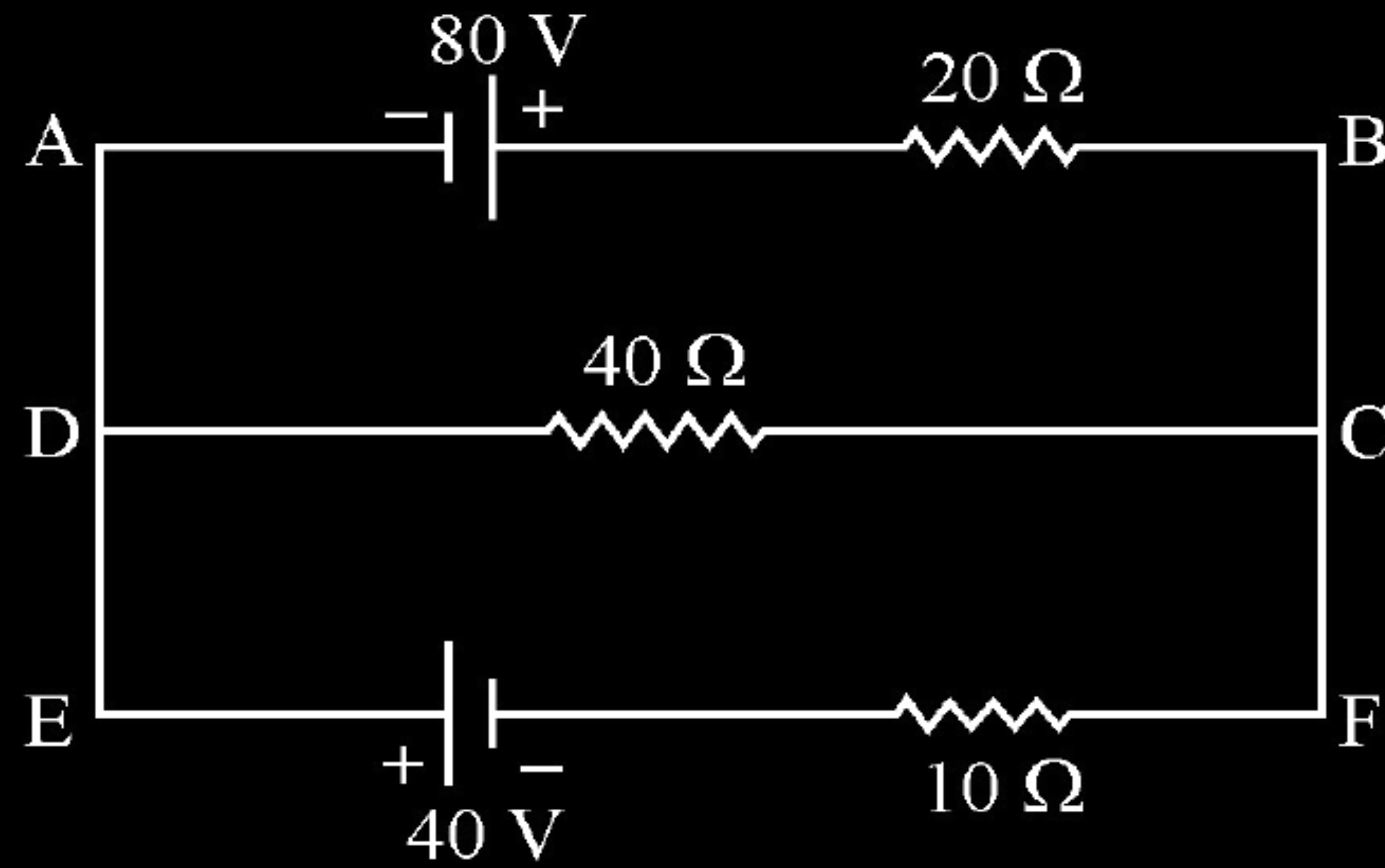
1

1

$\frac{1}{2}$

$\frac{1}{2}$

Using Kirchhoff's rules, calculate the current through the $40\ \Omega$ and $20\ \Omega$ resistors in the following circuit :



In loop ABCFA

$$+80 - 20 I_2 + 40 I_1 = 0$$

$$4 = I_2 - 2 I_1$$

In loop FCDEA

$$-40 I_1 - 10(I_1 + I_2) + 40 = 0$$

$$-50 I_1 - 10 I_2 + 40 = 0$$

$$5 I_1 + I_2 = 4$$

Solving these two equations

$$I_1 = 0A$$

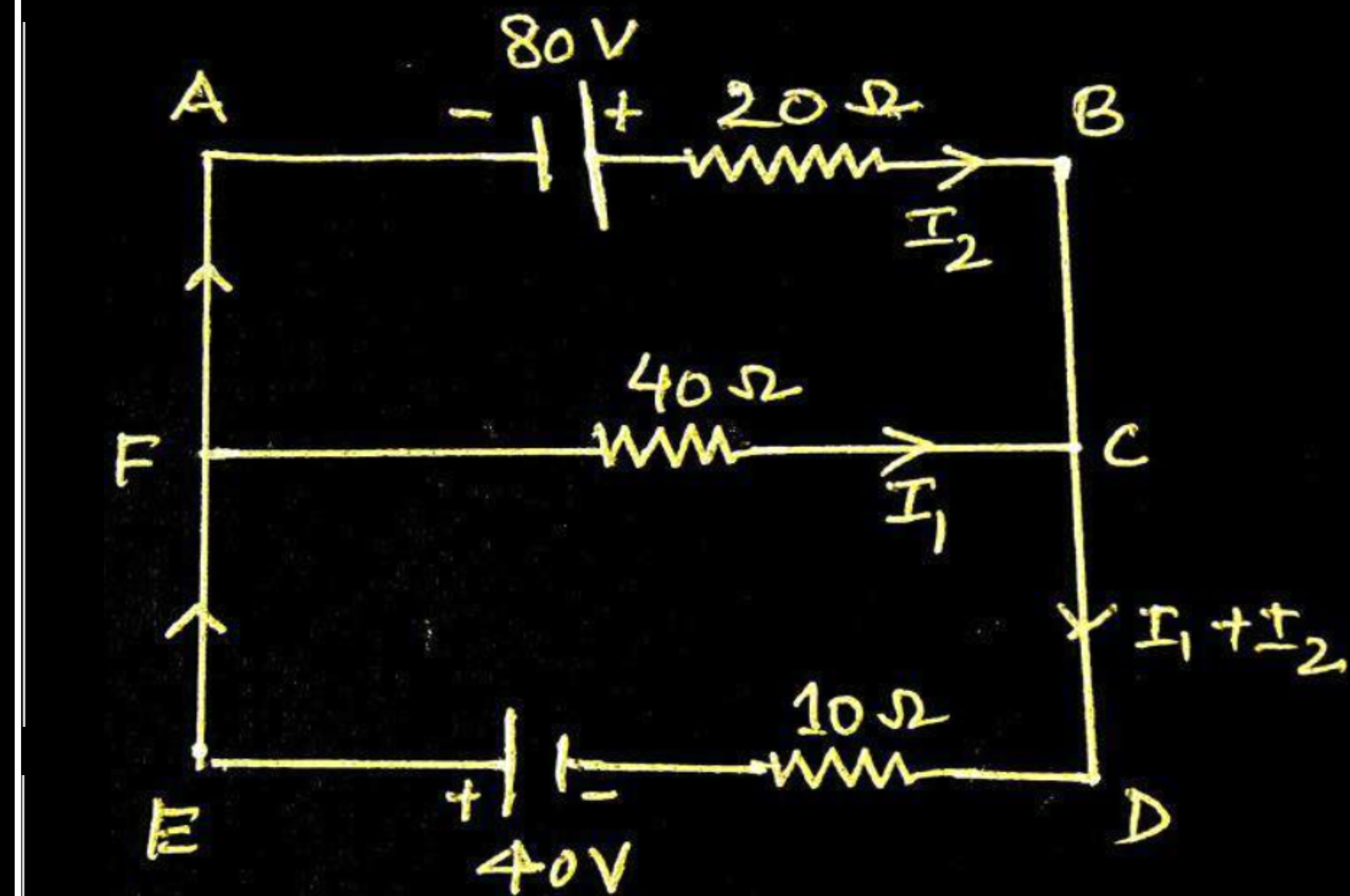
$$\& I_2 = 4A$$

1

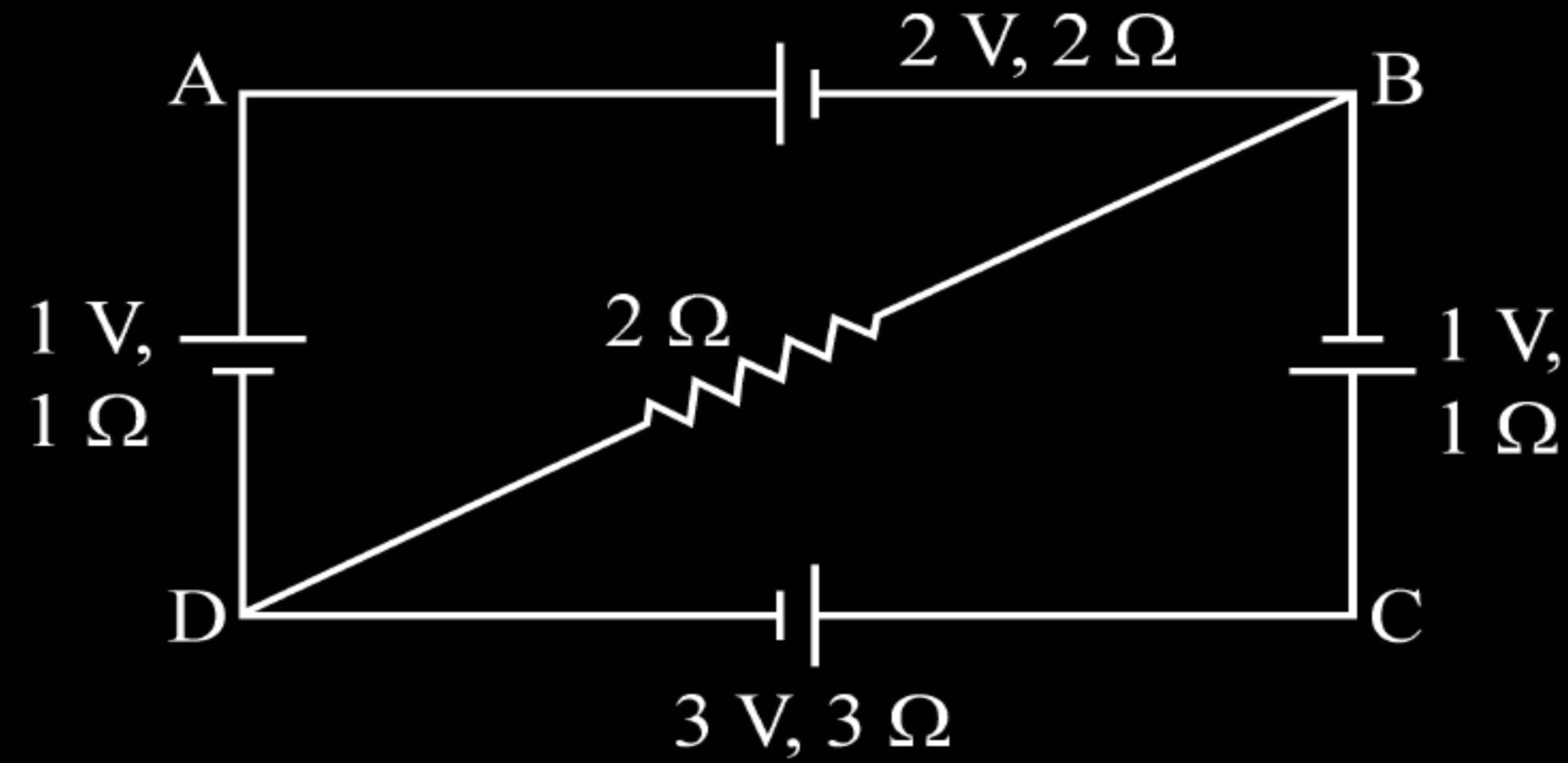
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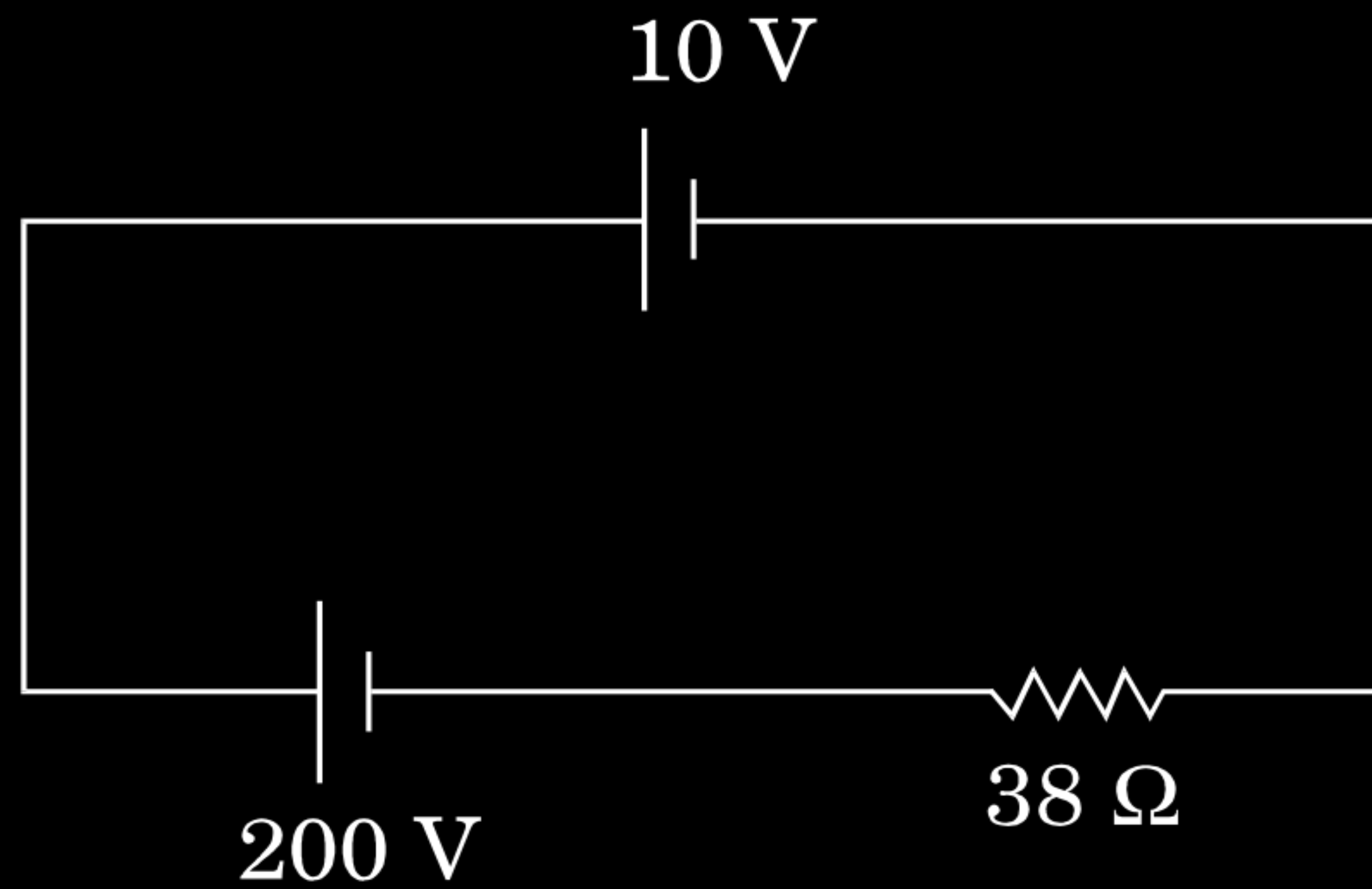
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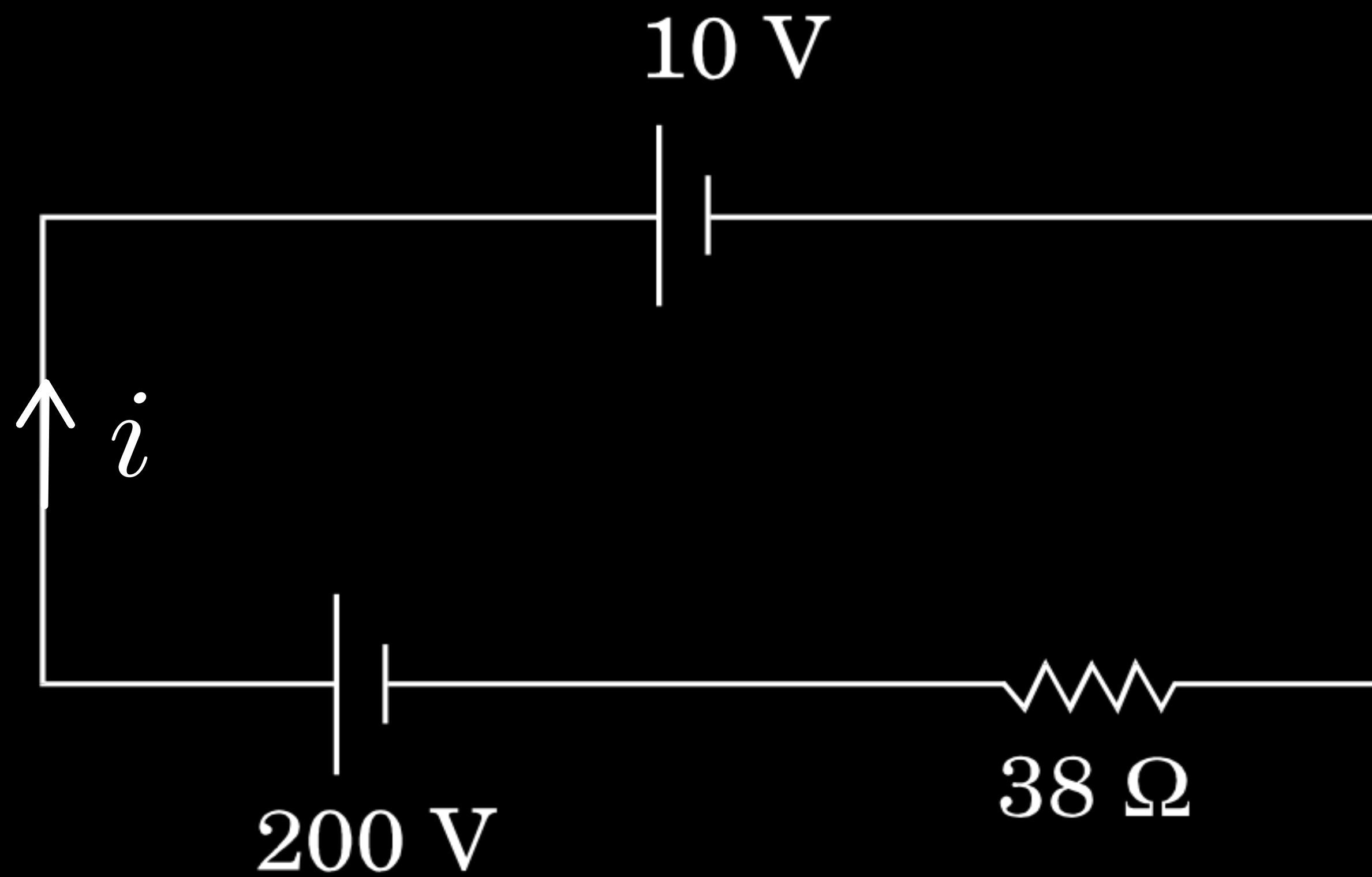
Using Kirchhoff's rules, calculate the potential difference between B and D in the circuit diagram as shown in the figure.



A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$ as shown in the figure. Find the value of current in the circuit.



By Kirchhoff's law, we have, for the loop ABCD,

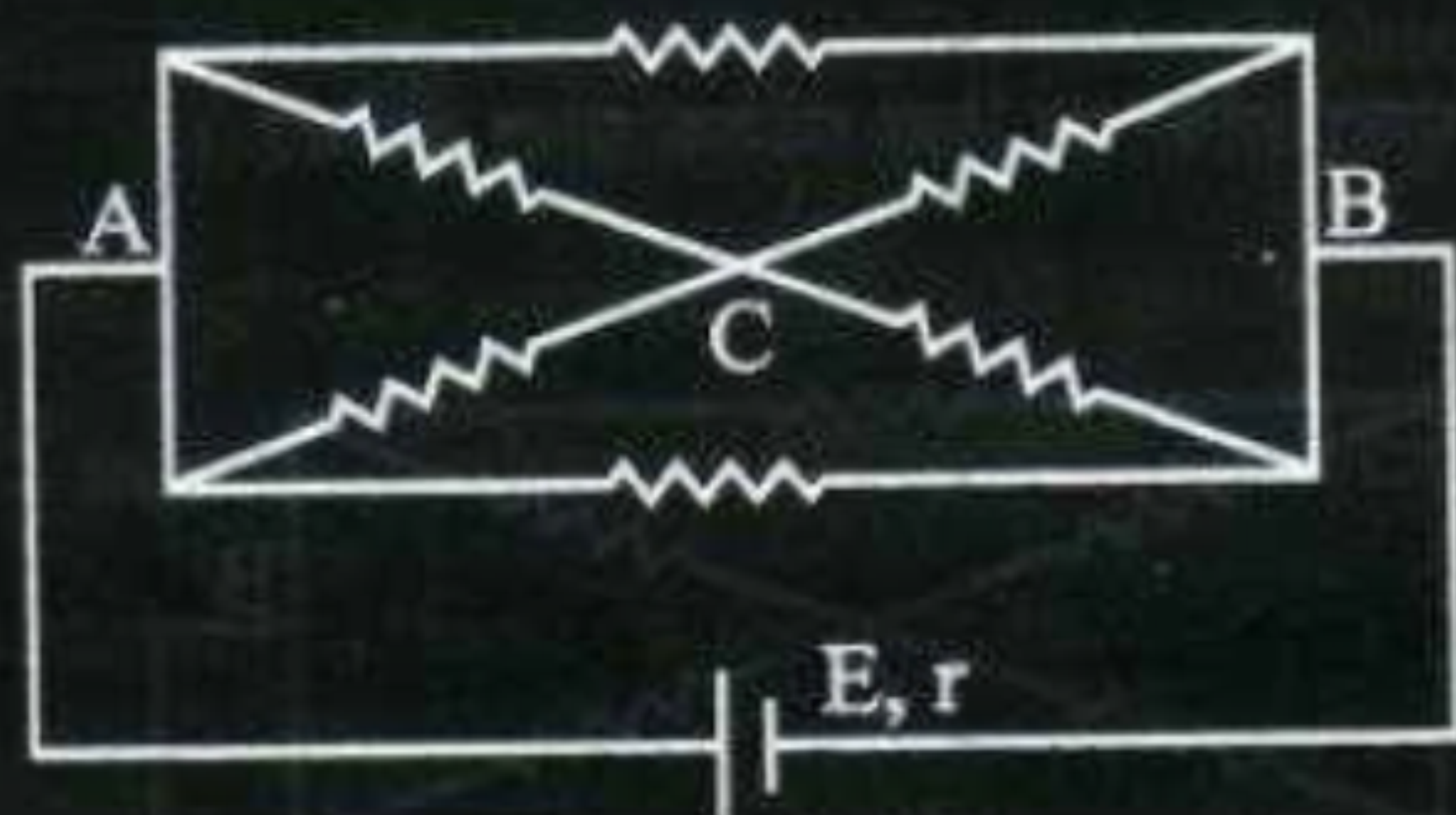
$$+200 - 38i - 10 = 0$$

$$\therefore i = \frac{190}{38} \text{ A} = 5 \text{ A}$$

1

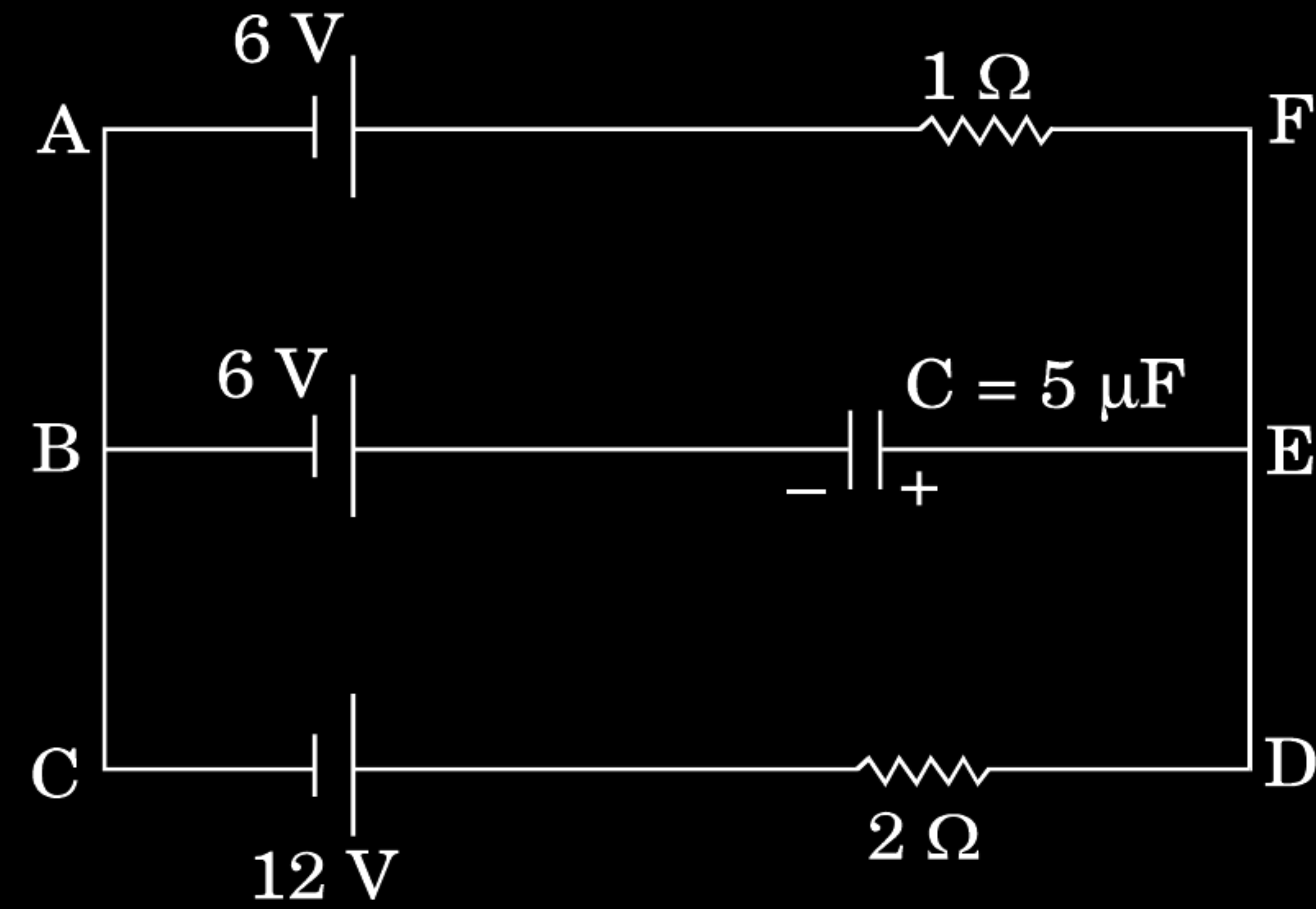
1

- (i) State the two Kirchhoff's laws. Explain briefly how these rules are justified.
- (ii) The current is drawn from a cell of emf E and internal resistance r connected to the network of resistors each of resistance r as shown in the figure. Obtain the expression for (i) the current draw from the cell and (ii) the power consumed in the network.

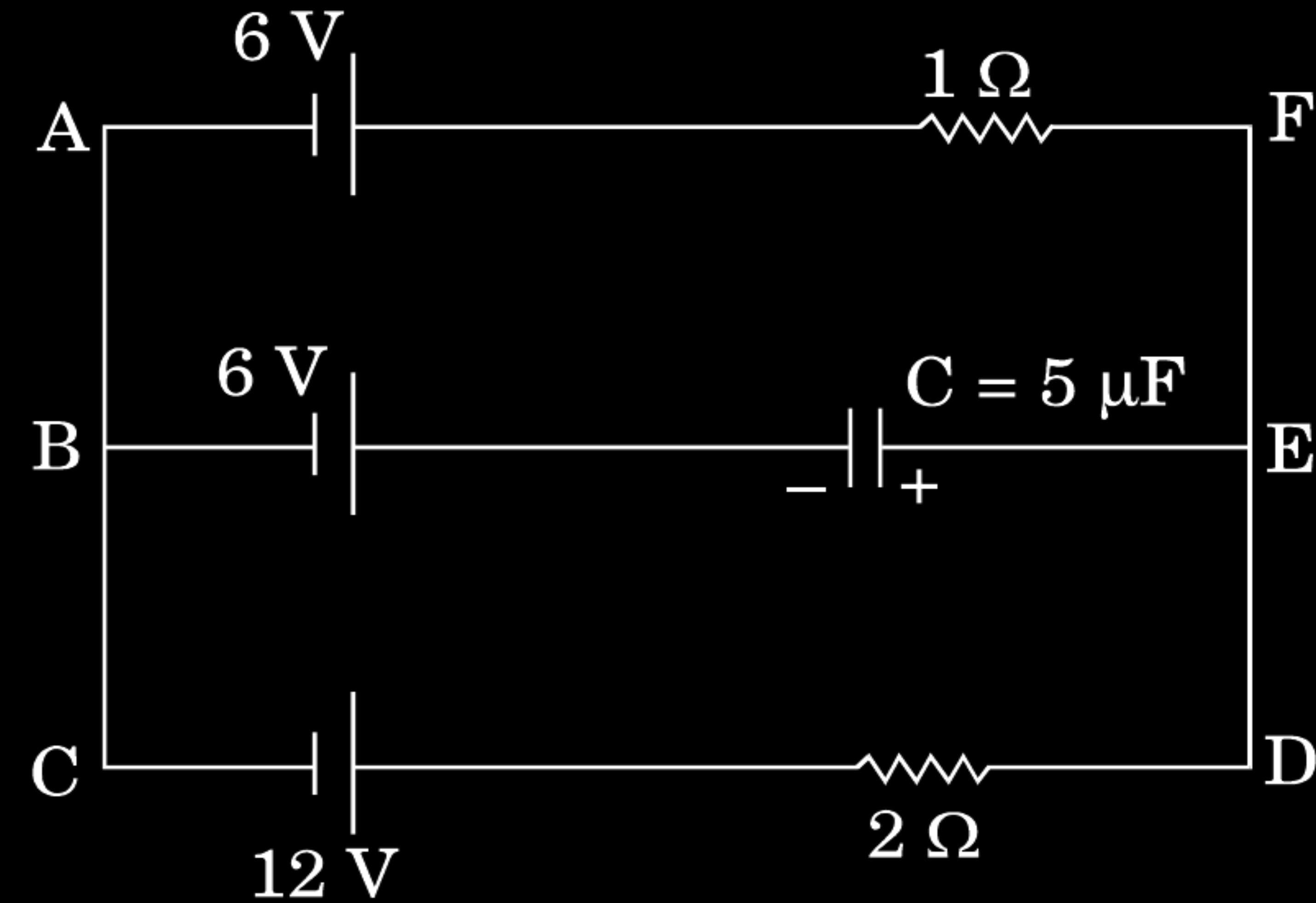


(i) Junction Rule: At any Junction, the sum of currents, entering the junction, is equal to the sum of currents leaving the junction. Loop Rule: The Algebraic sum, of changes in potential, around any closed loop involving resistors and cells, in the loop is zero. $\sum(\Delta V) = 0$ Justification: The first law is in accord with the law of conservation of charge. The Second law is in accord with the law of conservation of energy.	1 1 $\frac{1}{2}$
(ii) Equivalent resistance of the loop $R = \frac{r}{3}$ Hence current drawn from the cell $I = \frac{E}{\frac{r}{3} + r} = \frac{3E}{4r}$ Power consumed $P = I^2 \left(\frac{r}{3}\right)$ $= \frac{9E^2}{16r^2} \times \frac{4r}{3} = \frac{3E^2}{4r}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
[Note: Award the last 1 $\frac{1}{2}$ marks for this part, if the calculations, for these parts, are done by using (any other) value of equivalent resistance obtained by the student.)	

In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



In the given circuit, with steady current, calculate the potential difference across the capacitor and the charge stored in it.



In loop ACDEFA

$$I = \frac{12-6}{(1+2)} = 2A$$

$$V_{AF} = V_{BE}$$

$$\Rightarrow 6 + 2 = 6 + V_c$$

$$\Rightarrow V_c = 2V$$

$$\text{Charge } Q = CV_c = 5\mu F \times 2V = 10\mu C$$

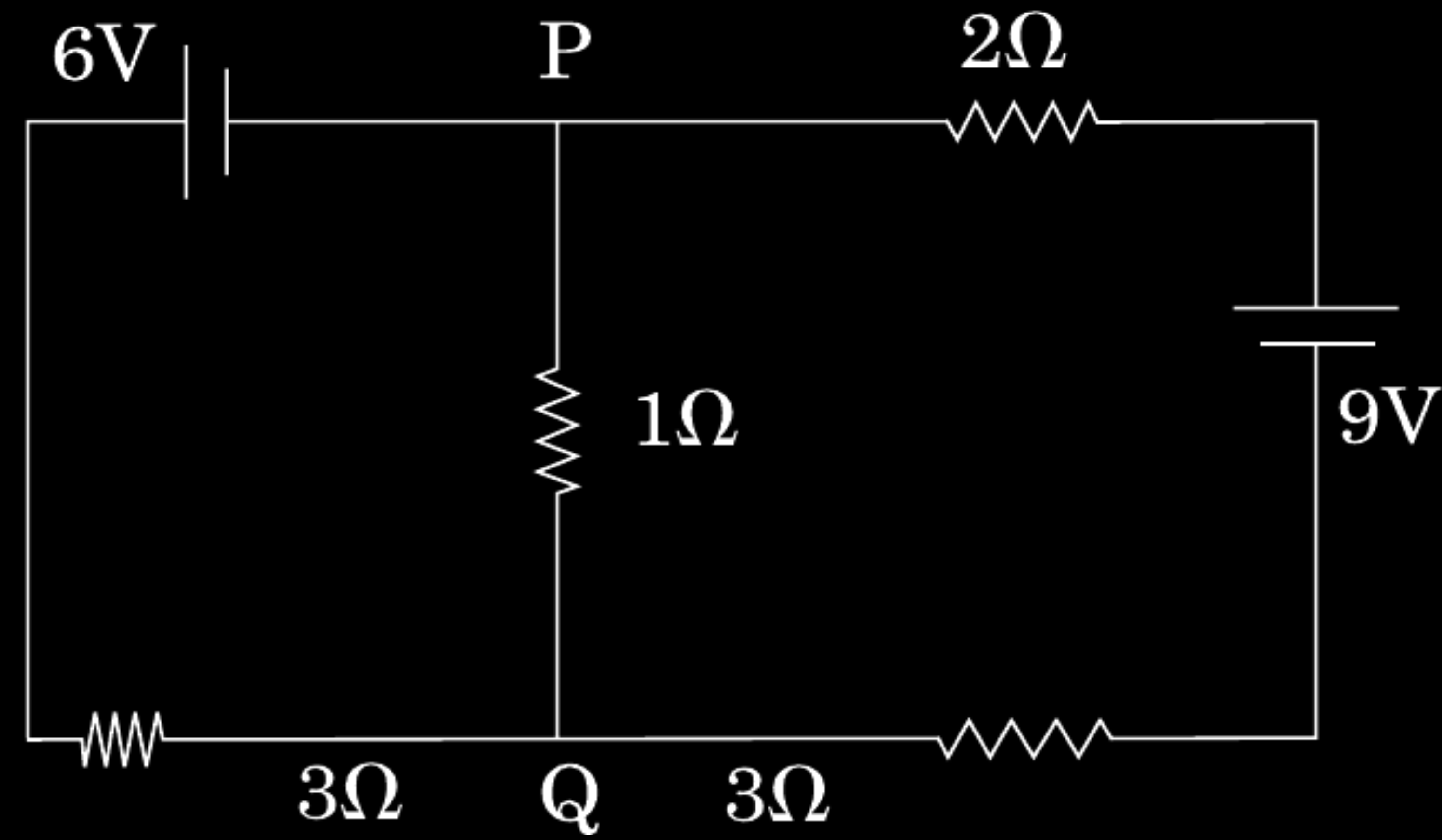
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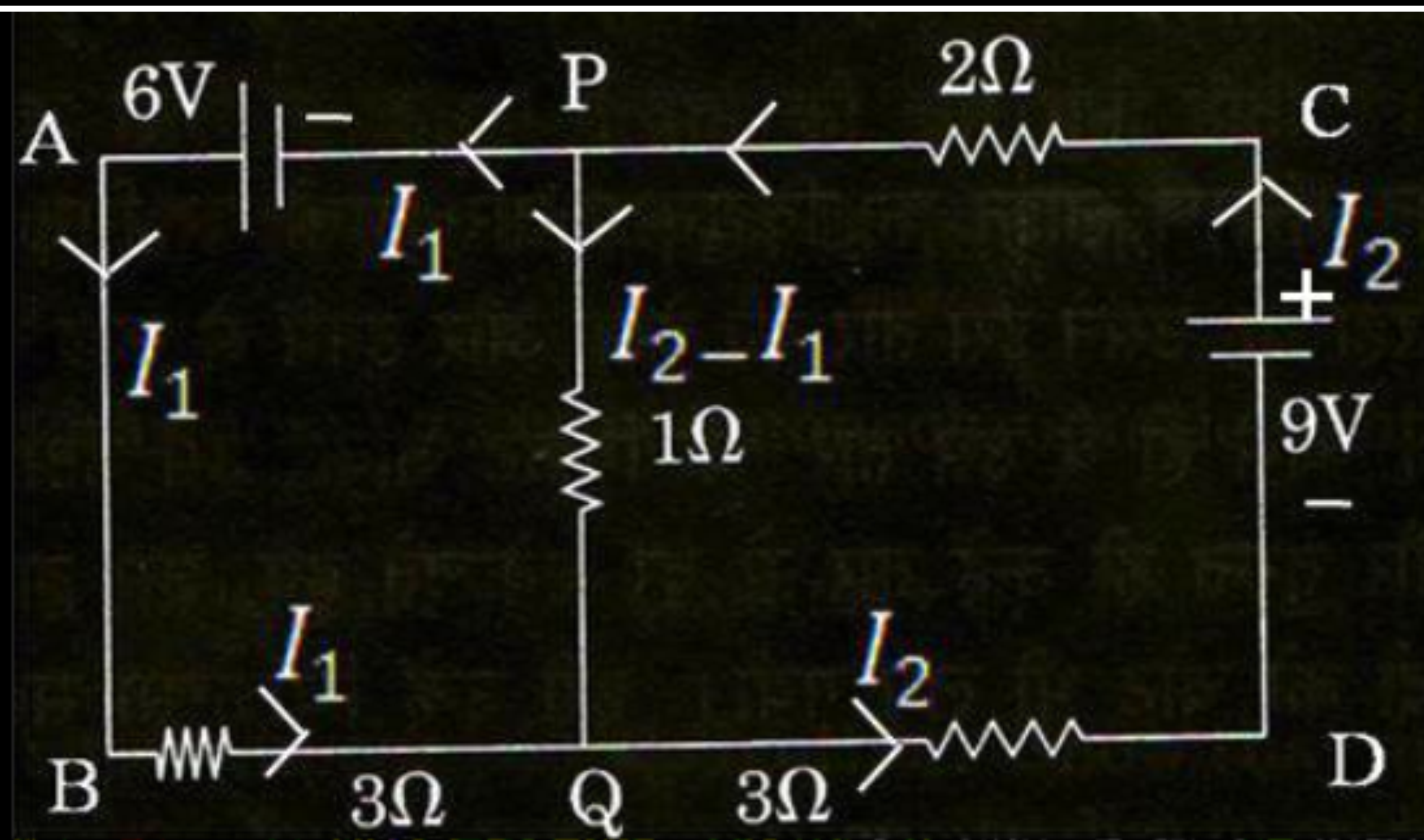
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1

- (i) Find the magnitude and direction of current in $1\ \Omega$ resistor in the given circuit.





For the mesh APQBA

$$-6 - 1 (I_2 - I_1) + 3 I_1 = 0$$

$$\text{Or } -I_2 + 4I_1 = 6 \dots\dots\dots(1)$$

For the mesh PCDQP

$$2I_2 - 9 + 3I_2 + 1 (I_2 - I_1) = 0$$

$$\text{Or } 6I_2 - I_1 = 9 \dots\dots\dots(2)$$

Solving (1) and (2) , we get

$$I_1 = \frac{45}{23} \text{ A}$$

$$I_2 = \frac{42}{23} \text{ A}$$

$$\therefore \text{Current through the } 1\Omega \text{ resistor} = \frac{-3}{23} \text{ A}$$

1

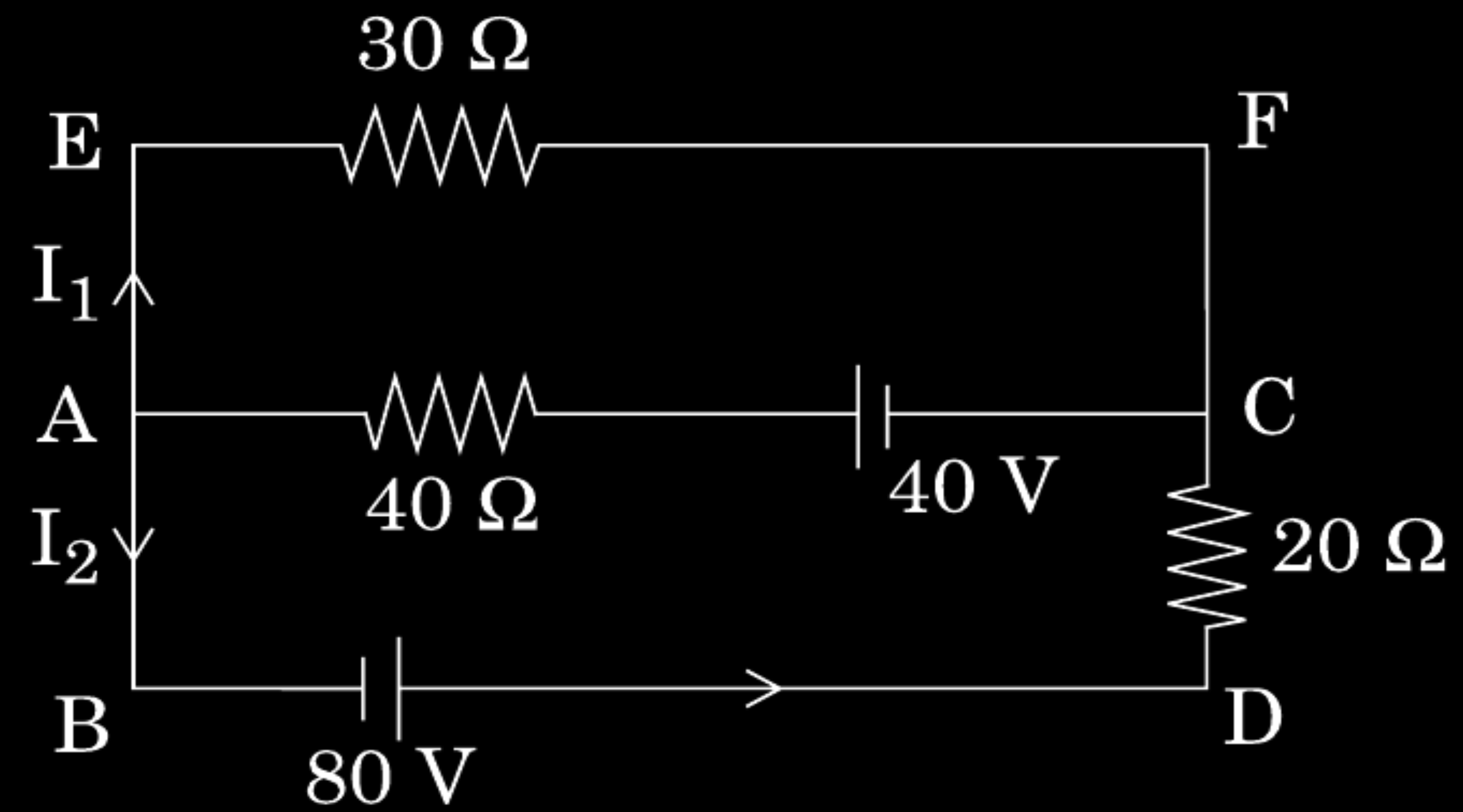
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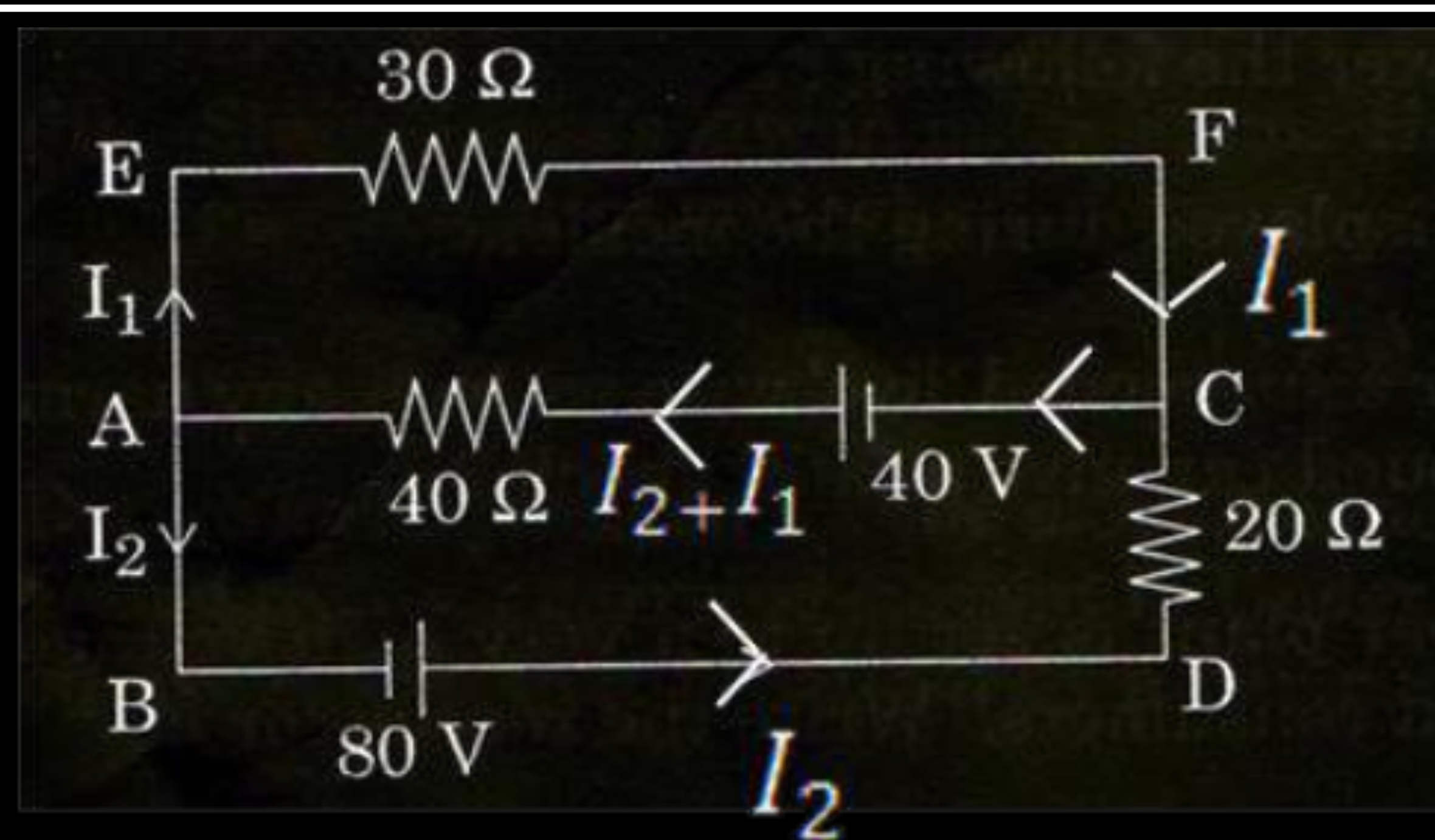
$\frac{1}{2}$

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$\frac{1}{2}$

- (a) Use Kirchhoff's rules, calculate the current in the arm AC of the given circuit.





For the mesh EFCAE

$$-30I_1 + 40 - 40(I_1 + I_2) = 0$$

$$\text{Or } -7I_1 - 4I_2 = -4$$

$$\text{Or } 7I_1 + 4I_2 = 4 \dots\dots\dots(1)$$

For the mesh ACDBA

$$40(I_1 + I_2) - 40 + 20I_2 - 80 = 0$$

$$\text{Or } 40I_1 + 60I_2 - 120 = 0$$

$$\text{Or } 2I_1 + 3I_2 = 6 \dots\dots\dots(2)$$

Solving (1) and (2), we get

$$I_1 = \frac{-12}{13} A$$

$$I_2 = \frac{34}{13} A$$

∴ Current through arm AC = $I_1 + I_2$

$$= \frac{22}{13} A$$

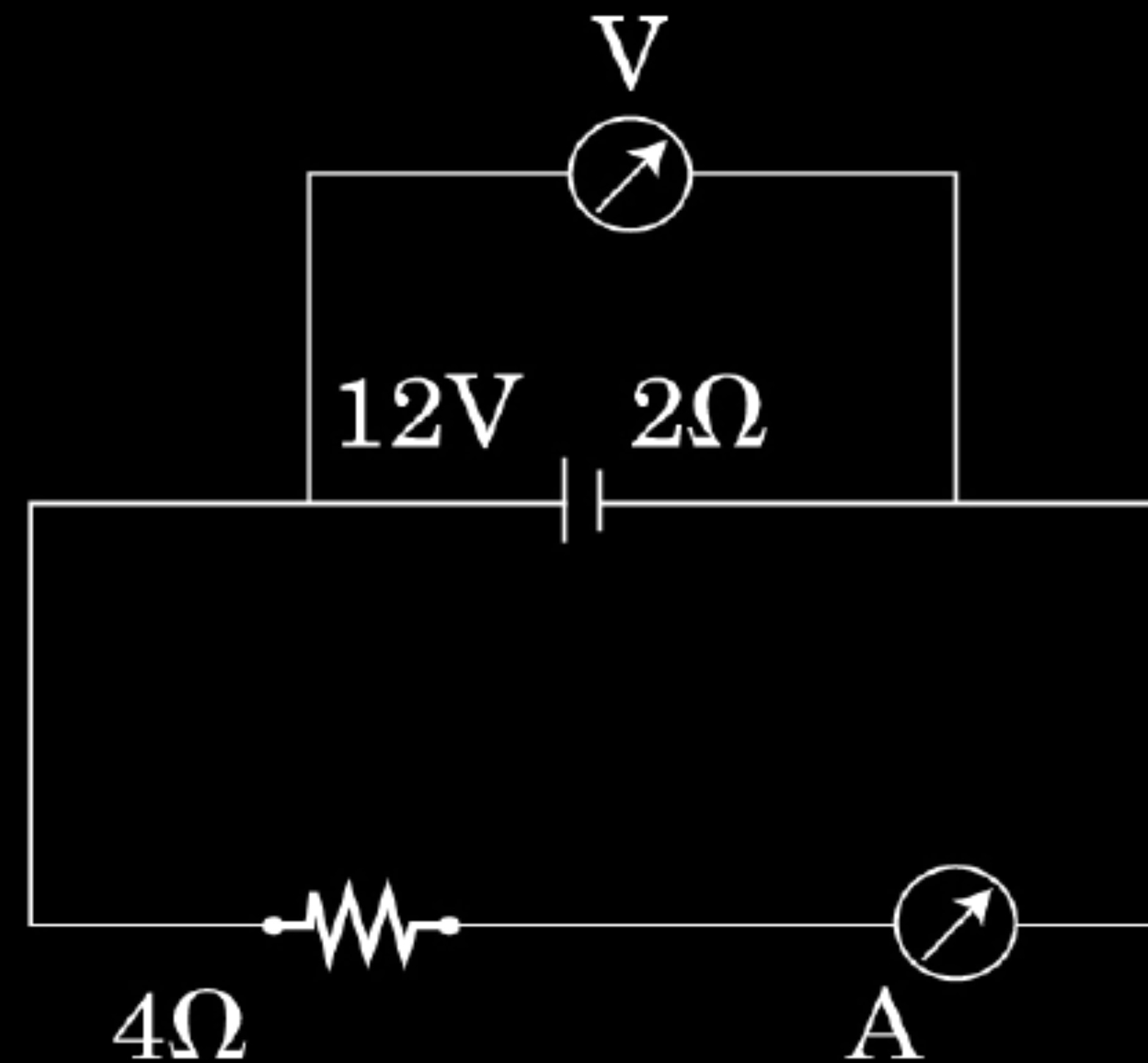
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1

1

A battery of emf 12V and internal resistance $2\ \Omega$ is connected to a $4\ \Omega$ resistor as shown in the figure.

- (a) Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the same reading.
- (b) To record the voltage and the current in the circuit, why is voltmeter placed in parallel and ammeter in series in the circuit ?



a) Effective resistance of the circuit $R_E = 6\Omega$

$$\therefore I = \frac{12A}{6} = 2A$$

Terminal potential difference across the cell, $V = E - ir$

$\frac{1}{2}$

Also p.d. across 4Ω resistor $= 4 \times 2V = 8V$

Hence the voltmeter gives the same reading in the two cases.

$\frac{1}{2}$

b) In series -current same

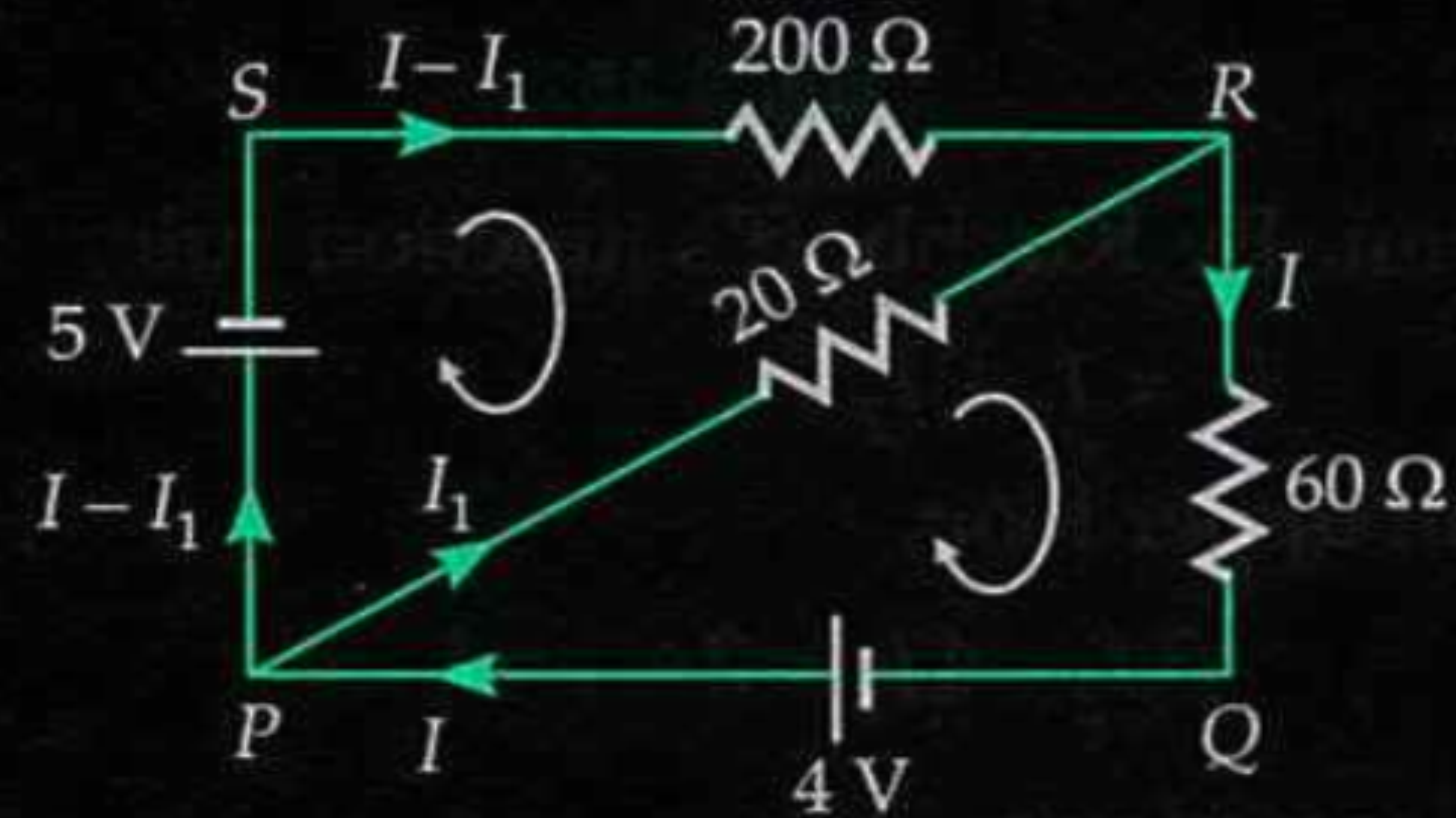
$\frac{1}{2}$

In parallel – potential same

$\frac{1}{2}$

Example 137. Network PQRS (Fig. 3.148) is made as under : PQ has a battery of 4 V and negligible resistance with positive terminal connected to P, QR has a resistance of $60\ \Omega$. PS has a battery of 5 V and negligible resistance with positive terminal connected to P, RS has a resistance of $200\ \Omega$. If a milliammeter, of $20\ \Omega$ resistance is connected between P and R, calculate the reading of the milliammeter.

[NCERT]



Solution. Applying Kirchhoff's second law to the loop $PRQP$, we get

$$20 I_1 + 60 I = 4 \quad \dots(i)$$

Similarly, from the loop $PSRP$, we get

$$200 (I - I_1) - 20 I_1 = -5$$

or $40 I - 44 I_1 = -1 \quad \dots(ii)$

Multiplying (i) by 2 and (ii) by 3, we get

$$120 I + 40 I_1 = 8 \quad \dots(iii)$$

and $120 I - 132 I_1 = -3 \quad \dots(iv)$

Subtracting (iv) from (iii), we get

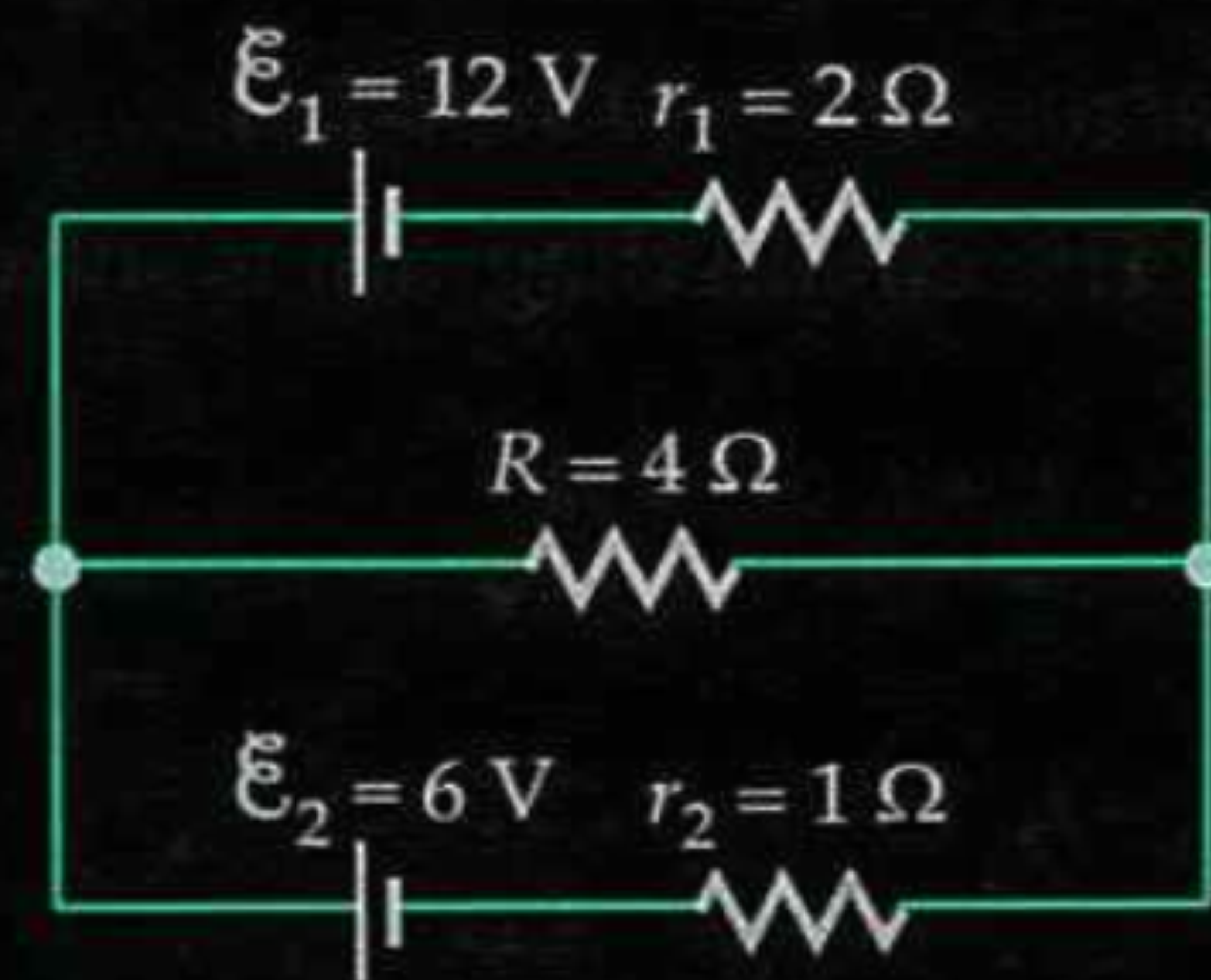
$$172 I_1 = 11$$

or $I_1 = \frac{11}{172} = 0.064 \text{ A}$

Thus the milliammeter of 20Ω will read 0.064 A .

Example 139. Find the potential difference across each cell and the rate of energy dissipation in R . [Fig. 3.150(a)].

[CBSE SP 11]



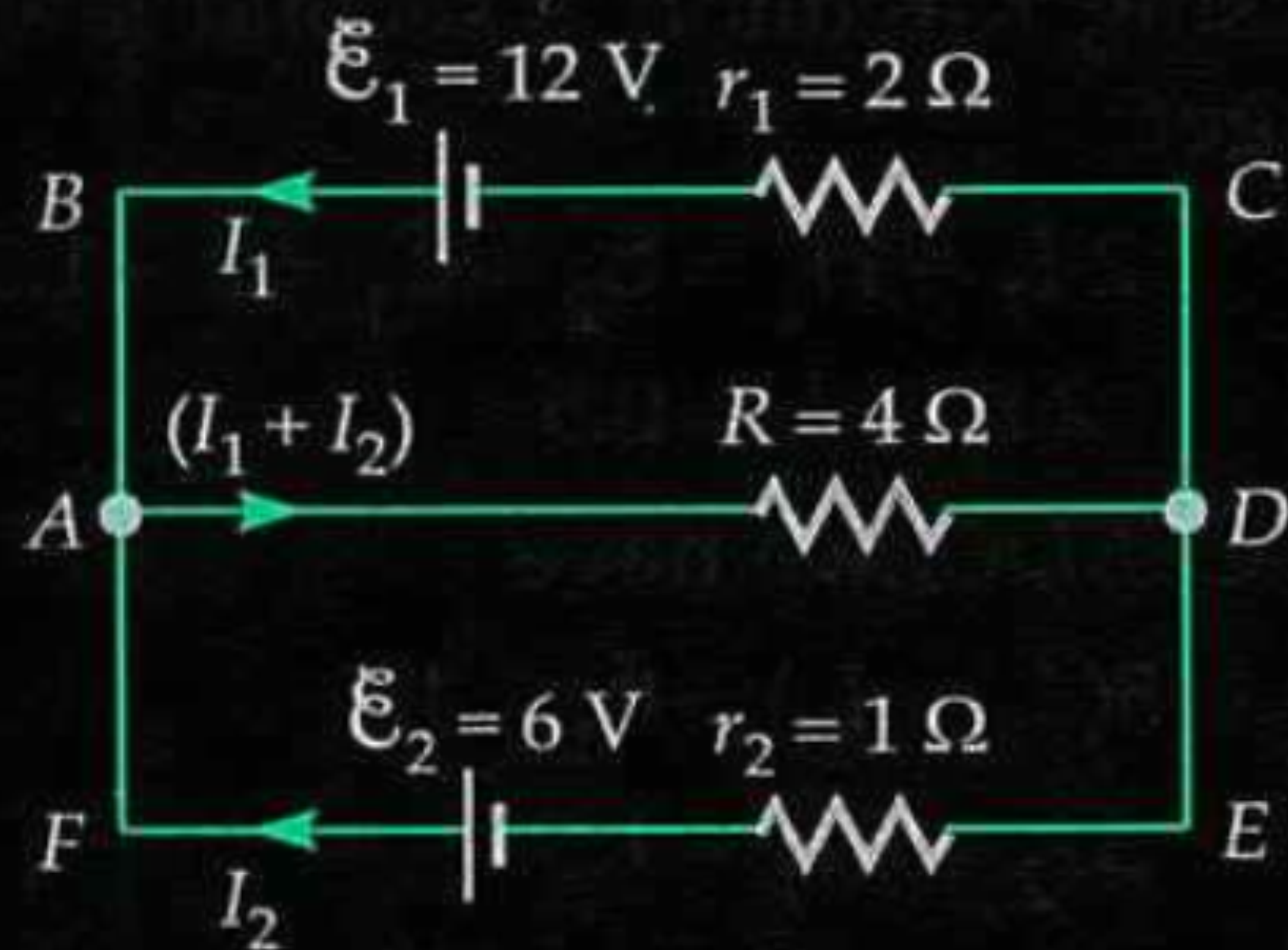
Solution. Applying Kirchhoff's laws,

For closed loop ADCBA

$$12 = 4(I_1 + I_2) + 2I_1 = 6I_1 + 4I_2 \quad \dots(i)$$

For closed loop ADEFA,

$$6 = 4(I_1 + I_2) + I_1 = 4I_1 + 5I_2 \quad \dots(ii)$$



Solving (i) and (ii), we get

$$I_1 = \frac{18}{7}\text{ A} \quad \text{and} \quad I_2 = -\frac{6}{7}\text{ A}$$

P.D. across $R = V$

$$= (I_1 + I_2)R$$

$$= \left(\frac{18-6}{7}\right) \times 4 \text{ volt} = \frac{48}{7} \text{ volt}$$

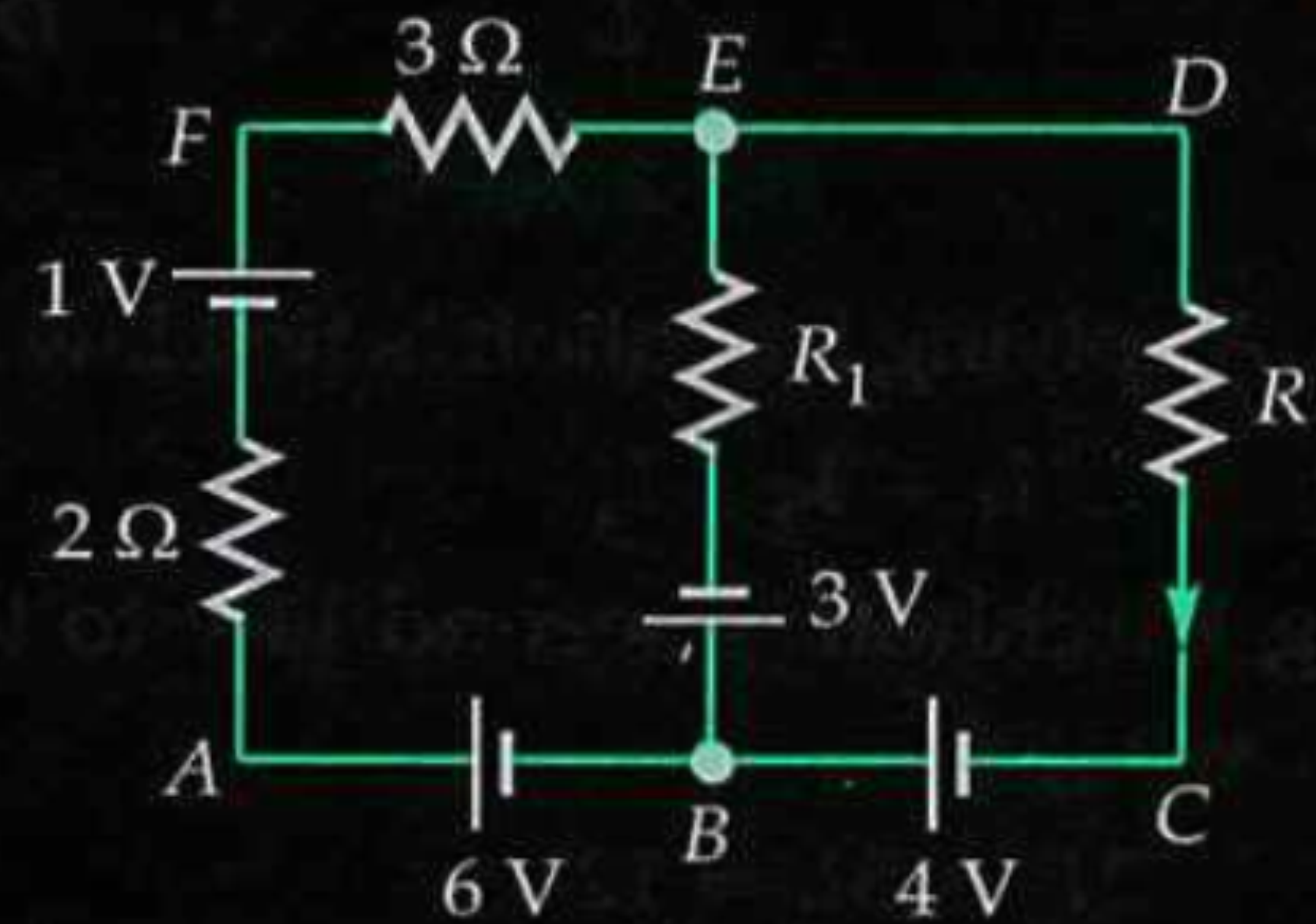
$$\text{P.D. across each cell} = \text{P.D. across } R = \frac{48}{7} \text{ V}$$

Energy dissipated in $R = 4\ \Omega$ resistor

$$= (I_1 + I_2)^2 R = \left(\frac{12}{7}\right)^2 \times 4 \text{ J}$$

$$= \frac{576}{49} \text{ J} = 11.75 \text{ J}.$$

Example 143. Use Kirchhoff's rules to determine the potential difference between the points A and D when no current flows in the arm BE of the electric network shown in Fig. 3.154(a). [CBSE OD 15]



Example 143. Use Kirchhoff's rules to determine the potential difference between the points A and D when no current flows in the arm BE of the electric network shown in Fig. 3.154(a). [CBSE OD 15]

Solution. No current flows through the arm BE .
Let I be the current along the outer loop as shown in Fig. 3.154(b).

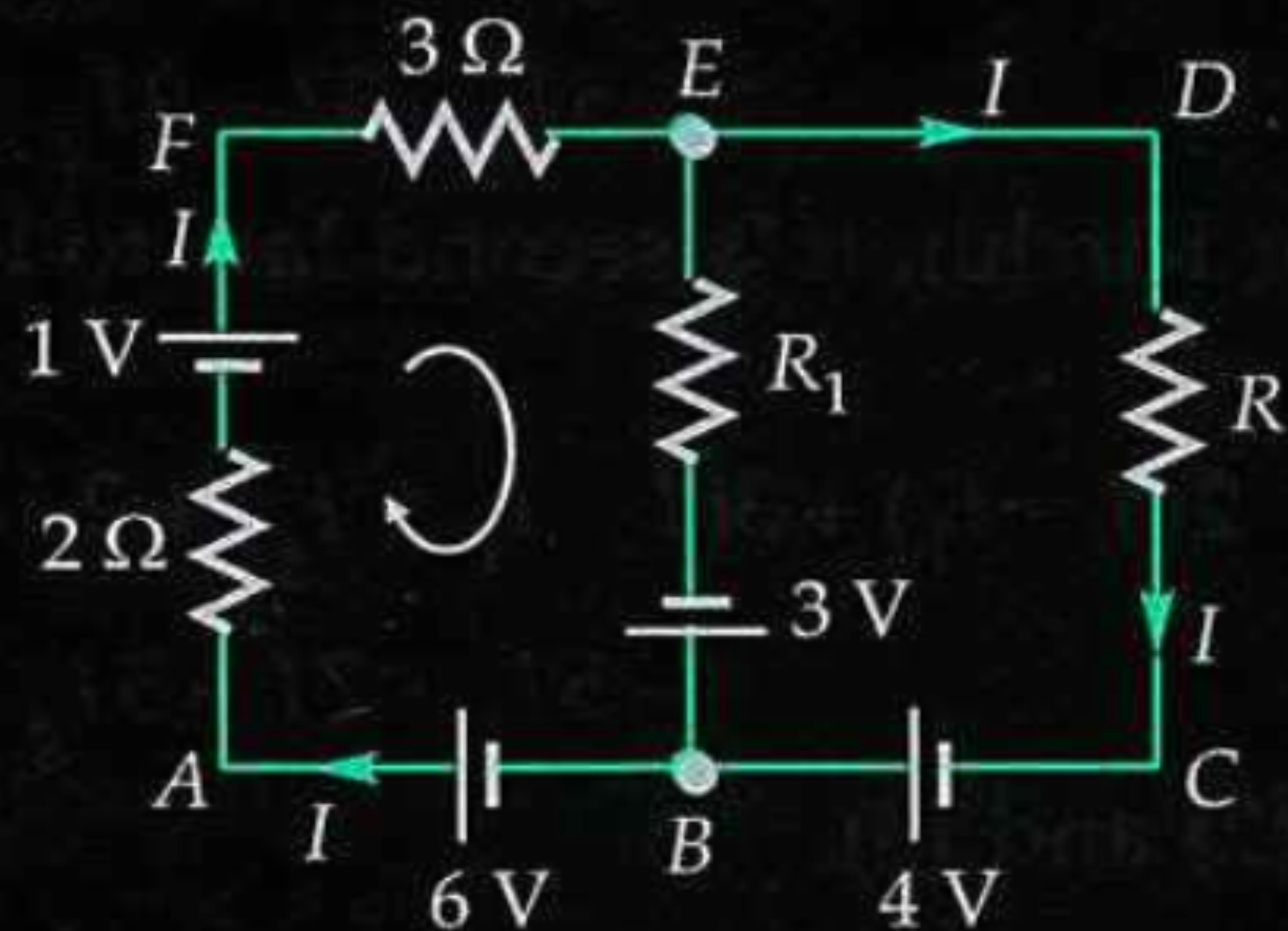


Fig. 3.154 (b)

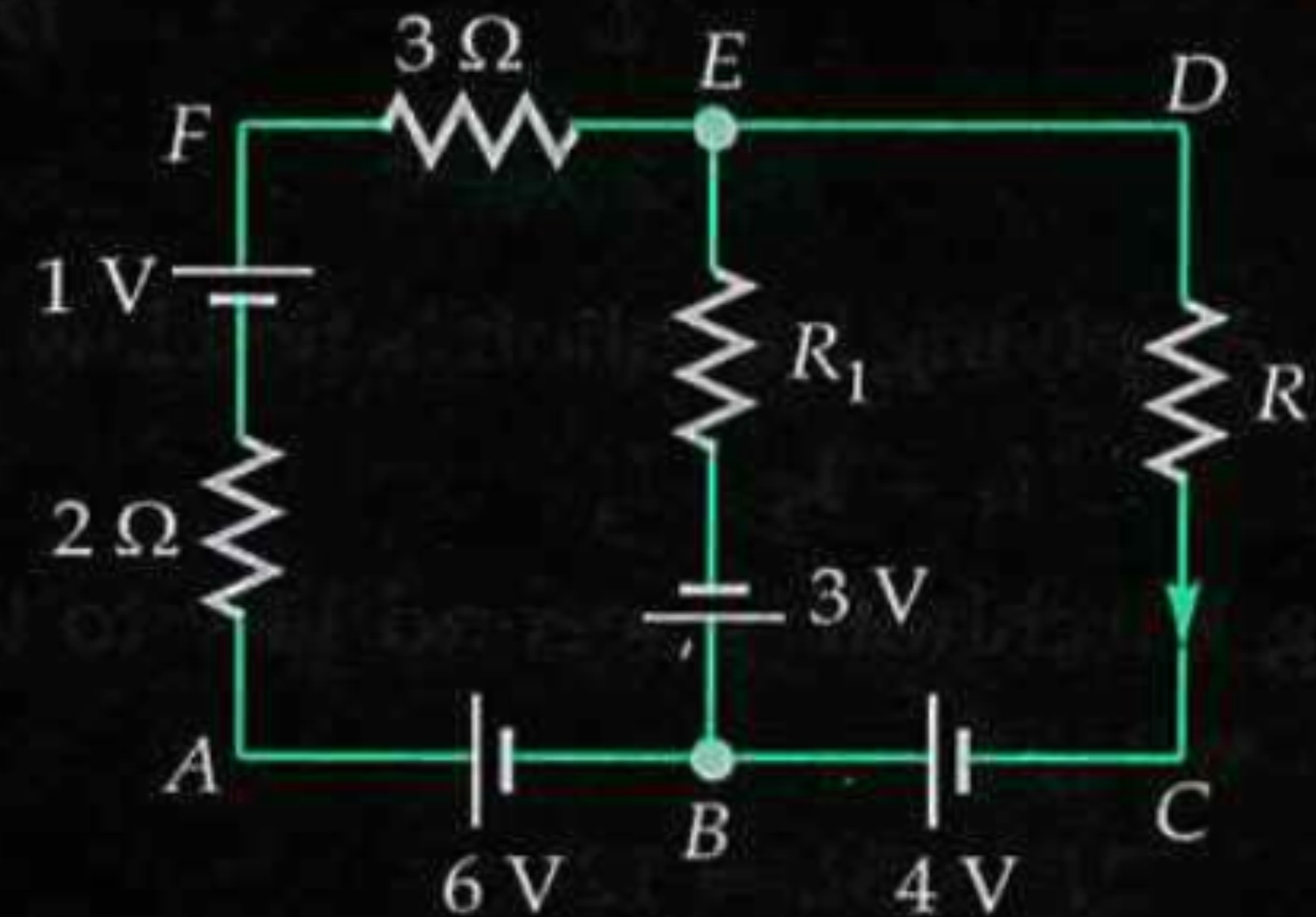
Applying Kirchhoff's loop rule to the loop $AFEBA$,

$$(2 + 3)I + R_1 \times 0 = 1 + 3 + 6$$

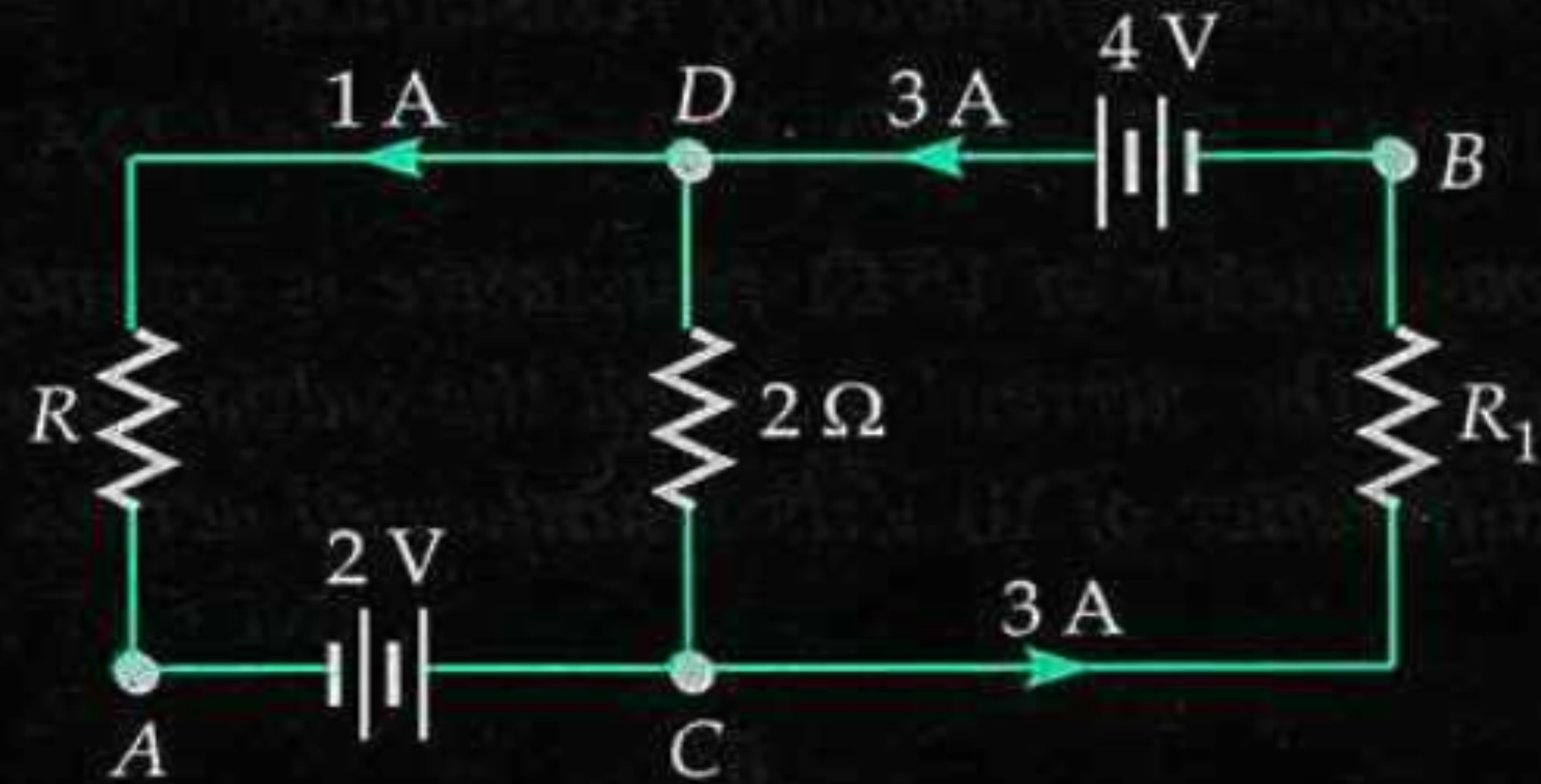
$$\therefore I = 2 \text{ A}$$

From A to D along AFD ,

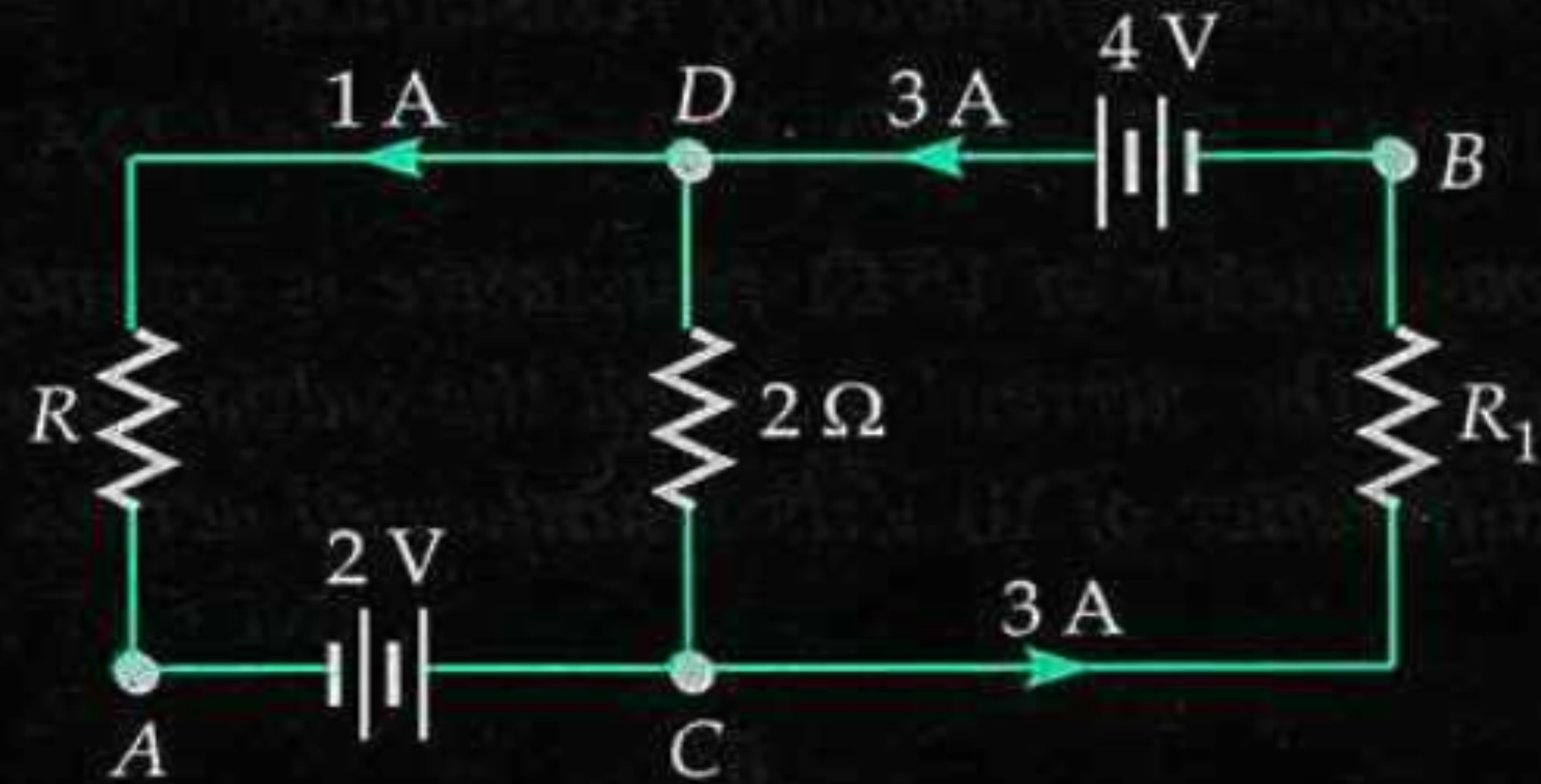
$$V_{AD} = 2 \times 2 - 1 + 3 \times 2 = 9 \text{ V}.$$



Example 144. In the circuit Fig. 3.155, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



Example 144. In the circuit Fig. 3.155, assuming point A to be at zero potential, use Kirchhoff's rules to determine the potential at point B.



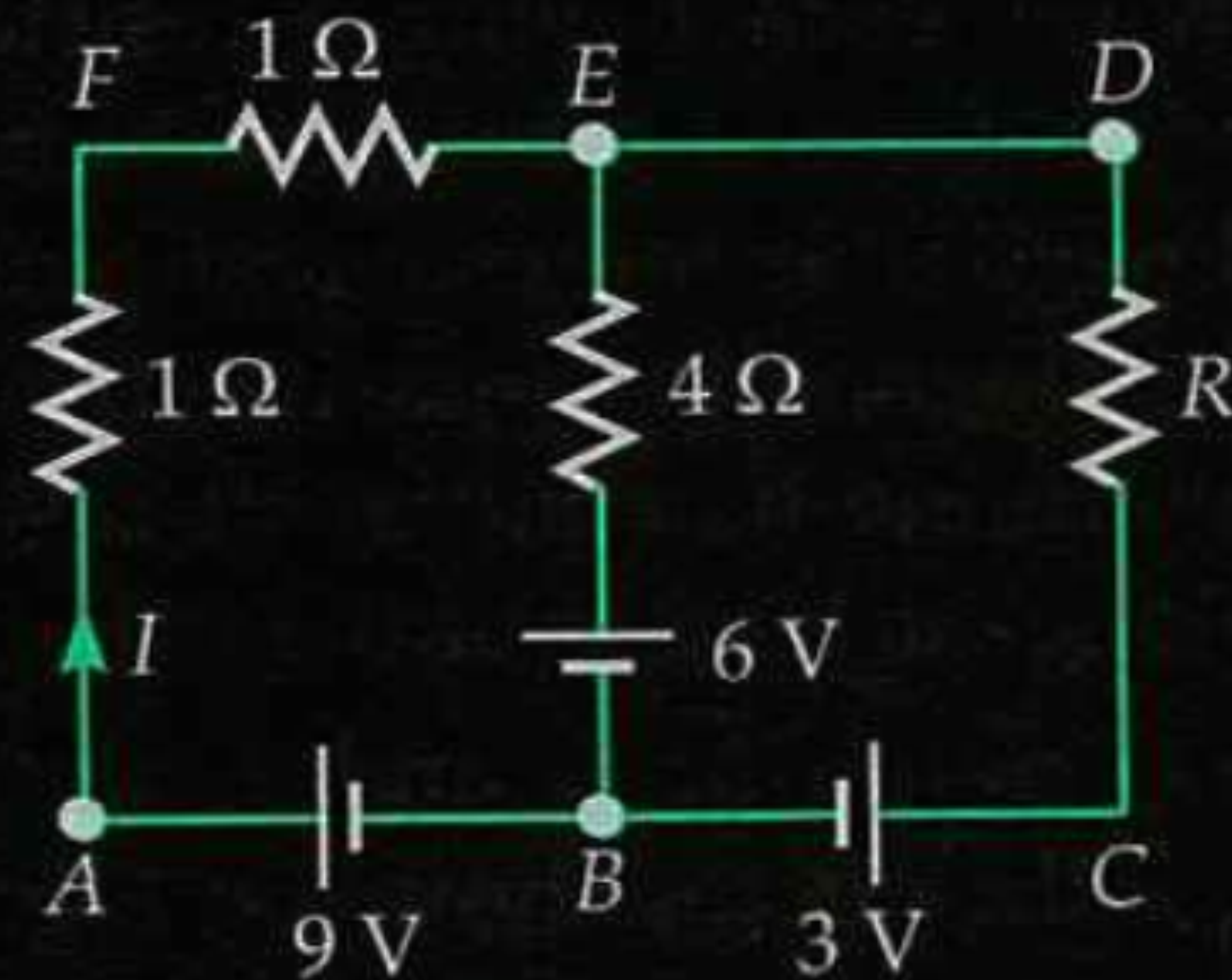
Solution. From the loop $BDCR_1B$, we get

$$2 \times 2 + 3 R_1 = 4 \quad \text{or} \quad R_1 = 0$$

$$\therefore V_B = V_C = 2 + V_A = 2 + 0 = 2\text{ V}.$$

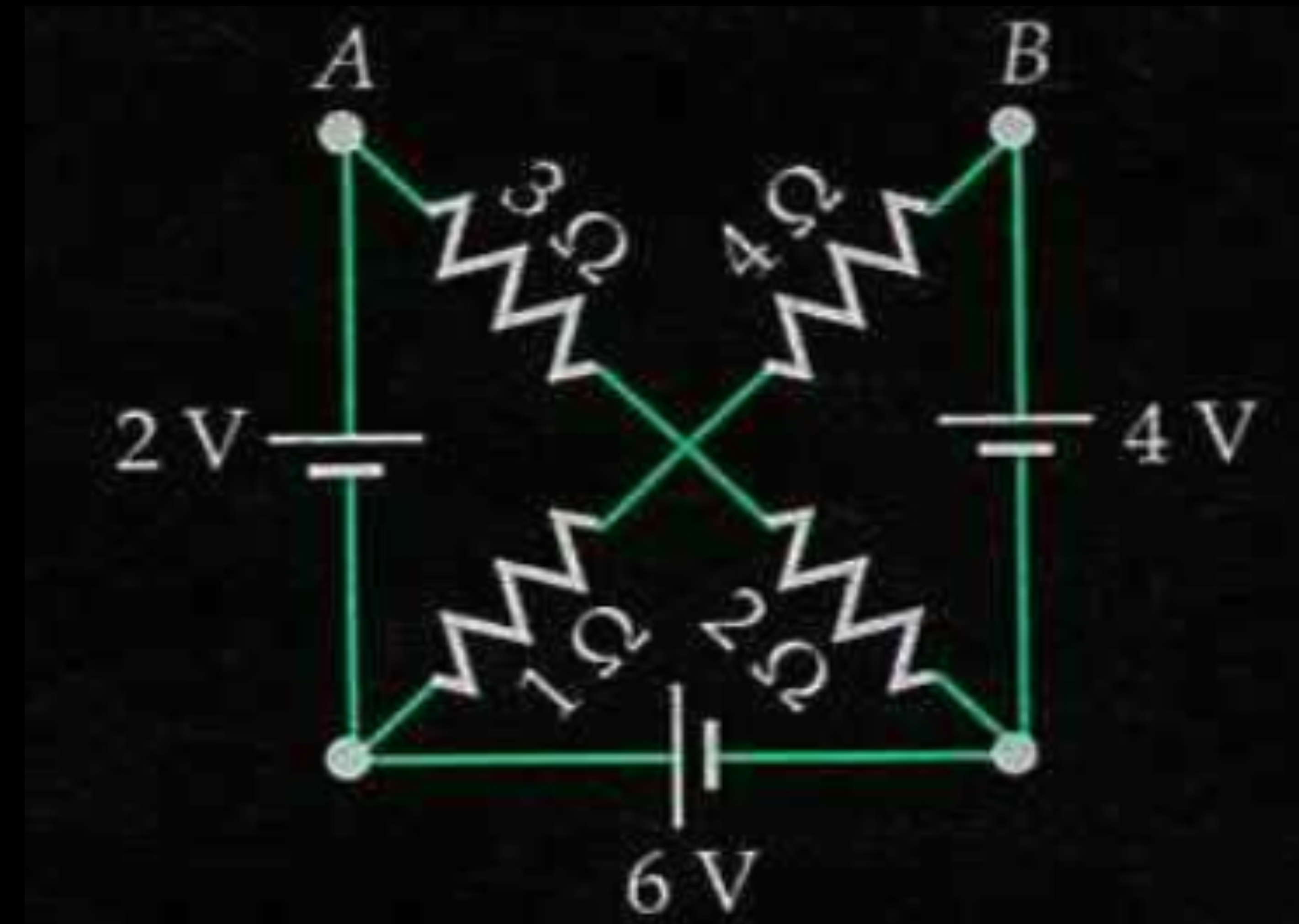
9. Using Kirchhoff's rules, determine the value of unknown resistance R in the circuit shown in Fig. 3.180 so that no current flows through 4Ω resistance. Also find the potential difference between A and D . [CBSE D 12]

(Ans. 3 V)



11. In the network shown in Fig. 3.182, (i) calculate the current of the 6 V battery and (ii) determine the potential difference between the points A and B .

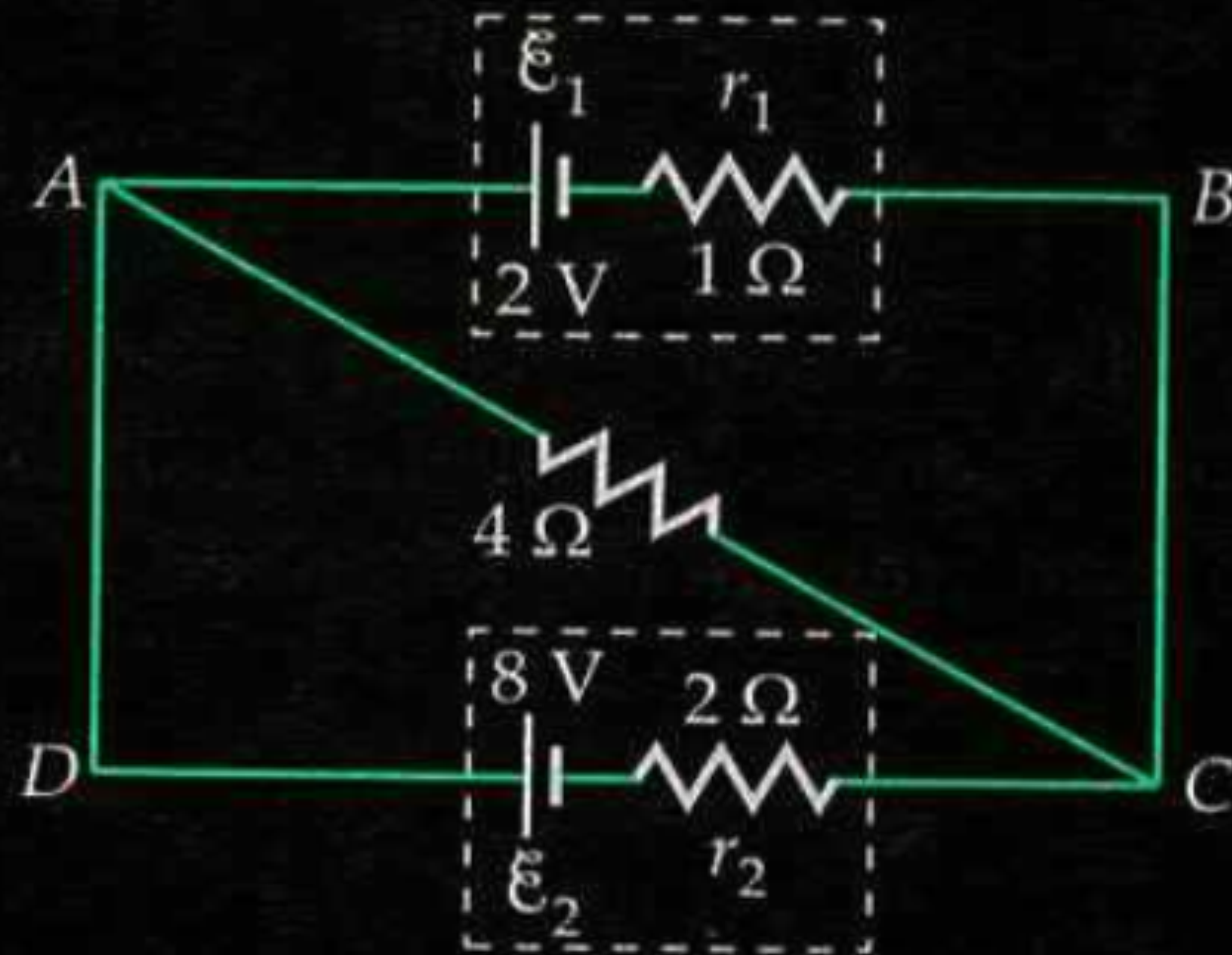
[Ans. (i) 2 A (ii) 4 V]



19. A cell of emf $\mathcal{E}_1 = 2 \text{ V}$ and internal resistance $r_1 = 1 \Omega$ is connected to another cell of emf $\mathcal{E}_2 = 8 \text{ V}$ and internal resistance $r_2 = 2 \Omega$ through an external resistance of 4Ω as shown in Fig. 3.189. Find the potential difference between point A and point C.

[CBSE 20C]

(Ans. $\frac{24}{7} \text{ V}$)



18. In Fig. 3.188, $ABCD$ is a uniform circular wire of resistance 2Ω . AOC and BOD are two wires along two perpendicular diameters of the circle, each having same resistance 1Ω . A battery of emf $\mathcal{E}$ and internal resistance r is connected between the points A and D . Calculate the equivalent resistance of the network.

(Ans. $\frac{15}{14}\Omega$)

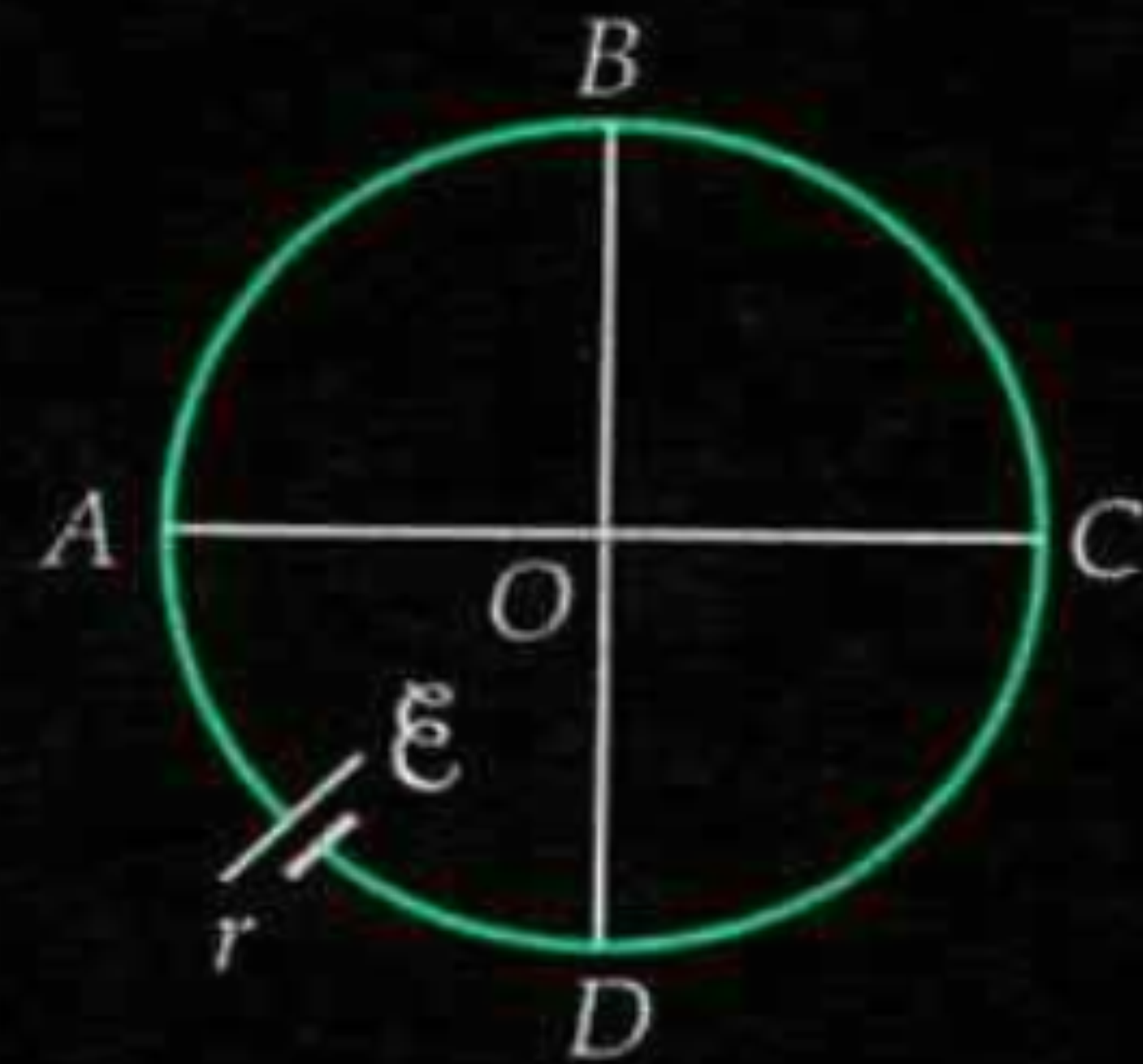
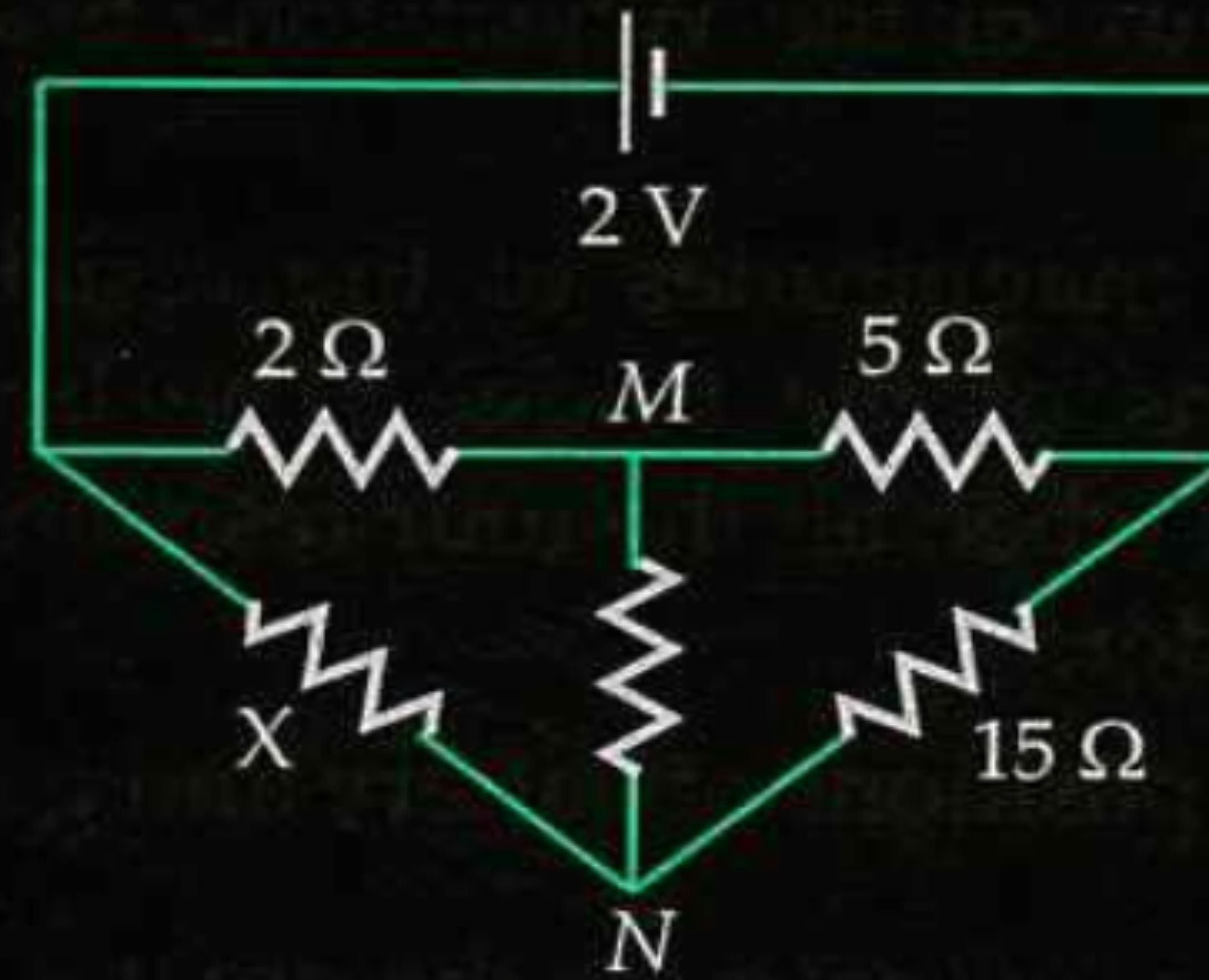
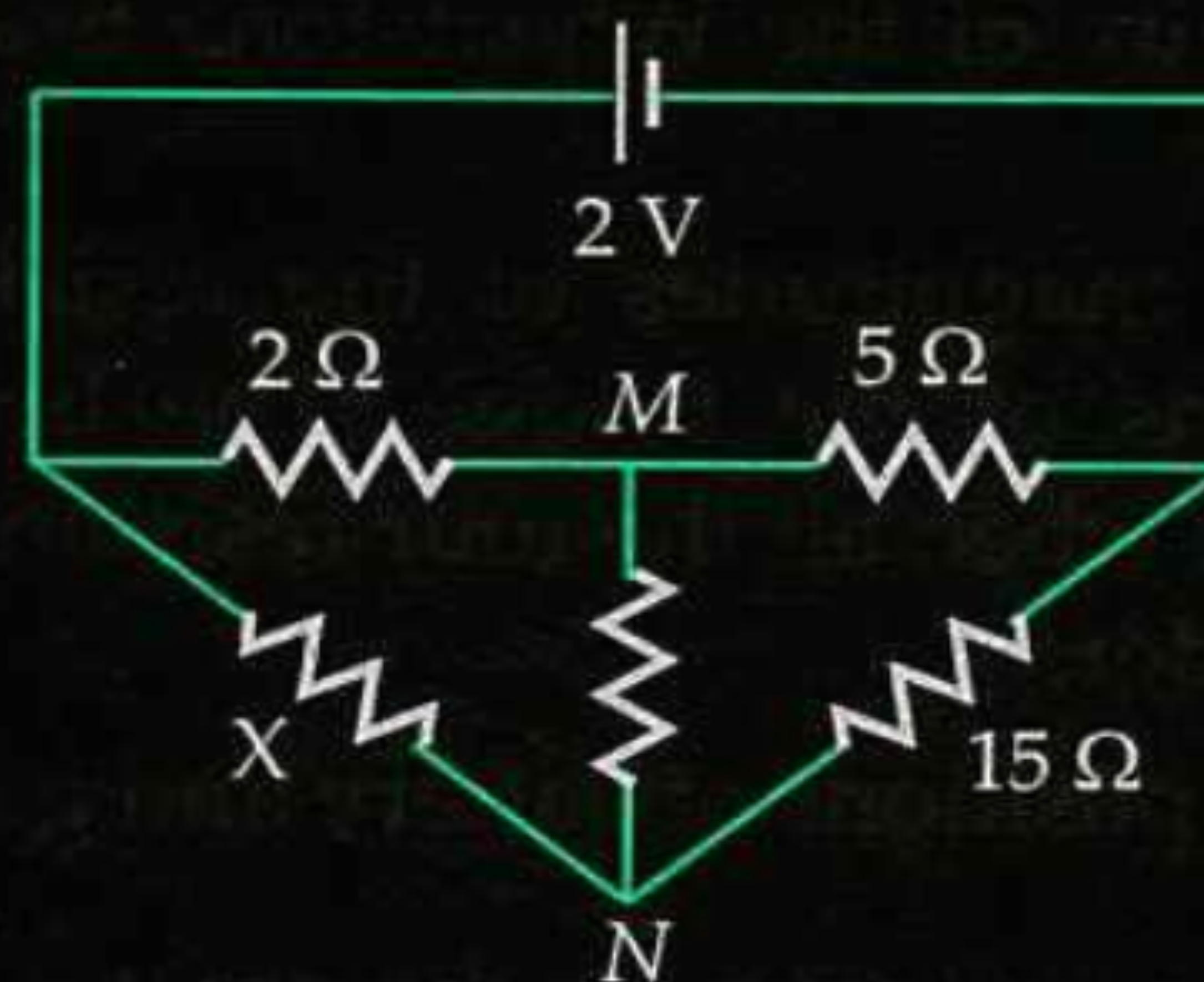


Fig. 3.188

Example 172. Find the value of resistance X in the circuit shown in Fig. 3.218 so that the junctions M and N are at the same potential. [ISCE 09]



Example 172. Find the value of resistance X in the circuit shown in Fig. 3.218 so that the junctions M and N are at the same potential. [ISCE 09]

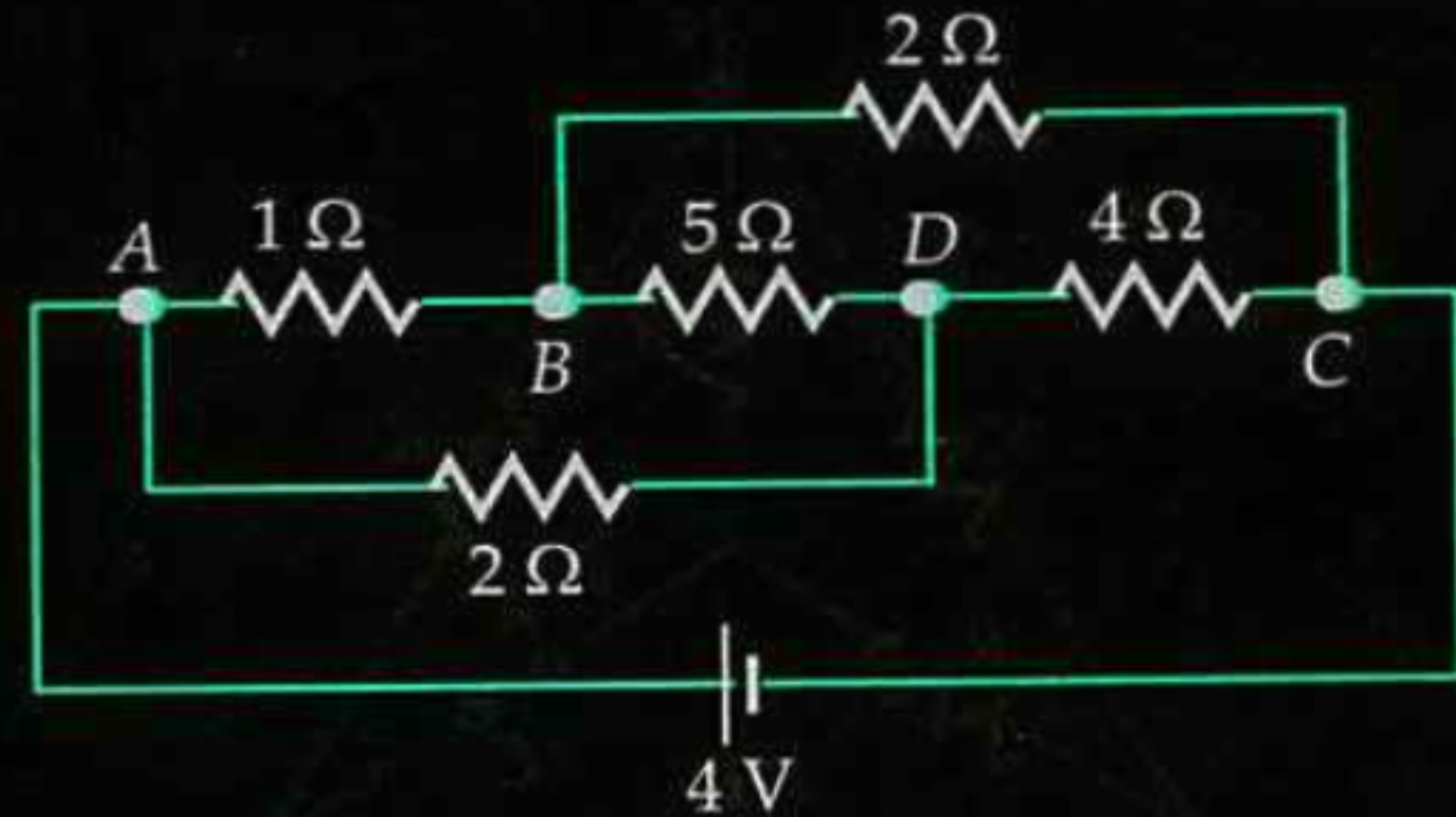


Solution. As no current flows through the middle arm MN , the circuit represents a balanced Wheatstone bridge.

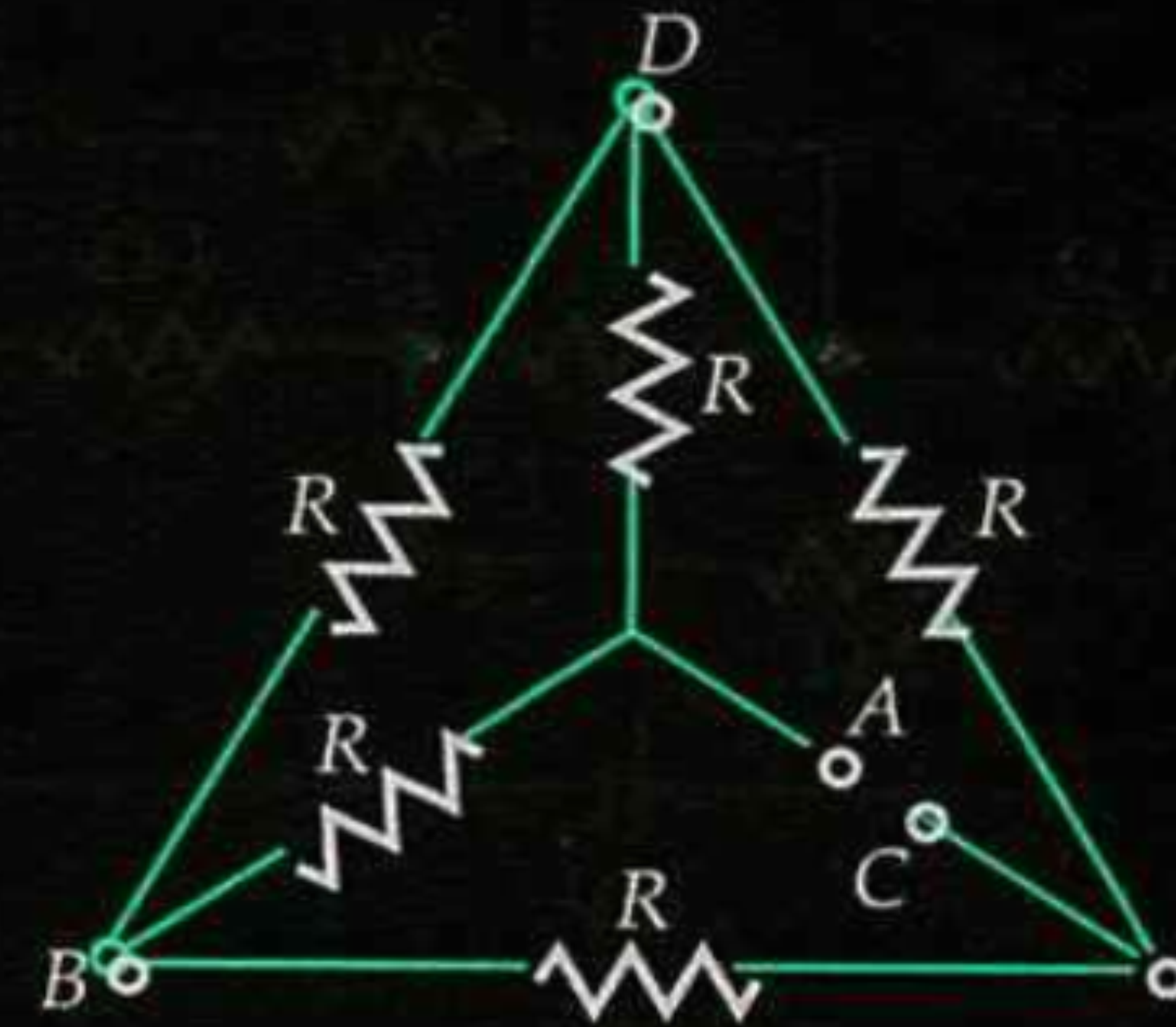
$$\therefore \frac{X}{15} = \frac{2}{5} \quad \text{or} \quad X = \frac{2}{5} \times 15 = 6\Omega.$$

Example 176. Calculate the current drawn from the battery by the network of resistors shown in Fig. 3.220.

[CBSE OD 09, 15C]

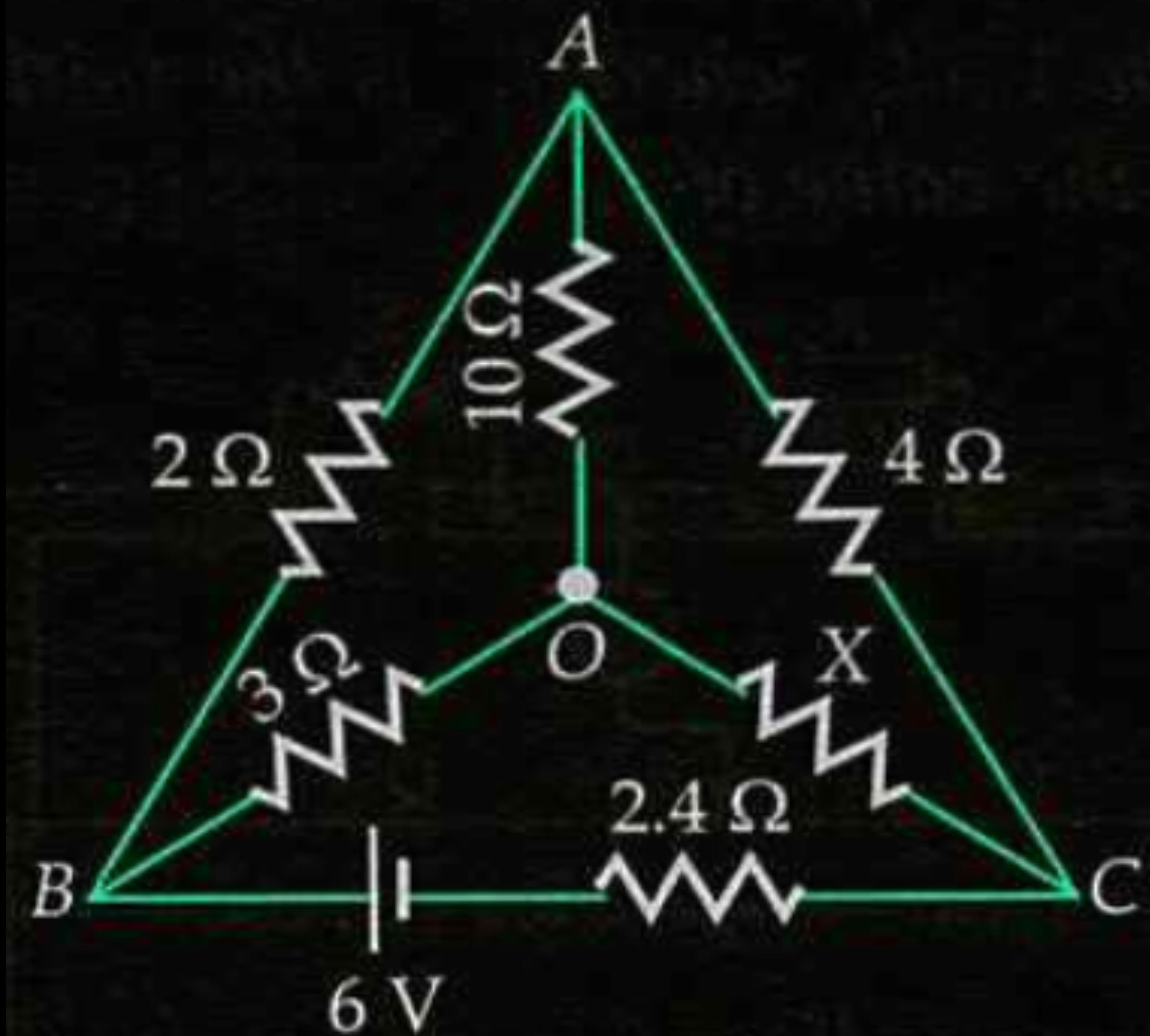


Example 177. Each of the resistances in the network shown in Fig. 3.222(a) equals R . Find the resistance between two terminals A and C . [JEE Main 22]

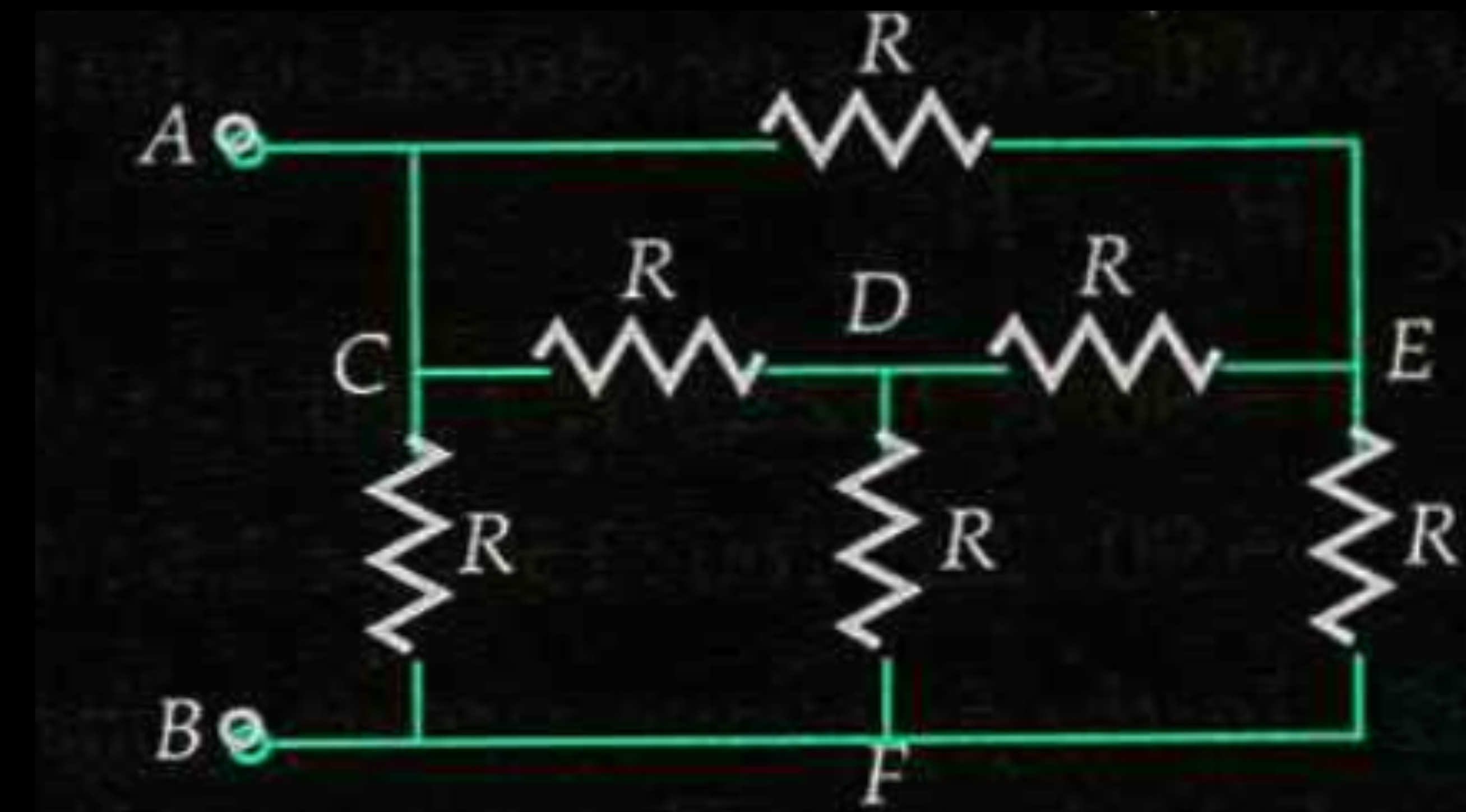


Example 179. Find the value of the unknown resistance X , in the following circuit, if no current flows through the section AO . Also calculate the current drawn by the circuit from the battery of emf 6 V and negligible internal resistance. [Fig. 3.225(a)]

[CBSE OD 02]

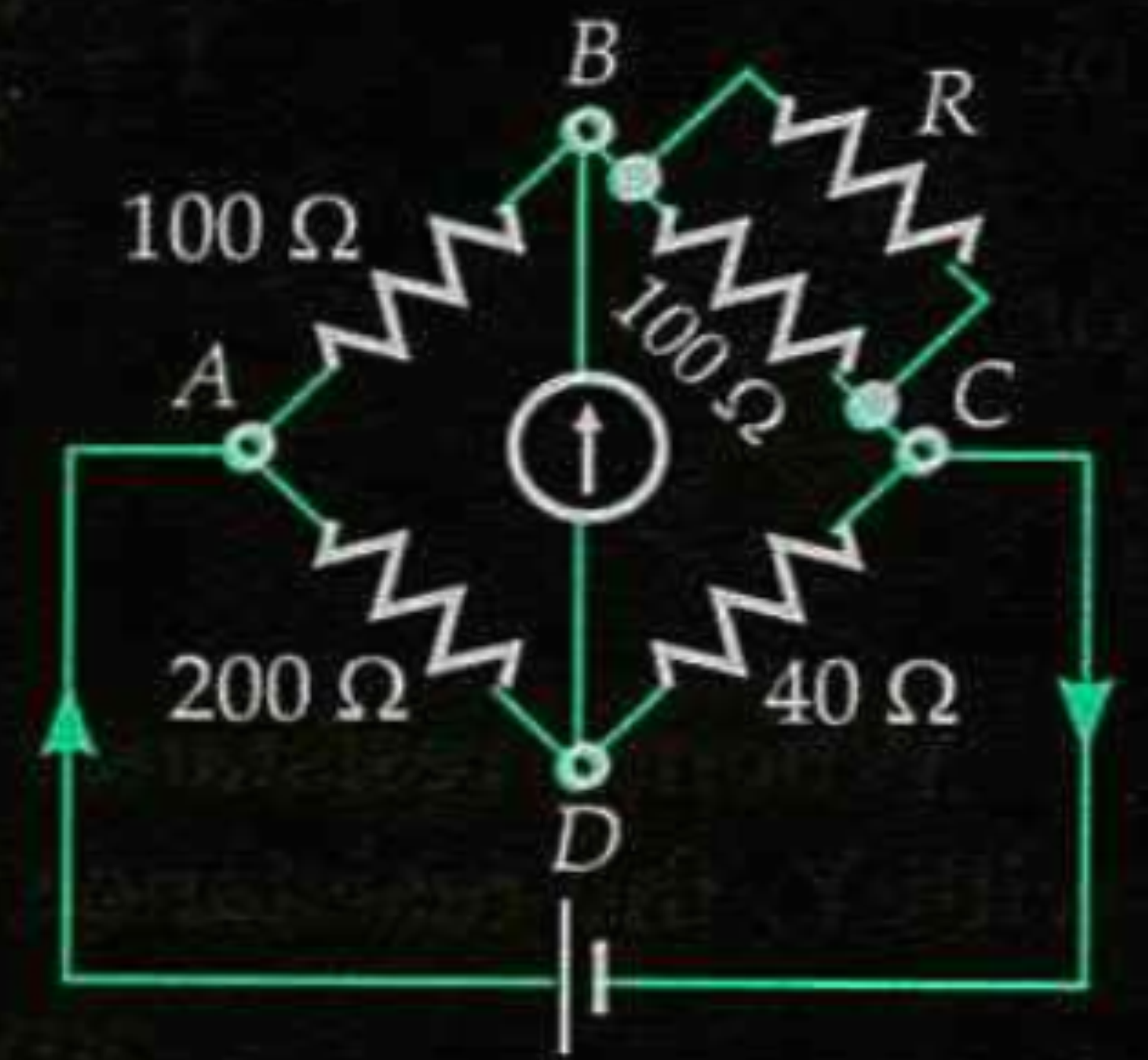


Example 180. Six equal resistors, each of value R , are joined together as shown in Fig. 3.226(a). Calculate the equivalent resistance across AB . If a supply of emf $\mathcal{E}$ is connected across AB , compute the current through the arms DE and AB . [CBSE SP 03]



1. The Wheatstone's bridge of Fig. 3.234 is showing no deflection in the galvanometer joined between the points B and D . Compute the value of R .

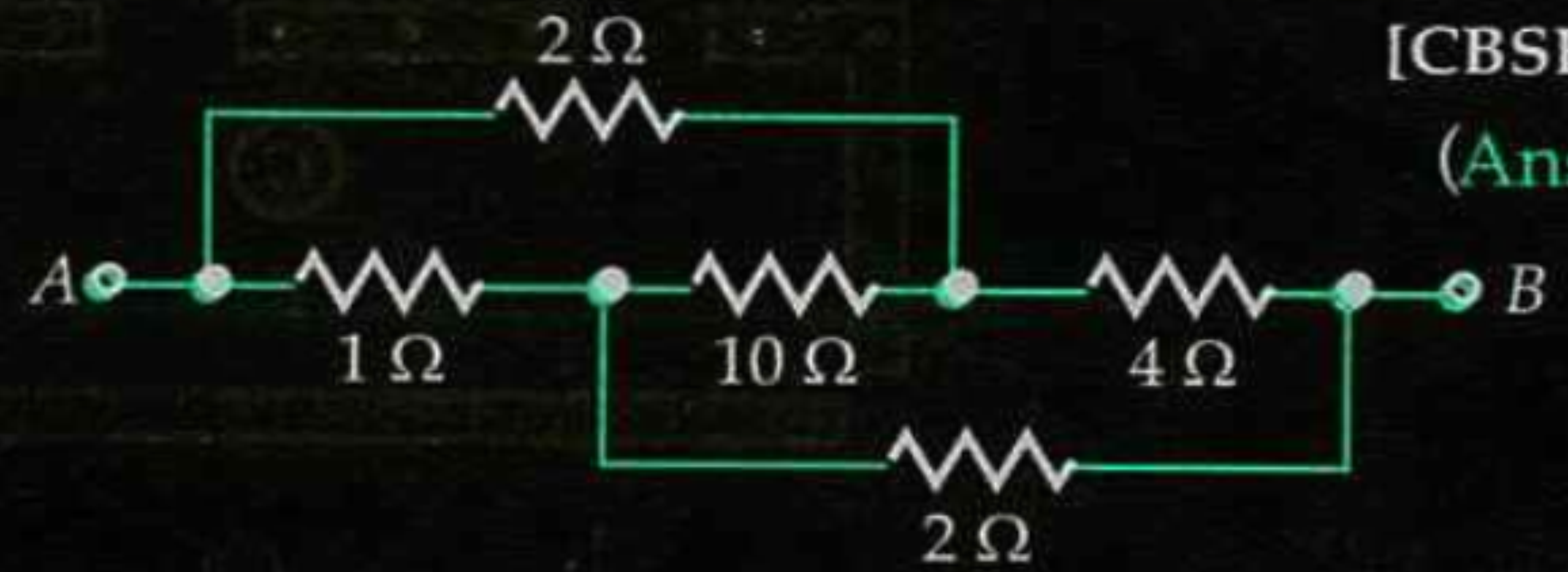
(Ans. $25\ \Omega$)



3. Calculate the equivalent resistance between points A and B of the network shown in Fig. 3.236.

[CBSE D 99]

(Ans. $2\ \Omega$)



12. Find the equivalent resistance between the points X and Y of the network shown in Fig. 3.245.

(Ans. $10\ \Omega$)

